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# 1 Welcome

Dear Scientist,

Thank you for choosing abberior for your microscope.

You invested in a versatile super-resolution and confocal imaging platform tailored to your imaging needs. This instrument has been designed and developed by scientists for scientists. We hope your imaging experience will be enjoyable and productive.

Your abberior team.

## 1.1 Super-resolution imaging with abberior

We offer innovative technologies packages to improve performance for confocal and STED super-resolution imaging. A list of the most important modules is listed in *TABLE 1.1-1*. Please note that this list may include modules not included or not available for your microscope and may not include custom solutions.

| STED 775 nm | STED 595 nm | Easy3D |
|---|---|---|
| Super-resolution for red, orange & long Stokes shift dyes | Super-resolution for green to yellow dyes | Super-resolution in all 3 dimensions |
| Fully pulsed lasers | Fully pulsed lasers | High stability via single-beam design |
| XY resolution down to 20 nm (<30 nm typical)* | XY resolution down to 25 nm (<40 nm typical)* | XYZ resolution down to 75 nm (90 nm typical)* |
| Standard/high-power version | Standard/high-power version | Compatible with many objective lenses |

| Adaptive Optics | Rainbow detection | MATRIX detector |
|---|---|---|
| Improved confocal & STED performance in deep samples (e.g., tissue) | Spectral detection range freely adjustable with two clicks | Background reduction for confocal and especially STED |
| Aberrations compensation for all beams (excitation, detection, STED) | Flexible range for higher signal and/or reduced crosstalk | Improved Z-sectioning and multiple processing options |
| Easy to use | Highest detection efficiency (up to 65%) | Easy to use |

| Adaptive Ilummination | TIMEBOW | Auto-alignment |
|---|---|---|
| Strongly reduced light dose | Easy FLIM imaging | One-click alignment |
| Lower phototoxicity | Easy phasor analysis tools | Alignment of all beams |
| Less bleaching | Lifetime images & overlays | Best super-resolution |
| Higher resolution | Lifetime species extraction | Best confocal images |

*Table 1.1-1: abberior modules for improved imaging*

*Depending on sample, dyes and laser models

## 1.2 Your microscope

**FACILITY:**

Our FACILITY platform offers you a workhorse instrument that boasts top-of-the-line super-resolution STED and confocal imaging. It combines cutting-edge microscopy with unprecedented ease-of-use. Our software allows beginners to intuitively arrive at a top-notch image within three clicks, while also giving experts full control over the instrument. Microscopy has never felt better.

Our FACILTY system is a truly cutting-edge micro- and nanoscope that is incredibly easy to use. It combines advanced features for high-end confocal and super-resolution imaging with foolproof operation. ADAPTIVE ILLUMINATION (DYMIN, RESCUE, MINFIELD), Adaptive Optics, EASY3D STED, confocal & STED autofocus, spectral RAINBOW detection, and full AUTOALIGNMENT.

**INFINITY / EXPERT LINE:**

The INFINITY platform was formerly known as EXPERT LINE. The INFINITY platform is the most customizable platform for all things microscopy. Whether it's STED and confocal microscopy, intravital microscopy, material science or optical trapping, the INFINITY is always up to the challenge. Just tell us what you need and we will build you a customized, continuously upgradable system specialized for your research.

The INFINITY is a highly flexible open platform that can be extended and upgraded, even years after purchase. It can be combined with all expert DEVICES and any custom developments to form a truly unique microscope that suits all your imaging needs. Get in touch with us and we will design and manufacture a tailor-made instrument that leaves nothing to be desired.

To protect your investment, we guarantee that your INFINITY grows with your science. And it grows with us. Your system could be a totally different machine a few years from now, because we at abberior make sure that everything we develop in the past, present or future can be retrofitted into your microscope. This way, you always stay at the forefront of science.

**MINFLUX:**

The MINFLUX platform offers an unprecedented array of imaging possibilities and allows you to resolve structures as small as a molecule, along all three dimensions. This unmatched resolution capability combined with unprecedented speeds reveals sample details never seen before. The MINFLUX is the world's most powerful fluorescence microscope.

Although impactful, previous super-resolution methods have failed to achieve resolutions on the length scales of the fluorescent molecules. Harnessing an entirely new and revolutionary localization principle, MINFLUX has finally accomplished this feat. Expect a resolution leap of 10-fold compared to other super-resolution methods and 100-fold compared to confocal fluorescence imaging.

MINFLUX is the most precise and most photon-efficient way of localizing fluorescent molecules. With its new MINFLUX system, abberior has produced the first commercial fluorescence microscope that can achieve localization precisions of 1-3 nm (3D) in biological samples on large fields of view (10 x 15 µm). MINFLUX imaging means getting the maximal localization information out of your sample.

## 1.3 Your microscope configuration

Your microscope configuration can be tailored for your research needs. Our support team and sales teams are happy to offer advice on further customization and compatibility of third-party components (***support@abberior-instruments.com*** *or* ***info@abberior-instruments.com***).

The following list (***TABLE 1.3-1***) shows a typical microscope configuration with many optional upgrades.

| Microscope body | |
|---|---|
| Type | inverted |
| Model | Olympus IX83 |
| Stage | motorized |
| Z drive | Z-piezo & nosepiece |
| Autofocus | Olympus ZDC |

| Objective lenses |
|---|
| 60X oil NA1.42 UPLXAPO |
| 60X water NA1.2 UPLSAPO |
| 60X silicone NA1.3 UPLSAPO |
| 20X dry NA0.8 UPLXAPO |
| - |
| - |

| Widefield Observation |
|---|
| Eyepiece (standard) |
| Camera (standard) |
| Brightfield |
| Epifluorescence (405/470/590/635) |
| DIC |
| Phase contrast |

| Illumination units |
|---|
| Brightfield lamp |
| Fluorescence (LED CoolLED) |
| - |
| - |

| Excitation lasers | Super-resolution lasers |
|---|---|
| 405 nm CW | STED 595 nm high power |
| 440 nm pulsed | STED 775 nm high power |
| 485 nm pulsed | - |
| 518 nm pulsed | - |
| 561 nm pulsed | - |
| 635 nm pulsed | - |

| Detectors |
|---|
| MATRIX (spectral) |
| MATRIX (spectral) |
| APD (spectral detection) |
| APD (spectral detection) |
| PMT (transmitted light) |
| PMT (reflected light) |
| - |
| - |

| Upgrades |
|---|
| Rainbow spectral Detection |
| MATRIX detection |
| easy3D STED (595 nm) |
| easy3D STED (775 nm) |
| TIMEBOW (FLIM) |
| Adaptive Optics (Deformable Mirror) |
| Adaptive Illumination (RESCUE, DYMIN & MIN-FIELD) |
| Autoalignment (full) |
| - |

| Software |
|---|
| Imspector (microscope control) |
| Lightbox (microscope control) |
| Huygens |
| SymPhoTime64 |

| Accessories |
|---|
| Incubation (cage, temp., humidity, CO2) |
| Multi-well plate sample holder |

*Table 1.3-1: Your microscope configuration*

## 1.4 Service contact

For assistance with super-resolution and confocal microscopes:

Call **+49 551 9995 4020**
or send an email to
**support@abberior-instruments.com**

For assistance with dyes and labels:

Call **+49 551 9995 4011**
or send an email to **support@abberior.com**

**Our team**

Our Support Department consists of microscopy experts, all with extensive experience in the field of super-resolution. This includes PhDs with a scientific background of 10-20 years in the development of STED, RESOLFT and related microscopy methods, and technicians with years of experience in building and customizing super-resolution microscope systems. They are actively involved in the continuous development of our products.

**Our philosophy**

Our focus is to cooperatively support our customers in carrying out their research and experiments efficiently and continuously. Every customer is of equally high importance to us and we pursue a “no downtime policy“. In the rare event of such an interruption, our customers can be sure that the system will be restored as fast as possible. To this end, our warehouses with replacement parts are continuously stocked. In case a component needs to be sent in for service, we supply every customer with an immediate replacement or offer a temporary exchange with a similar component, so the system’s availability is not compromised.

**The fastest remote support**

abberior offers immediate first-level support via remote assistance. For this, a member of our expert team connects to your system over the internet to check, calibrate, align, or maintain it remotely. Typically, 95% of all service cases are solved this way, because of the open platform concept of our systems, and because all critical components are motorized and can be aligned by software. This way, issues are fixed within minutes and experiments can be finished with hardly an interruption.

**Service worldwide**

To ensure fast support and short service times all around the world, we nurture a large network of certified partners and technicians trained by abberior. This enables on-site visits on short notice, wherever your microscope is. Additionally, our partners can be supported by our expert teams from Göttingen and Heidelberg, immediately via remote support, or on-site.

**Warranty and service extensions**

Beyond the general warranty of our systems, we offer a variety of different warranty extensions and service contracts for all our products. Please do not hesitate to contact us, if you have questions or want to have more detailed information about our service portfolio.

# 2 Imspector Image Acquisition & Analysis Software

The abberior microscope control software Imspector is the main platform for the operation of abberior MINFLUX, INIFINITY and EXPERT LINE microscopes. On FACILITY microscopes, the user can choose between the intuitive Lightbox interface or the versatile Imspector software. The Imspector software is proprietary and designed by abberior for control and gathering of measurement data as well as data analysis and data storage.

In the following sections, the Imspector software components will be explained to enable users ready for first measurements.

> ***Note:*** *Additional information is available online at* **www.imspector.de**.

## 2.1 Imspector Full version and BASE version

Imspector is available as a full version, which allows operation of a microscope. This includes drivers and dialogs for control of microscope hardware and settings.

Imspector is also available as a BASE version, which is intended to open, display and analyze data offline. Here, many dialogs for controlling hardware and settings are not available. However, image metadata (settings) can be exported or checked in specialized dialogs.

## 2.2 The Graphical User Interface (GUI)

The graphical user interface of Imspector features several core elements, that will be introduced in the following (***Figure 2.1***): Most elements fall into the class of measurement windows containing data , live dialogs (windows) for control, or Menu Bar items.

Each measurement may contain one or several measurement windows with data (e.g., images as shown here on the left). The live dialog Configurations summarizes the structure of measurement files and their respective measurement windows.

Visualization of the data in measurement windows can be controlled via buttons within the measurement window (***0***) as well as the Stack Display live dialog (***2.10.4***) which acts as a hub for all types of visualizations.

The basic imaging settings are defined and displayed in the easyCommander live dialog shown here on the right. For further options, a range of live dialogs can be selected (Menu Bar on top >Window > Live Dialogs…).

The live dialogs and measurement windows can be arranged freely in the program workspace. The selection and arrangement of live dialogs can be saved as a personalized workspace (Menu Bar on top >Window > Save Workspace..). The menu bar itself can be customized and saved as well.

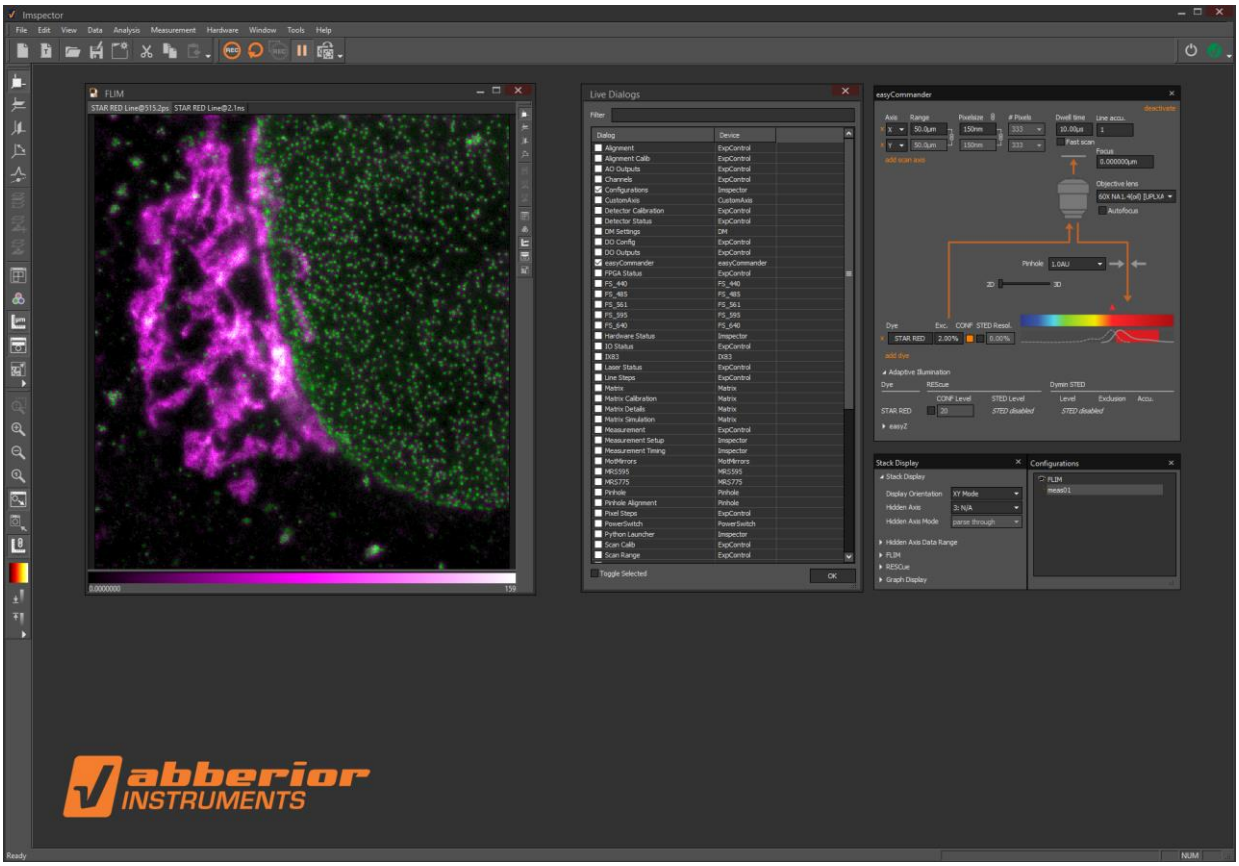



*Figure 2.1: General layout of the Imspector GUI*

Imspector software window with control and visualization buttons. Sub-windows include…

- easyCommander for creating/controlling measurements based on dye selection (top right)
- Measurement windows containing images, plots, metadata and visualization buttons (left)
- A list of available live dialogs for detailed parameter control (center)
- Stack Display live dialog acting as a hub for detailed visualization options
- Configurations live dialog providing a tree view of files and their measurements

## 2.3 Starting Imspector

Imspector is started by executing the file Imspector.exe, using the shortcut being placed on the desktop of the measurement PC. (The default installation directory is C:\Imspector\.)

During the startup procedure, all software modules will be loaded. A dialog will ask whether the user would like to power up the microscope. If you would like to acquire new images, please ensure all devices are ready and confirm. Decline if you would like to open previously acquired data only. Make sure to familiarize yourself with the complete startup procedure (see section ***4.1***).

On standard MINFLUX, INFINITY and EXPERT LINE microscopes, Imspector will load directly. On FACILITY microscopes and other setups with the Lightbox software, initially only the Lightbox interface will be displayed. The user can switch to the Imspector GUI via the key combination **“Ctrl+Shift+Alt+F11”**. The Imspector GUI can be hidden again via the same key combination.

### 2.3.1 Imspector Configurations

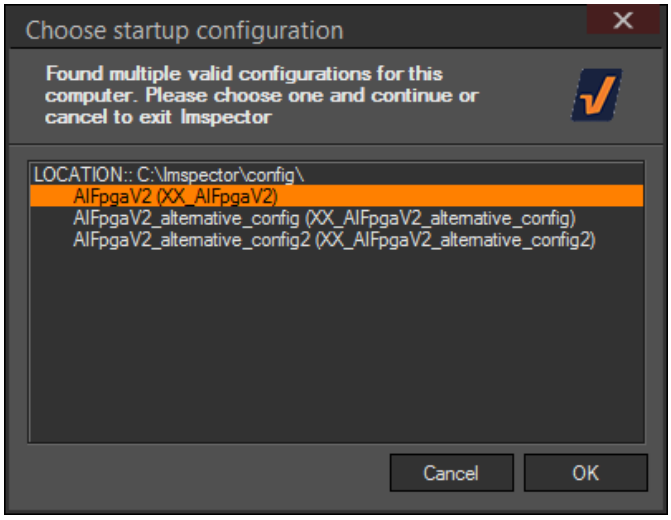



*Figure 2.2: 'Choose startup configuration' dialog*

Imspector allows working with several configurations that may be saved for individual users and/or usage scenarios. The configurations contain required hardware settings, display options, default measurements settings, etc.

The configuration (sub-)folders can be found in the folder C:\Imspector\Config\.

If a single configuration is in this folder, it will be automatically loaded (default setting after delivery of the instrument). If more configurations e.g., for individual users are placed in that folder, the software will start up with a screen in which you can choose between the user configurations (***Figure 2.2***). The configuration which was used during the last measurement session will be highlighted. After choosing the desired configuration and startup, Imspector proceeds to the main screen.

In case several different versions of Imspector are installed on the system, the following dialog (see ***Figure 2.3***) may appear. In this case, usually the first choice is preferable as this will load the configuration last saved by the currently open Imspector version.

In some cases, it can be advantageous to load a configuration generated more recently by an older Imspector version. This is true only if the more recent configuration includes major improvements to the alignment or settings that are lacking in the configuration generated with the current Imspector version.

In rare cases, e.g., after testing an early software release, users may want to load a configuration generated with a newer version of Imspector and downgrade this configuration.

> ***Note:*** *Upgrading a configuration to a newer Imspector version works reliably, while downgrading is not always successful.*

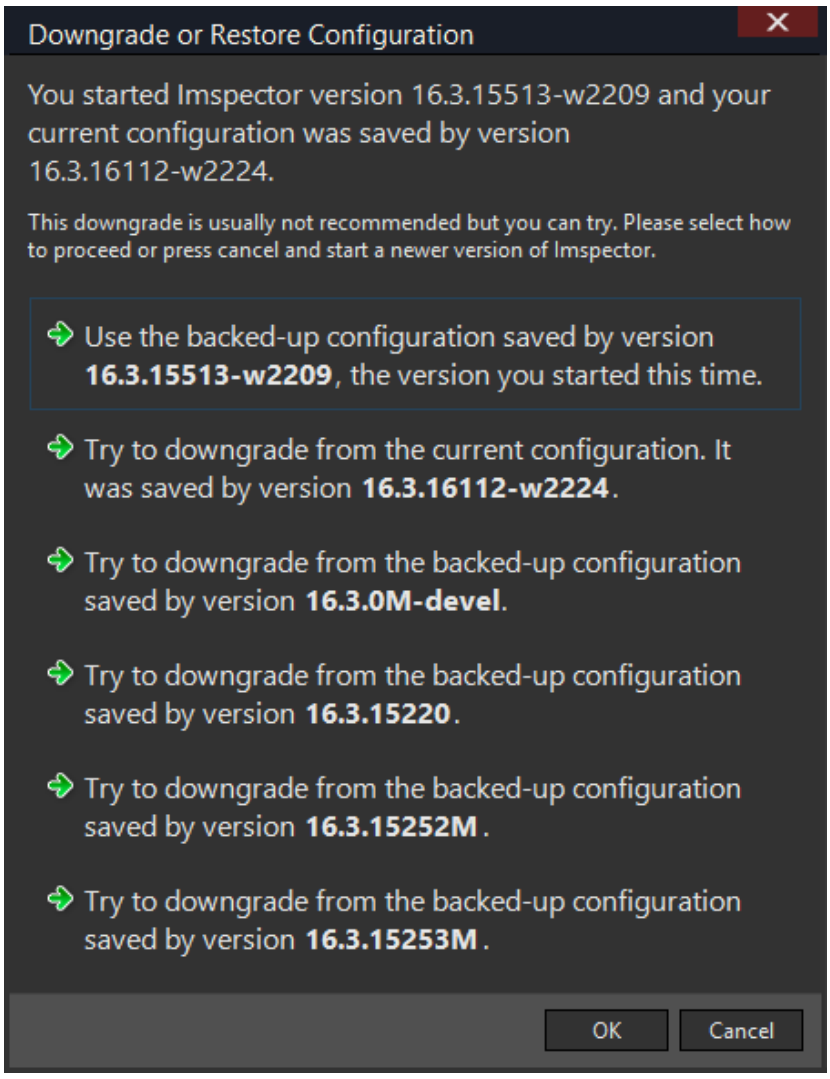



*Figure 2.3: Configuration options for multiple software versions*

## 2.4 Imspector Measurements

Imspector measurements may contain one or multiple windows. Each window is an individual dataset which can contain several individual measurement data stacks and/or analyzed data. The instrument parameters/meta data related to the measurement data are also attached to the same measurement window. A measurement window can be activated with a single click (on the frame or contents), this will display the attached imaging parameters in the relevant live dialogs (e.g., easyCommander).

All windows within a measurement will be saved in a single file in the '.msr' file format. The file and window structure is displayed in detail in the Configurations Live Dialog (***2.7.2***). The names of all individual stacks within a measurement window are displayed in tabs towards the top of the measurement window (double-click to rename).

> ***Important note:*** *By default, Imspector does not use any "auto-save" function. Starting an acquisition will always overwrite the data in the currently active window. Therefore, users should always save a measurement and its respective windows under a new file name as soon as relevant data has been recorded.*

The '.msr' file format can be read by Imspector, Imspector Base, Python and Matlab with the necessary extensions (see ***2.12***) as well as Fiji (or ImageJ with the BIOFORMATS plugin). Further, data can be exported into several common data formats (see ***2.13***).

In Imspector, the instrument parameters for performing a measurement can be defined in several ways. Parameters can be set and changed in an intuitive manner using the easyCommander live dialog (section ***2.4.1***) for new and existing measurements. Opening a predefined measurement template (section ***2.5***) or re-opening an existing measurement (section ***2.5.1***) offers the opportunity to re-record images with exactly the same settings as before. Opening a completely new measurement (Menu Bar > Open > New Measurement…) will start with a measurement with default settings (commonly a simple confocal image with one channel). Finally, expert users can adjust settings fully individually via a large range of additional live dialogs (here it is still recommended to start with standard parameters set up via easyCommander).

### 2.4.1 The Smart Way to Start: easyCommander

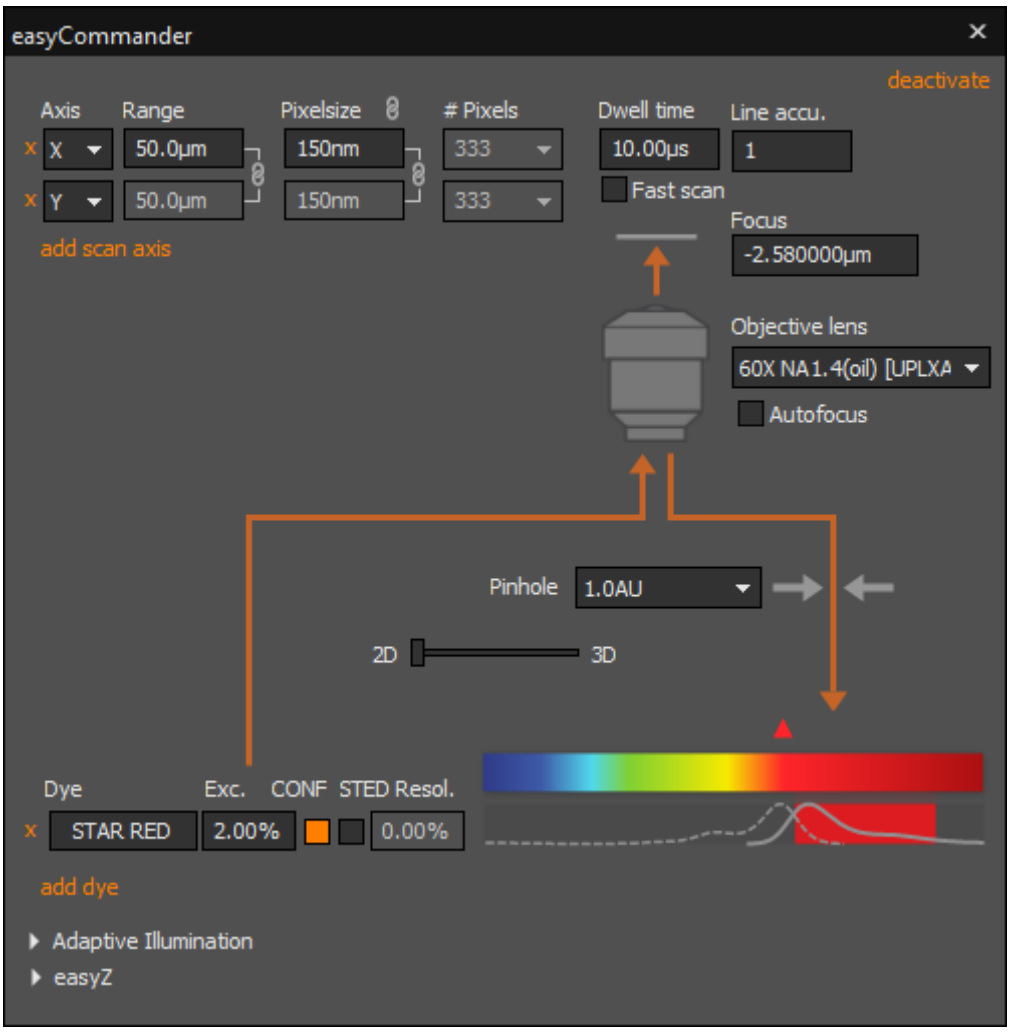



Figure 2.4: 'easyCommander' live dialog

**Introduction**

abberior easyCommander is a simplified, central control dialog to define a measurement in Imspector (*FIGURE 2.4*) and to improve ease-of-use in multi-user environments such as imaging core facilities.

easyCommander allows the user to choose the dyes used in the sample which automatically generates appropriate settings. The user can then focus on navigation and operation via a small selection of relevant scan parameters.

Most standard measurement scenarios are implemented in easyCommander. To maintain the simplicity of easyCommander, more advanced functions can be defined using additional control elements found mostly in the individual live dialogs.

Once users become more experienced, they are encouraged to use more of the specialized live dialogs to tap into the complete functionality and flexibility of the microscope. Here, we still recommend starting with a measurement set up via easyCommander followed by modifications made in live dialogs.

### EasyCommander details

The easyCommander live dialog consists of the following components:

**'Dye' and 'Add Dye':** This is the best place to start creating any measurement. In easyCommander, depending on the selected dyes, most measurement parameters are set automatically, including excitation lasers, STED lasers, time gating, detectors and spectral range. A dye can be replaced by clicking into the name e.g., STAR RED or removed from the measurement by clicking the “x” left to each dye. New dyes can be added via the “add dye” link.

When adding or replacing dyes, the 'Dye selection' dialog (*FIGURE 2.5*) will open. Here the user can pick the correct dye or a spectrally similar dye from a large selection. The 'Dye Selection' is connected to the abberior dye database that contains information about individual dyes. This allows automatic selection of the best detection window, excitation laser and STED laser.

> ***Note:*** *By default, the “Dye Selection” dialog displays a selection of common dyes suitable for STED or dyes that can be important as confocal reference channels. For a broader selection of dyes, please tick the checkboxes “Show extended list” and/or “Show incompatible dyes”.*

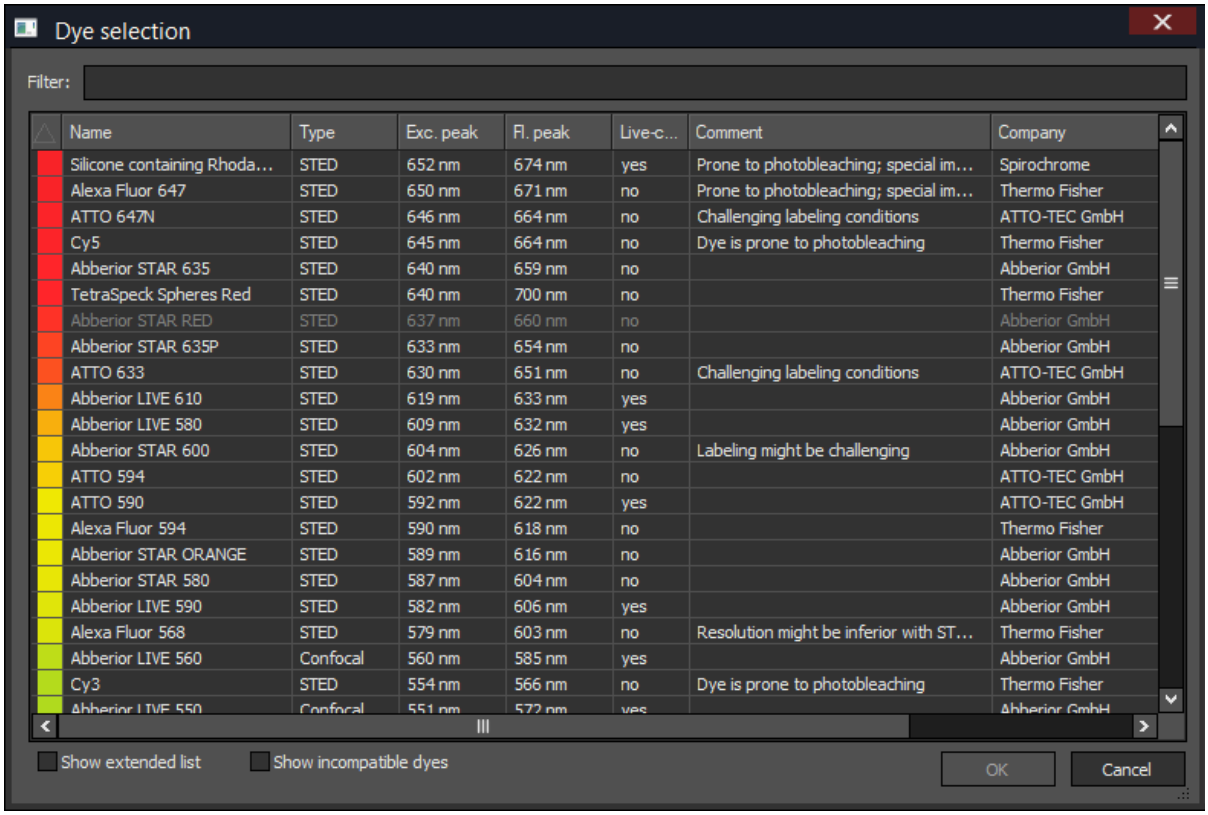



*Figure 2.5: 'Dye selection' dialog*

**'Axis':** Here the type of image or measurement can be determined by selecting the Scan Axis (X, Y, Z, T). This allows scanning of a single axis (e.g., just X) or the scanning of multiple axis (e.g., standard XY images, XZ images and more). There are additional axis types that are available only from specialized live dialogs, e.g., spectral scans (see *SCAN RANGE LIVE DIALOG*) and custom axis (see *CUSTOM AXIS LIVE DIALOG*) and will be displayed in easyCommander as “λ” and “?” axis respectively.

**'Add scan axis':** This option allows the addition of further axis to acquire e.g., a volume (XYZ) or timeseries (T). There is currently a maximum number of four axis. The axis will be scanned in the order displayed in easyCommander. Generally, it is recommended to scan the fastest axis first, e.g., XZ instead of ZX. The exact timing (syncing) of all scan axis can be modified in the Measurement Setup Live Dialog.

**'Range':** The range defines the size of the acquired image for each axis. The knot next to the input fields can be used to link or unlink the two axis of the range. If the X and Y axis are linked, the image aspect ratio is fixed, i.e. when the X axis is changed, the Y axis is automatically adjusted by the same factor. The maximum range is defined by the current objective (e.g., 133 µm x 133 µm FOV for a standard 60x objective).

**'Pixelsize' and # 'Pixels':** The pixelsize can be set for every axis independently. The knot to the right hand side of the input field can be used to link the pixelsize, hence creating an image with isometric pixels. The knot above the input field locks the number of pixels. If this is unlocked, it is possible to change the number of pixels and the pixelsize is adapted correspondingly.

**'Dwelltime':** The dwelltime defines the acquisition time per pixel in a single line scan. The default value is usually 5 or 10 µs per pixel.

| | **Range [µm]** | | **Pixel size [nm]** | | | **Auto Rep.** | **STED** | **Line Accu.** |
|---|---|---|---|---|---|---|---|---|
| | X | Y | X | Y | Z | | | |
| **Overview** | 133 | 133 | > 250 | > 250 | | Yes | No | 1 |
| **Closeup** | (10)* | (10)* | > 150 | > 150 | | Yes | No | 1 |
| **Fine (confocal)** | (10)* | (10)* | 50 | 50 | 150 | No | No | >2 |
| **Fine (2D STED)** | (10)* | (10)* | 20 | 20 | 150 | No | Yes | >3 |
| **Fine (Confocal & 2D STED)** | (10)* | (10)* | 20 | 20 | 150 | No | Yes | >3 |
| **Fine (Confocal & 3D STED)** | (10)* | (10)* | 50 | 50 | 50 | No | Yes | >3 |

*Table 2.4-1: Recommended scan settings (60x objectives)*

* Depending on the ROI

**'Line accu. ':** This field is a master switch controlling how many times a line of pixels in the image is scanned. In each pixel, the signal of the repeated line scans is summed up.

When a measurement is set up via easyCommander (without changes in additional live dialogs), the field “Line accu.” simply sets the number of consecutive scans of the same line for each channel before moving on to acquisition of the next channel. For each channel, the signal is accumulated (summed) over the respective number of consecutive line scans. For all STED channels, the actual number of line accumulations is 3x the value displayed in easyCommander. Hence, setting the line accumulations to 3 in easyCommander will generate a confocal image with 3 line accumulations and a STED image with 9 line accumulations. This usually generates more comparable signal levels in confocal and STED channels as STED switches off a large number of fluorophores to generate a smaller scanning spot.

It is important to note that Imspector allows the creation of far more complex line scan patterns via additional live dialogs. For a complex line scan pattern, the easyCommander field “Line accu.” only sets the number of repetitions of this complex line pattern. For detailed information on the organization of channels and line scan patterns, please see sections ***Channels and Measurement Live Dialog*** and ***Line Steps Live Dialog***.

**'Fast Scan':** The fast scan checkbox activates a different scan mode optimized for fast acquisition of small images. This works best for images up to 10 x 10 µm. Avoid this scan mode for large images, as it can lead to lateral image shifts or edge artifacts.

**'Focus':** This input field allows to change the current focus position by typing a number or by clicking into the field and scrolling with the mouse wheel. The shown focus position is either the position of the microscope nosepiece (identical to the z-position in the Olympus Touch panel control (TPC), usually between 0 and 10 mm) or, if available, shows the z-position of the installed z-piezo (Usually between -110 and +110 µm).

**'Objective Lens':** Here, installed objective lenses can be selected from a drop-down menu. The lens is only changed once a measurement is actively recorded. Upon start of the recording a dialog appears, and the user needs to confirm the changing of the objective to start the measurement (see ***Figure 2.6***) e.g., in case the immersion medium is changed).

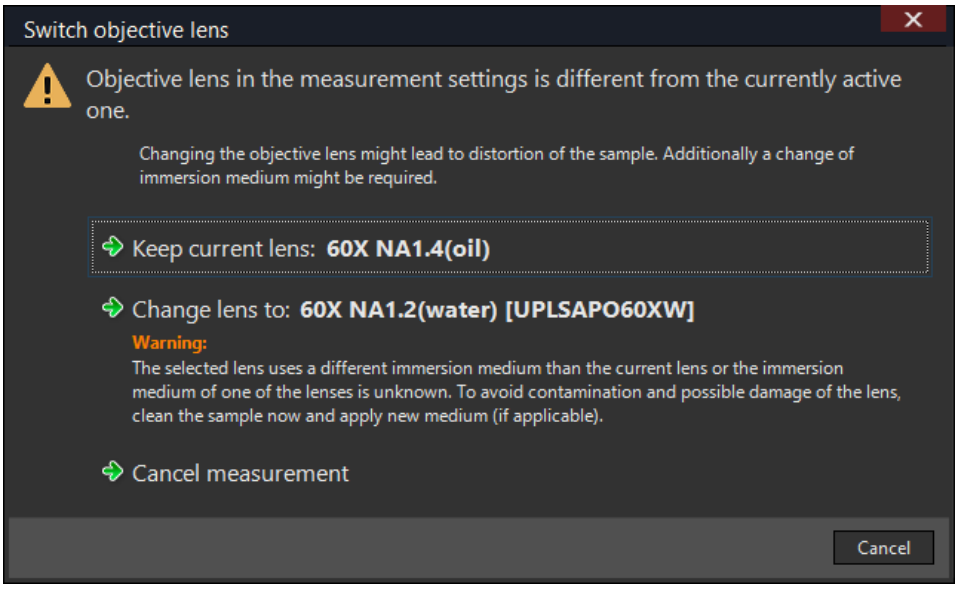



Figure 2.6: 'Change Objective' dialog

**'Autofocus':** In case an Olympus autofocus unit (ZDC) is installed, the autofocus can be activated by ticking this checkbox. Upon activation, the autofocus will perform a focus search. Once it has detected a focus position it will start to lock onto this position. The autofocus has a total working range of approximately 35 µm (from the interface of coverslip and sample medium) within which it can compensate axial drift. For detailed information please see the hardware quick guide section ***Olympus ZDC: Focus Search / Continuous*** Autofocus and ***Microscope body Live Dialogs***.

**'Pinhole':** A dropdown menu allows the selection of the pinhole used for this measurement. Based on pinhole size the corresponding AU is shown for each used objective.

**'2D-3D' STED Slider:** This slider determines whether STED enhances the resolution either only in the XY dimensions (2D-STED), nearly equally in all 3 dimensions (3D-STED) or using a mixture of both extremes. An easy3D STED module (optional upgrade) is required for 3D-STED and the seamless power distribution between 2D-STED and 3D-STED.

> ***Note:*** *The slider is displayed only when one of the STED lasers is activated in the measurement and only if the setup is equipped with an easy3D module.*

While 2D-STED increases the resolution in the XY plane dramatically (<30 nm depending on setup configuration), the resolution along the Z-axis remains the same as for confocal images (~600 nm depending on settings). 2D-STED is best suited for single XY slices requiring optimum XY resolution and Z-projections of volume images.

3D-STED increases resolution in all three dimensions (XYZ) nearly isotropically (<80 nm depending on setup configuration). However, 3D-STED increases the XY resolution much less than 2D-STED does. 3D-STED is best suited for vertical slices or volume images for rendering and 3D analysis.

The slider allows mixtures between 2D-STED and 3D-STED allowing users to find their preferred balance between best XY resolution and best Z-sectioning. For more information see section ***2.7.12***.

**'Exc':** In this input field the excitation of the confocal channel and the STED channel can be set as percent of available excitation laser power. A standard step corresponds to 1%, pressing CTRL can be used for small incremental steps (0.1% per step) and SHIFT for large steps (10% per step). The excitation for the STED channel will be automatically increased relative to the used STED laser power for optimal performance.

**'Conf.' / 'STED' Selection:** These checkboxes allow activation or deactivation of acquisition of the confocal and STED channel. A filled orange box indicates that a channel will be acquired, a dark box indicates that this channel is currently inactive. Allowed options are Confocal active, STED active or Confocal and STED active.

**'Resolution':** The STED resolution is the STED laser power used in the current acquisition. This is given as percentage of the available maximal STED power of the given STED laser.

> ***Note:*** *Resolution increases with STED laser power, albeit in a non-linear fashion. Very low STED laser power may already provide a significant resolution increase compared to standard confocal imaging. Moving to higher resolutions, more and more STED laser power is required for further increases and photobleaching will eventually limit the achievable resolution.*

**Detection Spectrum:** The spectrum representation (*FIGURE 2.7*) displays the information of the currently selected dye according to the dye database. Information includes: The excitation spectrum (dotted line), the emission spectrum (solid line), the automatically selected excitation laser (up-facing arrowhead), the automatically selected depletion laser (down-facing arrowhead) and the selected detection window (colored box).

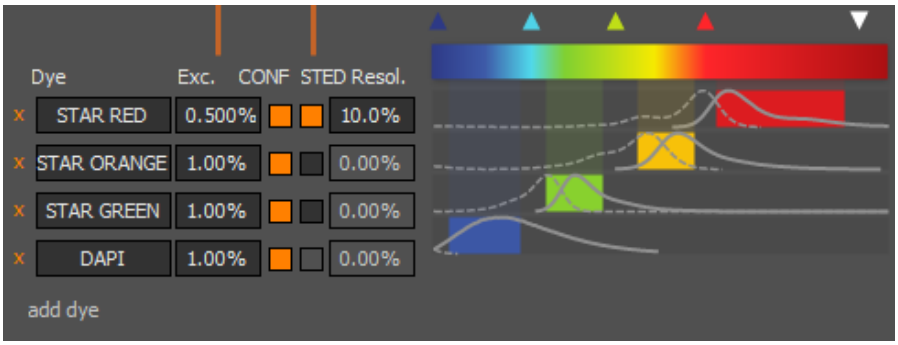



Figure 2.7: Selected excitation parameters and spectral windows in ‘easyCommander’

## 2.4.2 easyZ volume imaging

For volume images (or any measurement involving a Z axis), the user needs to define the Z axis range and position.

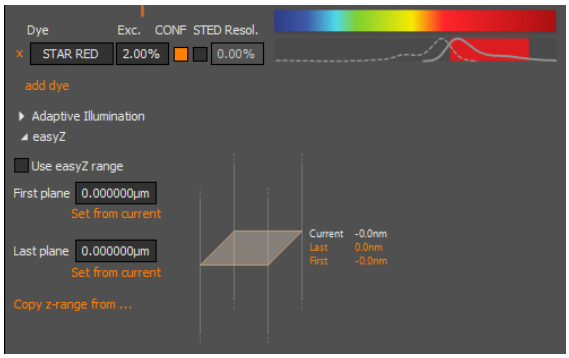
### Quicker simple solution

A quick solution is simply setting the Z axis range in the easyCommander axis options. This produces an image with the chosen Z-size (“Range”) centered onto the current focal position along the Z axis.

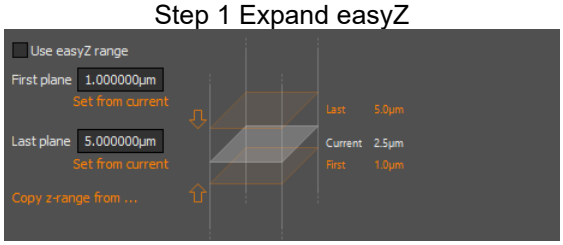
### Well-defined Z-limits via easyZ

A more advanced solution is the easyZ tool, allowing users to identify and set the lower and upper Z limits of the image. The ‘easyZ’ tool is part of ‘easyCommander’ and can be expanded via the easyZ drop down menu located at the bottom of easyCommander (see first panel in

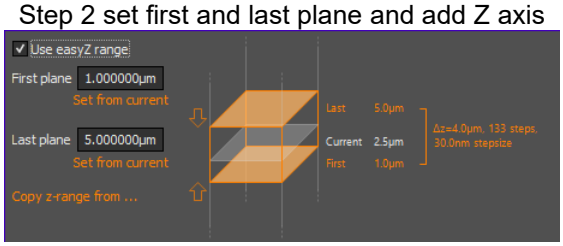
*FIGURE* 2.8). There are two options to define the Z coordinates via easyZ:




Step 1 Expand easyZ




Step 2 set first and last plane and add Z axis




Step 3 Activate easyZ range

Figure 2.8: 'easyZ' live dialog

The first option is the ‘Copy z-range from’ command. This can be used if there is already another measurement window containing the correct easyZ settings. The upper and lower Z limits are then copied from the selected measurement window.

The second option is to manually select the first and last plane of the volume while scanning the intended XY ROI live (in auto repetition mode). During the live acquisition, adjust the focus position in ‘easyCommander’ via the mouse wheel. The position of the current focal plane can be seen as grey plane in the dialog and is labelled ‘current’. The order of selecting the lower and upper limit is important:

- **First the lower Z limit** (lower focus value, first plane) needs to be set. For this purpose, bring the current focal position to the intended lower Z limit and click ‘set from current’ just below the easyCommander field ‘first plane’ .
- **Then the upper Z limit** (larger focus value, last plane) can be approached by adjusting the current focus position. Set the upper Z limit by clicking ‘set from current’ just below the easyCommander field ‘last plane’.

Once the planes are set the checkbox ‘use easyZ range’ can be activated. This transfers the selected upper and lower limits to the ‘easyCommander’ main menu and shows the size of the volume in micrometer, the number of planes and the step size in the easyZ dialog. This also adds a Z-axis to the scan parameters in easyCommander. The Z-pixel size can then be set in ‘easyCommander’. Upon hitting record, the defined z-stack will then be acquired.

## 2.4.3 Adaptive Illumination

### *2.4.3.1 Adaptive illumination concept*

Adaptive illumination is a set of patented technologies developed by abberior to reduce laser illumination of the sample. The reduction in illumination allows the user to optimize the photon budget (maximum available fluorescence signal), minimize photo-bleaching and to reduce phototoxicity. This enables high-quality confocal images and higher-resolution STED images even in challenging and live samples.

The basis of adaptive illumination is blanking (switching off) the lasers whenever sample areas are scanned that do not require further illumination, i.e. background areas or empty areas without any structures of interest. Blanking is particularly helpful in STED imaging mode to reduce illumination with the strong depletion (STED) lasers but can also help to reduce the light dose in simple confocal mode. The decision, when and where to blank the lasers can be made via the following mechanisms:

- **Probe step:** Prior to scanning the sample with high laser power (e.g., high resolution STED imaging), the sample is scanned with much lower laser power (e.g., confocal mode or low-resolution STED mode). In the low-power scan, areas with and without structures are identified based on the fluorescence signal and a user-defined signal threshold. This information is used to create a laser-banking mask which is applied to the following high-power scan, blanking the high-power lasers in those areas without structures of interest. Typically, probing is done line-wise, scanning each image line first in probe mode and then in high power mode (applying the blanking mask) before moving to the next line.
- **RESCue (lower threshold):** This mechanism requires only one imaging channel/step and the decision to blank lasers is made during acquisition of this channel. Each pixel is scanned for a certain amount of time, called the pixel dwell time. RESCue allows abortion of laser illumination after only a fraction of the pixel dwell time, reducing total laser illumination. The decision whether to continue or abort laser illumination is again based on whether the current pixel contains structures of interest. For this purpose, the fluorescence signal is checked at several time points before completion of the pixel dwell time. At each checkpoint there is an estimate of the chances of reaching a user-defined minimum fluorescence signal that should be obtained at completion of the dwell-time. If the chances to are too low, laser illumination is aborted immediately. This RESCue mechanism can be applied to confocal and STED imaging.
- **RESCue (upper threshold):** Just as described above for RESCue (lower threshold), this mechanism requires only one imaging channel/step and the decision to blank lasers is made during acquisition of this channel. Here, total laser illumination can be reduced by aborting illumination after only a fraction of the pixel dwell time, as soon as enough fluorescence signal has been collected in the current pixel. The software will then extrapolate the signal for this pixel, calculating the signal level that would have been reached by illuminating for the duration of the full pixel dwell time. The fluorescence signal level (upper threshold) at which illumination is aborted, is derived from the minimum fluorescence signal defined by the user for RESCue (lower threshold) as mentioned above. By default, the upper threshold is defined as 1.5x the value of the lower threshold. This RESCue mechanism can be applied to confocal and STED imaging and is automatically activated with RESCue.

The laser-blanking is performed individually for each pixel via fast FPGA-based technology enabling line-by-line decisions (and even pixel-by-pixel). This renders the process fast enough even for moving objects in live cell imaging.

> ***Note:*** *For adaptive illumination, we recommend using pixel dwell times above 5-10 µs to allow completion of laser power adjustments for each pixel.*

Importantly, blanking lasers in one pixel also protects dyes in neighboring pixels from bleaching. This is due to the fact that the focal spots of the lasers are usually much larger than a single image pixel (e.g., ~200 nm for excitation lasers, >600 nm for STED lasers). In fact, the donut shape of the STED laser results in a higher protection of neighboring pixels (at the intensity crest of the donut) compared to the current pixel (at the intensity minimum in the donut center).

The definition of areas that do not require further illumination is done automatically on the fly during scanning, generating a laser-blanking mask. The input required by the user is the definition of simple thresholds for the fluorescence signal, assuming that pixels with signal below threshold do not contain structures of interest and therefore do not need further illumination. Optionally, the user-defined thresholds can also set a maximum signal level at which illumination is aborted (see 'RESCue' above). To achieve optimal use of the photon budget and highest resolution, the user can select from several adaptive illumination modes:

- ***STED* WITH CONFOCAL PROBING**
- ***DYMIN: STED* WITH DUAL PROBING**
- ***RESCUE STED* WITH CONFOCAL PROBING**
- ***CONFOCAL RESCUE***

Adaptive illumination reduces laser illumination resulting in less photo-bleaching and less photo-toxicity. This makes adaptive illumination a valuable option whenever sensitive samples that suffer too much damage from laser illumination during conventional (super-resolution) imaging. The improved photon-budget can be utilized to…

- …record longer super-resolution live-cell movies or movies with fewer artifacts from photo-toxicity.
- …acquire super-resolution images with higher fluorescence signal and/or higher resolution than possible in conventional mode.
- …acquire full volume images in high-resolution STED mode, especially 3D-STED mode.

The efficiency of adaptive illumination can be seen in the results and the actual reduction of laser illumination can be checked via the 'Stack Display' live dialog (see section ***2.10.4.6***).

> ***Note:*** *Adaptive illumination requires a minimum level of fluorescence signal to be able to make laser-blanking decisions. For best results, try to adjust scan parameters (excitation power, dwell time, accumulations) in a simple confocal image until a fluorescence signal of 50-100 photons is reached in the brighter image pixels. After that, start with these parameters in more complex measurements with adaptive illumination.*

### 2.4.3.2 STED with confocal probing

**Principle**

This simple adaptive illumination mode significantly reduces the strong laser illumination during STED imaging based on the information from a standard confocal image channel. It has the following features:

- Easy to use via a single parameter (threshold)
- Required image channels: Confocal + STED (good speed)
- Protection against bleaching: Medium
- Overall reduction of illumination: Good (reduced phototoxicity)
- Perfect for live-cell imaging (reduced phototoxicity and easily adapted to fluctuating signal levels)

The fluorescence signal level in simple confocal imaging mode is the basis for estimating which image pixels contain structures of interest based. This estimation process is referred to as 'confocal probing' (see *ADAPTIVE ILLUMINATION CONCEPT*). In STED mode, all lasers are blanked (switched off) in pixels without structures of interest.

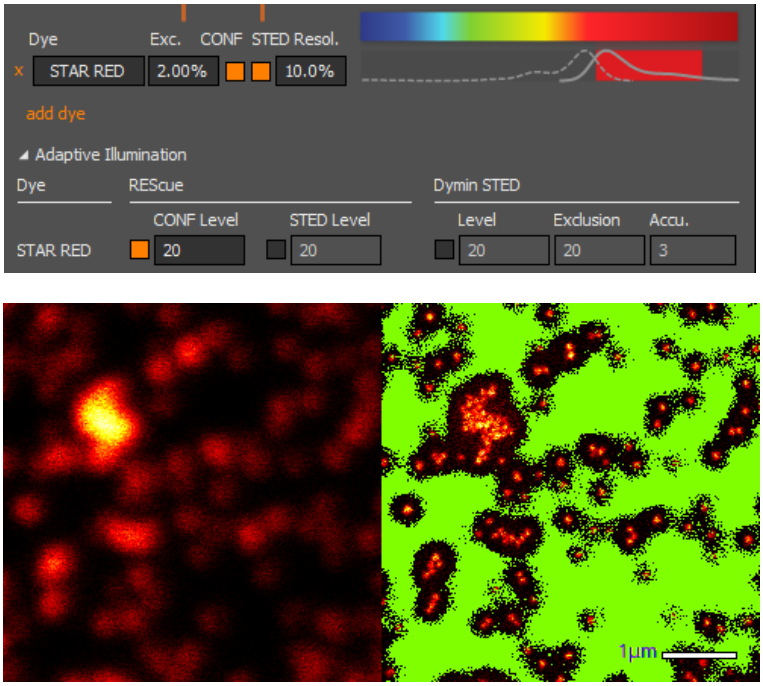



Figure 2.9: Adaptive Illumination - STED with confocal probing

Top: Activating STED with confocal probing in 'easyCommander' requires ticking the parameters check box 'Rescue – CONF Level'.

Bottom left: Confocal probing image.

Bottom right: High-resolution STED image acquired with high STED laser power. The laser-blanking mask (green) is derived from the confocal probing step and marks all pixels where the lasers are blanked in STED mode.

**Process**

During image acquisition, each image line is first scanned in confocal mode, then in STED mode before moving to the next line. The initial confocal scan 'probes' each pixel in an image line, determining whether the signal is above the user defined threshold (set via the parameter 'CONF Level', see Figure 2.9). The FPGA electronics remember which pixels in the current line are above or below threshold. Subsequently the line is scanned in STED mode with much stronger illumination, blanking (switching off) all lasers in pixels below confocal probe threshold (considered devoid of structures of interest). Effectively, a laser-blanking mask (see green area in FIGURE 2.9) is created line-by-line on the fly during image acquisition.

**Parameters**

**CONF Level:** This parameter sets the signal threshold value for the confocal probing image, generating a laser blanking mask for pixels below threshold (see green mask in Figure 2.9). Laser exposure in STED mode is then blanked in those pixels covered by the blanking mask (i.e. below threshold). Pixels not covered by the mask (i.e. above threshold) are illuminated fully in STED mode.

This threshold is given in photon counts per pixel and needs to adjusted carefully by the user. The threshold should be at least the background fluorescence level found in the image, as background pixels do not require illumination in STED mode. Setting the threshold too low will illuminate more pixels than needed in the powerful STED mode, increasing phototoxicity and bleaching. The threshold should also not be set too high as to avoid blanking pixels containing structures of interest with weak fluorescence signal. If weakly fluorescent structures are blanked, they will simply not appear in the final STED image as they are not illuminated. The user can quickly check whether any weakly fluorescent structures are blanked by showing/hiding the blanking mask in the confocal image via the 'Stack Display' live dialog (see *2.10.4.6*).

> ***Note:*** *The blanking mask is generated based on information from the confocal image channel and applied to lasers in the STED channel. The mask can be optionally displayed in the confocal image (showing pixels below threshold) and/or in the STED image (showing pixel-by-pixel laser-blanking based on the previous confocal probe step) as described in section* **2.10.4.6**.

### *2.4.3.3 DYMIN: STED with dual probing*

**Principle**

This adaptive illumination mode provides the largest reduction in photobleaching. DYMIN builds on the mode ***STED WITH CONFOCAL PROBING***. It introduces a second probing step (exclusion step) using very low STED laser power to identify additional pixels that do not require illumination with high-power STED laser light in the final high-resolution STED imaging step. The exclusion step itself requires only very little extra illumination, as a medium resolution can be obtained at very low STED laser power due to the non-linear relationship between resolution and depletion power.

- Medium complexity as it requires control of 2-3 parameters, each of which is relatively simple to control.
- Required image channels: Confocal + low-power STED + high-power STED (low speed)
- Protection against bleaching: Best
- Overall reduction of illumination: Good (reduced phototoxicity)
- Perfect for volume imaging in 3D-STED mode or highest-resolution 2D-STED mode.

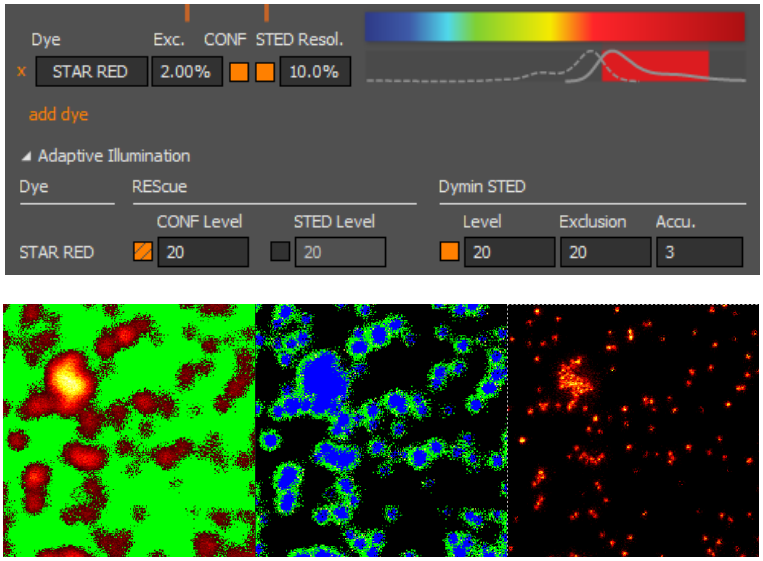



Figure 2.10: Adaptive Illumination - DYMIN STED

Top: 'easyCommander' with 'Dymin STED' activated. 'RESCue CONF Level' is automatically activated as well.

Bottom Left: Confocal image with confocal probing mask derived from pixels below threshold (RESCue - CONF Level) in this channel. This mask determines which pixl are blanked in the next channel.

Bottom centre: Exclusion image acquired with very low STED laser power providing a medium resolution. The exclusion mask (green) is derived from pixels below threshold (Dymin - STED Level) in this image. Blue marks very bright pixels. Many pixels show no fluorescence signal (black) as lasers were blanked based on the confocal probing mask.

Bottom right: Final high-resolution STED image acquired with high STED power. Only pixels not covered by the blanking masks from the confocal probing step and the exclusion step were illuminated.

**Process**

The process starts in confocal probing mode, just as for ***STED WITH CONFOCAL PROBING***. During image acquisition, an image line is first scanned in confocal mode (confocal probing), then in medium-resolution STED mode with very low STED laser power (exclusion step) and finally in high-resolution STED mode with high STED laser power. Only then does the scanner move to the next line in the image.

In the initial confocal probing step, an image line is scanned in simple confocal mode, determining for each pixel whether the signal is above the user defined confocal probing threshold (set via the parameter 'RESCue - CONF Level', see ***FIGURE*** 2.10). All pixels below this threshold are considered devoid of structures of interest and are added to the first laser-blanking mask (see green area in bottom left image in ***FIGURE*** 2.10).

In the subsequent exclusion step (an additional probing step), the same line is scanned in medium-resolution STED mode with slightly stronger laser illumination (low-power STED laser illumination). During the exclusion step, the lasers are blanked in those pixels below covered by the blanking mask from the previous step (confocal probing), avoiding unnecessary illumination with the STED laser. Thanks to the slightly improved resolution, the fluorescent structures can now be discriminated from background areas in more detail. For this purpose, the resolution of the exclusion step needs to be significantly better than in the confocal probing step, but not too high as to avoid unnecessary strong illumination with the STED laser. The resolution in the exclusion step can be adjusted via the parameter 'Dymin STED – Exclusion' (see ***FIGURE 2.10***). In the exclusion step, all pixels with a fluorescence level

below the second, user-defined threshold (set via the parameter 'Dymin STED – Level', see *FIGURE 2.10*) are also considered devoid of structures of interest and added to the second laser-blanking mask (green area of bottom-center image in *FIGURE* 2.10).

In the final, high-resolution STED imaging step, the same image line is scanned with high-power STED laser illumination. This step can cause the most photo-damage. However, only those pixels are illuminated which are not covered by the two laser-blanking masks generated in the previous confocal probing and exclusion steps. Especially the second laser-blanking mask (exclusion mask) can avoid much of the strong laser illumination near the fluorescent structures, reducing photo-bleaching dramatically. The user can set the number of line-accumulations for the final high-resolution STED image via the parameter 'Dymin STED – Accu.' (see top image in *FIGURE 2.10*). If bleaching is reduced, more line accumulations may be used for improved signal. Alternatively, the reduced bleaching may allow raising the resolution of the image.

**Parameters**

**RESCue - CONF Level:** This parameter sets the fluorescence signal threshold for the confocal probing step (see section *2.4.3.2* for more details). Pixels in the confocal image with fluorescence signal below this threshold will be added to the first laser-blanking mask. The first laser-blanking mask is applied to the exclusion step and the final high-resolution step to reduce illumination with the STED laser.

**DyMIN STED - Level:** This parameter sets the fluorescence signal threshold for the exclusion step (second probing step with low-power STED). Pixel in the exclusion image with fluorescence signal below this threshold will be added to the second laser-blanking mask. The first and second laser-blanking masks are applied to the final high-resolution STED imaging step to reduce laser illumination.

**Dymin STED - Exclusion:** This parameter sets the resolution in the exclusion image. The value is given in percent of the resolution of the final high-resolution STED image. The value does not linearly translate into a STED laser power due to the non-linear relation between resolution and STED laser power. Users need to adjust the 'Exclusion' parameter carefully as the resolution in the exclusion step should be significantly higher than in the confocal image but also significantly lower than in the final, high-resolution STED image. Typically, users choose a value around 40-50% in measurements with a medium resolution in the final STED image or 30-40% in measurements with a very high resolution in the final STED image.

> ***Note:*** *To show and check the resolution of the exclusion image properly, the user may need to hide the masks in this image (see* ADAPTIVE ILLUMINATION: DISPLAYING THRESHOLDS*) and adjust the look-up-table.*

**Dymin STED – Accum.:** The number of line accumulations in the final, high-resolution STED image. More line accumulations result in higher signal up to the point where more line accumulations cause more bleaching and therefore lower signal.

> ***Note:*** *The line accumulations set via the parameter 'Dymin STED – Accu.' need to be multiplied with the general line accumulations parameter in 'easyCommander' (see top right in* FIGURE 2.4*) for the total number of line accumulations in the final STED image.*

#### 2.4.3.4 RESCue STED with confocal probing

**Principle**

This adaptive illumination mode builds on the mode ***STED WITH CONFOCAL PROBING***. It significantly reduces the strong laser illumination during STED imaging based on information from a standard confocal image channel as well as dynamically blanking whilst in STED imaging mode (RESCue STED). It has the following features:

- Requires adjustment of two parameters (one of which is complex to control)
- Required image channels: Confocal + STED (good speed)
- Protection against bleaching: Good
- Overall reduction of illumination: Very good (reduced phototoxicity)
- Good for live-cell imaging: Stronger reduction of phototoxicity than with the mode ***STED WITH CONFOCAL PROBING***. However, it is more complex to adjust, especially when fluorescence signal levels decrease during live-cell imaging)

**Process**

The first step of this adaptive illumination mode is confocal probing which is performed and controlled in the same way as for the adaptive illumination mode ***STED WITH CONFOCAL PROBING***. Confocal probing identifies those pixels without structures of interest which are added to a laser-blanking mask (displayable via the Stack Display dialog, see section ***2.10.4.6***) and will not be illuminated in the following high-resolution STED step. However, the identification process offers only limited accuracy due to the limited resolution of confocal imaging.

In the second step, each line is scanned in high-resolution STED mode using the STED (depletion) laser at high power. This imaging mode can reveal additional pixels without structures of interest due to its higher resolution. Often those empty pixels can be identified on the fly, before a pixel has been completely scanned, i.e. after a fraction of the pixel dwell time. This allows blanking the powerful STED laser after a fraction of the pixel dwell time reducing total illumination. The mechanism is also described in the subsection RESCue (lower threshold) in the section ***ADAPTIVE ILLUMINATION CONCEPT***. Identification of empty pixels is done based on a user-defined minimum expected fluorescence level for pixels containing fluorescent structures (set via the parameter 'RESCue – STED Level'). While scanning a single pixel, the chances of reaching the minimum fluorescence level are estimated at multiple checkpoints. If the chances are too low, then laser illumination can be aborted early. Pixels with incomplete illumination are reported via a second laser-blanking mask (displayed via show the 'Stack Display' live dialog option 'Show Below Threshold', see section ***2.10.4.6***).

In the second step, illumination of each pixel can also be aborted early if the pixel shows particularly bright fluorescence signal. This mechanism is described in the sub-section 'RESCue (upper threshold) ' in the section ***ADAPTIVE ILLUMINATION CONCEPT***. Pixels with incomplete illumination due to high signal are reported via a third laser-blanking mask (displayed via show the 'Stack Display' live dialog option 'Show Stopped at Max', see section ***2.10.4.6***). The final pixel value is extrapolated from the shortened duration of illumination, the full pixel dwell time and the signal level at the time of abortion.

> ***Note:*** *The second step (RESCue STED) may also be used by itself without a probe step. While this process is faster, it provides far less reduction of laser illumination.*

**Parameters**

**RESCue - CONF Level:** This parameter sets the fluorescence signal threshold for the confocal probing step. Pixel in the confocal image with fluorescence signal below this threshold will be added to the first laser-blanking mask. The first laser-blanking mask is applied to the high-resolution STED imaging step to reduce illumination with the high-power STED laser (for more details see ***STED WITH CONFOCAL PROBING***).

> ***Note:*** *The blanking mask is generated based on information from the confocal image channel and applied to lasers in the STED channel. The mask can be optionally displayed in the confocal image (showing pixels below threshold) and/or in the STED image (showing pixel-by-pixel laser blanking based on the previous confocal probe step) as described in section* **2.10.4.6**.

**RESCue - STED Level:** This parameter sets the minimum fluorescence signal expected for pixels containing structures in the high-resolution STED image. Laser illumination is aborted early if the minimum signal is unlikely to be reached at full pixel dwell time. For details see subsection 'RESCue' (lower threshold) in section ***ADAPTIVE ILLUMINATION CONCEPT***. Laser illumination is also aborted as soon as the fluorescence signal of a pixel reaches 1.5x the minimum expected signal. Pixels with incomplete illumination due to high signal are reported via additional laser-blanking masks (displayed via the 'Stack Display' live dialog, see section ***2.10.4.6***).

> ***Note:*** *To reveal the final STED image and to adjust the look-up-table, the laser-blanking masks may need to be hidden (see section* **2.10.4.6***).*

### 2.4.3.5 Confocal RESCue

**Principle**

This adaptive illumination mode is only suitable for confocal measurements without STED channels. Usually confocal imaging is very mild using abberior microscopes due to the high detection efficiency. However, confocal RESCue may reduce bleaching and/or phototoxicity for extremely sensitive samples or when applying extremely long dwell times and/or high numbers of accumulations. It offers the following features:

- Requires adjustment of only one parameter (which is complex to control)
- Required image channels: Confocal (fast speed)
- Protection against bleaching: Low
- Overall reduction of illumination: Low
- Only suitable if the sample is extremely sensitive or when applying extremely long dwell times and/or high numbers of line accumulations.

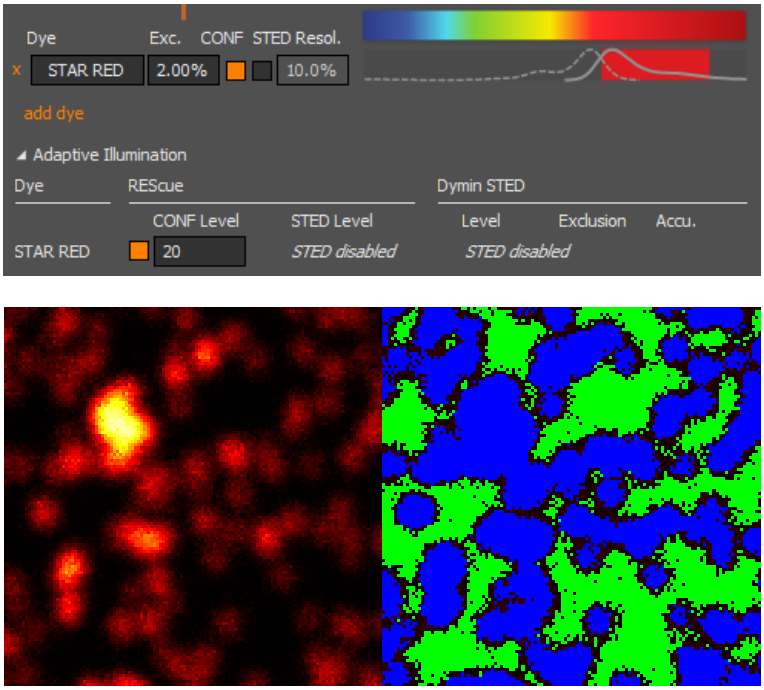



Figure 2.11: Adaptive Illumination - Confocal RESCUE.

Top: Adaptive Illumination dialog in ‘easyCommander’.

Bottom left: Confocal image without masks.

Bottom right: Confocal image with laser blanking masks showing pixels with reduced illumination due to low signal (green) or due to high signal (blue).

**Process**

Confocal RESCue utilizes no probe steps, only the RESCue mechanism with lower and upper thresholds (see Adaptive illumination concept). Often pixels that contain no structures of interest can be identified on the fly, before a pixel has been completely scanned, i.e. after a fraction of the pixel dwell time. This allows early laser blanking after a fraction of the pixel dwell time reducing total illumination. Identification of empty pixels is done based on a user-defined minimum expected fluorescence level for pixels containing fluorescent structures (set via the parameter 'RESCue – CONF Level', see FIGURE

*2.11*). While scanning a single pixel, the chances of reaching the minimum fluorescence level are estimated at multiple checkpoints. If the chances are too low, then laser illumination can be aborted early.

In addition, illumination of each pixel can also be aborted early if the pixel shows particularly bright fluorescence signal. The final pixel value is extrapolated from the shortened duration of illumination, the full pixel dwell time and the signal level at the time of abortion.

Pixels with incomplete illumination due to low or high signal are reported via two laser-blanking masks that can be displayed via the 'Stack Display' live dialog (see section *2.10.4.6*). In *FIGURE* 2.11, the laser-blanking mask based on low-signal pixels (lower threshold) is shown in green and the laser-blanking mask for high-signal pixel (upper threshold) is shown in blue.

**Parameters**

**RESCue - CONF Level:** For the Confocal RESCue mode, this parameter acts differently compared to other adaptive illumination modes (*STED WITH CONFOCAL PROBING*, *DYMIN: STED WITH DUAL PROBING*, *RESCUE STED WITH CONFOCAL PROBING*). For Confocal RESCue, this parameter sets the minimum fluorescence signal expected for pixels containing structures in the confocal image. Laser illumination is aborted early if the minimum signal is unlikely to be reached at full pixel dwell time. For details, see subsection 'RESCue' (lower threshold) in section *ADAPTIVE ILLUMINATION CONCEPT*. Laser illumination is also aborted as soon as the fluorescence signal of a pixel reaches 1.5x the minimum expected signal. Pixels with incomplete illumination due to high signal are reported via additional laser-blanking masks (displayed via the 'Stack Display' live dialog, see section *2.10.4.6*).

> ***Note:*** *To reveal the final confocal image and to adjust the look-up-table, the laser-blanking masks may need to be hidden (see section 2.10.4.6).*

## 2.4.4 Modification of default settings and resetting manual changes

The user can open additional live dialogs to make manual changes to the measurement settings. This is recommended for more advanced measurements. However, some of these manual settings may differ from the settings typically created by easyCommander, e.g., the default spectral detection range for a given dye. The manual settings will overwrite the typical easyCommander settings and easyCommander will notify the user with a notification (displayed at the bottom of the easyCommander live dialog). This message can be ignored if the user intended the manual changes. There is the options to reset to the typical easyCommander settings via the “reset” link next to the error message.

Depending on the extend of the measurement setting modifications, easyCommander may still be used for basic controls. If not, the respective control fields are greyed out or highlighted red (e.g., laser power settings when modified in the live dialog Channels, see *2.7.8*).

## 2.5 Measurement Template

In Imspector, a template-driven workflow is built in.

Measurement templates are ready-to-use parameter sets that enable a quick start into the use of Imspector and the microscope without the need for the user to be familiar with each and every detail of the microscope and the software. They contain the acquisition parameters (e.g., field of view, pixel size, scan speed, scan direction, dimensionality, pinhole size, activated lasers, activated detectors....) that are required for a type of measurement. As in an Imspector measurement, multiple windows may be included.

To open a Template, select 'File' > 'New' > 'File from Template'...

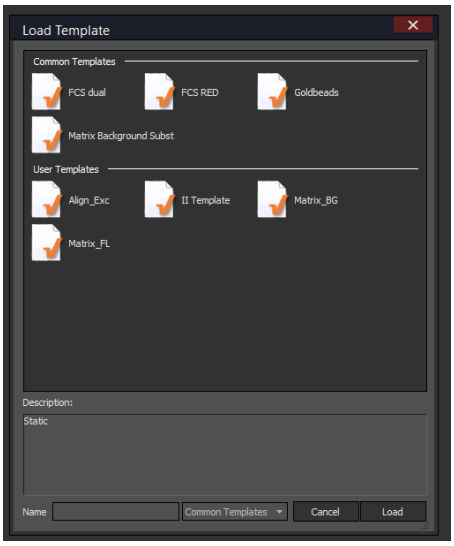



*Figure 2.12: 'Load Template' dialog.*

During installation of the system by abberior, a set of standard measurement schemes is pre-defined (*FIGURE 2.12*). Based on these templates users can start with several basic measurements. Later, the given templates can be either adapted for the users measurement of interest or new templates can be created by the user.

In contrast to easyCommander driven work-flows, measurement templates are not restricted to standard measurement settings, but may contain most parameters that are available in Imspector at a given setup.

### 2.5.1 Use of Existing Measurements as Templates

In Imspector existing measurements can be used as templates because they contain required instrument parameters/ meta data for starting a new measurement.

After opening a saved measurement, all experiments can be done in the exact same way by restarting the saved measurement with 'REC'.

> ***Note:*** *When several measurements are imported into Imspector, the scan settings of the active measurement window will be applied, when the 'REC' button is pressed. This allows a rapid change between measurement modes.*

## 2.6 Toolbars

Toolbars allow quick access to important functions of the Imspector software via simple buttons. Toolbars can generally be found either towards the edge of the Imspector main window or at the edge of a measurement window. The location of toolbars can be changed via left-click drag&drop. Right-clicking on a toolbar allows customization and addition of new functions.

### 2.6.1 File Toolbar




Figure 2.13: The 'File' toolbar.

The 'File' toolbar (see Figure 2.13) can be found at the edge of the Imspector main window. Following buttons are available in the toolbar:

1. Button 1 creates a **new measurement** with default imaging parameters (which are given during the installation of the microscope).
2. Button 2 creates a new **measurement from a template** (containing the imaging parameters defined by the template.
3. Button 3 **open**s a measurement from a file. Imspector measurements can be run again to repeat the acquisition with identical parameters. Files from other compatible imaging formats can be opened as well but cannot be run again.
4. Button 4 of the file toolbar **save**s a measurement with a given file name (**save as**).
5. Button 5 of the file toolbar saves a measurement under the same file name the current measurement is entitled with. If the 'overwrite warning' is activated Imspector will ask if the current measurement should be overwritten. Otherwise, it saves it with the same file name. ***Important note:*** *An already existing file may be overwritten.*
6. Button 6 of the file toolbar creates a **new empty window** within the current measurement.
7. Button 7 of the file toolbar **cut**s out the selected data window. The data is purged to the clipboard.
8. Button 8 of the file toolbar **copies** the selected data to the clipboard.
9. Button 9 of the file toolbar **paste**s the selected data window to the measurement or graph window.

### 2.6.2 Zoom Toolbar




*Figure 2.14: The 'Zoom' toolbar.*

The 'Zoom' toolbar (see **FIGURE** 2.14) allows the user to adjust the image display size and to zoom in/out of images. The toolbar can be found at the edge of the Imspector main window and/or at the edge of each measurement window.

A detailed description of the 'Zoom' toolbar functions can be found in the section

Image visualization, processing and analysis (*0*).

### 2.6.3 Display Toolbar




*Figure 2.15: The 'Display' toolbar*

The 'Display' toolbar (*FIGURE 2.15*) allows to quickly change the way the data is displayed. This toolbar can be found at the edge of the Imspector main window and/or at the edge of each measurement window.

A detailed description of the 'Display' toolbar functions can be found in the section (*2.10.3*).

### 2.6.4 Measurement Toolbar




*Figure 2.16: The 'Measurement' toolbar*

The 'Measurement' toolbar (*FIGURE 2.16*) toolbar allows starting, stopping and copying of measurements. This toolbar can be found at the edge of the Imspector main window.

1. The 'REC' button starts acquisition of the currently selected measurement with its given parameters. Only a single acquisition will be performed.

   ***Note:*** *If the measurement already contains unsaved data, the previous data will be overwritten and will be lost!*
   ***Note:*** *A measurement is selected by left-clicking on its respective window.*

2. The 'Auto-repetition' button starts the currently selected measurement in continuous (auto-repetition) mode. Clicking the button a second time deactivates auto-repetition mode and the measurement stops after completing the current frame.

   ***Note:*** *In auto-repetition mode, only the last frame can be saved after the acquisition stopped. All other frames are discarded. For movie acquisitions see the timelapse controls in sections 2.4.1, 0 and 2.7.5)*
   ***Note:*** *For STED measurements, auto-repetition may cause additional bleaching.*

3. The 'Scan in ROI' button creates a new measurement from a ROI selection in the image. The new measurement has the dimensions of the ROI while all other parameters remain identical to the original measurement.
4. The 'Pause' button pauses a measurement immediately even if the current frame has not yet been completed. However, the partially acquired frame can still be saved. The 'Pause' button pauses measurements acquiring in single-recording mode as well as those in continuous acquisition mode.
5. The 'Clone' button duplicates or clones a measurement creating a new measurement with identical dimensions, position and imaging parameters. The new measurement will contain empty images until its acquisition is started.

   ***Note:*** *If the 'overwrite warning' option is activated Imspector, it will ask if the current measurement should be overwritten. However, this options may be cumbersome in most workflows as it will generate a multitude of confirmation requests.*

   *If the 'overwrite warning' option is deactivated, most workflows are faster. However, care needs to be taken as an already existing image may be overwritten.*

### 2.6.5 Power Switch and Microscope Status Toolbar



*Figure 2.17:'PowerSwitch and Microscope Status' toolbar*

The 'Power Switch and Microscope Status' toolbar allows quick hardware status control as well as switch ON/OFF control. ***FIGURE 2.17*** shows three possible states of the toolbar. This toolbar is usually found towards the top edge of the Imspector main window.

The left 'Power Switch' button is used to switch the microscope hardware on and off.

> ***Note:*** *Please refer to section for complete hardware switch ON and shutdown procedures. Incorrect switch On or shutdown procedures may damage the hardware.*

The Microscope Status button on the right of the toolbar displays the microscope status via different symbols:

- **Green icon:** Everything is operating correctly, and measurements can be done without limitations.
- **Orange icon:** One or more device shows a notification. Measurements can typically be performed. Please check status of devices.
- **Red icon:** Error status. Measurements cannot be performed. Please check status of devices.

Upon pressing the status button, the 'Device Status' window will open, indicating the status of the individual devices (***FIGURE 2.18***).

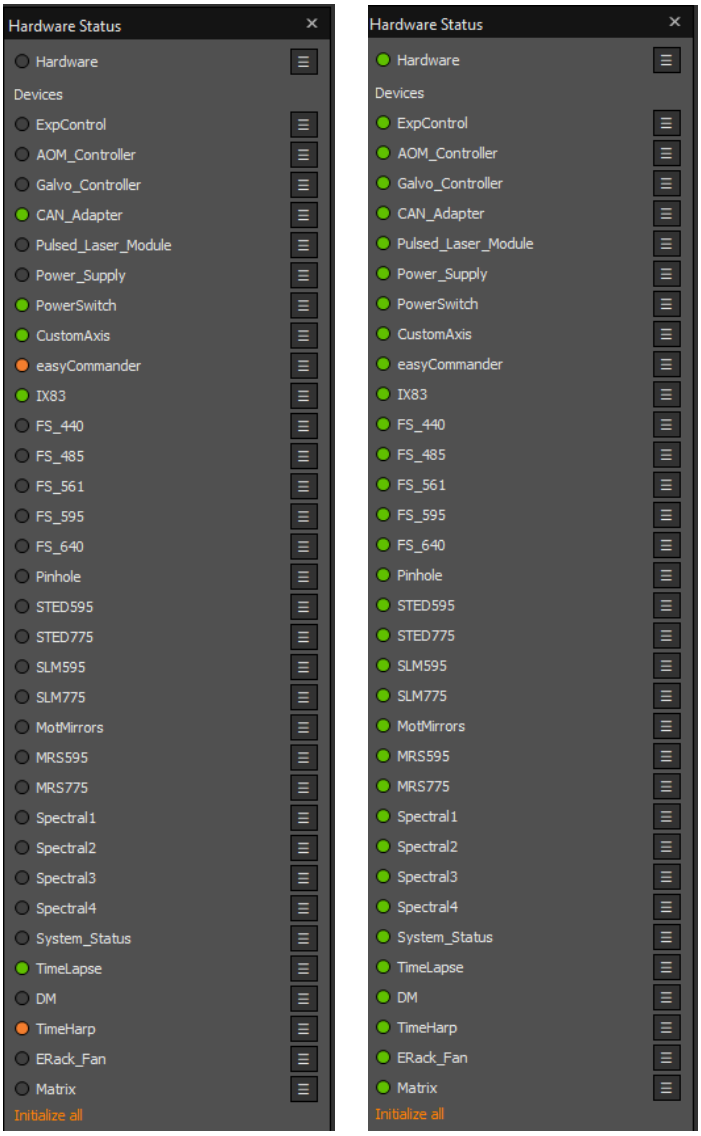



*Figure 2.18: The 'Device Status' window*

Left: Several devices are not initialized, and the measurement cannot be started.

Right: All devices are running without error notifications and the measurement can be started.

## 2.7 Live Dialogs

The easiest way to set up a measurement is via the easyCommander which provides all the required standard settings based on a dye selection. For more experienced users, live dialogs offer more advanced measurement designs, settings beyond the standard parameters and direct control of individual parameters (e.g., laser selection, laser powers, detection range, time gating, extended timing controls, alignment details and many more).

In most cases, it is recommended to start with parameters generated via easyCommander followed by manual changes via live dialogs. For more information on easyCommander and how it handles manual changes, see the section The Smart Way to Start: easyCommander.

### Opening live dialogs

Live dialogs can be chosen from a list (*FIGURE 2.19*) that can be opened via “Menu bar > Window > Live Dialogs”. The list can be filtered via the filter field and ordered alphabetically or according to devices by clicking on the respective fields on top of the list. Individual live dialogs can be hidden again by closing the respective live dialog window or unticking the respective checkbox in the live dialogs list. All open live dialogs can be hidden or revealed simultaneously via the <TAB> key. The full list of live dialogs can be seen in *FIGURE 2.19: 'LIVE DIALOGS' LIST*.

### Live dialogs and their specific functions

An overview of Imspector functions and their respective live dialogs is shown on the following pages in Table *2.7-1* and *TABLE 2.7-2*. You can use the list to jump to the respective manual sections.

### Workspaces: Saving live dialog selection and arrangement

Users may open a large number of live dialogs simultaneously for immediate access to various settings. The selection and arrangement of live dialogs can be saved by the user as a “workspace”. This allows one-click loading of different workspaces for different tasks or users, without the need of opening and arranging all live dialogs again. Workspaces can be saved via “Menu bar>Window>Save Workspace…” and loaded again via “Menu bar>Window>Load Workspace…”. For more details see Imspector Multiuser Management System.

| *easyCommander* |
|---|
| Automatic settings via dye selection |
| Scan axis (simplified) |
| Scan parameters (simplified): Range, pixel size |
| Dwell time |
| Scan modi (simplified): Fast Scan ON/OFF |
| Line accumulations (simplified) |
| Continuous autofocus (ON/OFF) |
| Objective selection |
| Pinhole |
| Laser power |
| Resolution (STED laser power) |
| Resolution distribution (2D-STED vs 3D-STED) |
| easyZ volume imaging |
| Adaptive Illumination control |

| *Stack Display* |
|---|
| Data vizualisation |
| Viewing order of axis<br>(e.g., XYZ, orthogonal views) |
| Hidden axis<br>(Analysis e.g., max projection, summation) |
| Hidden axis range<br>(Select axis range for analysis) |
| FLIM display (fractional lifetime analysis) |
| Adaptive Illumination<br>(Display of threshold masks) |
| Graph display (line width, symbols…) |

| *Configurations* |
|---|
| Measurement list and data structure |
| Select individual measurement windows |
| Hide measurements |

| *Microscope Body (e.g., IX83, NikonTi)* |
|---|
| Observation mode / light path |
| Escape function of objective nosepiece |
| Readout stage and nosepiece position |
| Transillumination lamp control |
| Shutters |
| Autofocus: Dichroic mirror position |
| Autofocus: Focus search |
| Autofocus: Continuous autofocus (ON/OFF) |

| *Scan Range* |
|---|
| Scan axis (detailed):<br>Single plane, movie, volume, FCS, spectral... |
| Scan range (detailed): Range, pixel size, offsets |
| Local and global offsets (positions) |
| Dwell time |
| Scan modi (detailed):<br>Standard, Fast Scan, bidirectional, elliptical… |
| Line accumulations (simplified) |
| Scan field rotation |
| Stepwise scanning |
| Frame triggers |
| Autofocus: Continuous autofocus (ON/OFF) |

| *Measurement* |
|---|
| Overview of channels, detectors, spectral range |
| Overview of spectral ranges |
| Overview of laser powers |
| Activate line / pixel steps |
| Laser syncing information |
| Laser pulse delay adjustment |
| Activate custom PSFs (donut shapes) |
| Detector streams<br>(Write single-photon data to disk) |
| Activate intermediate image mode |

| *Channels* |
|---|
| Channels: Create & modify |
| Detector selection |
| Spectral detection range |
| Time gated detection |
| Laser selection and settings |
| FLIM settings |
| Adaptive Illumination details (RESCUE, DyMIN) |

| *Alignment* |
|---|
| Autoalignment start |
| Autoalignment options |
| Manual alignment controls |

*Table 2.7-1: List of live dialogs and their functions (part 1)*
*Click on live dialog headings in the table to jump to the relevant manual section.*

| *Time Lapse* |
|---|
| Timelapse / Movie control |
| Timer overflow (check if scan speed achievable) |
| Acquisition sequence scheduling |

| *Camera* |
|---|
| Start/stop camera acquisition |
| Simple camera settings (gain, exposure time) |
| ROIs |

| *Camera Settings* |
|---|
| Advanced camera settings |
| Bit depth & binning |
| Hot/Cold pixels |

| *Workspace* |
|---|
| Advanced metadata access |

| *Pinhole* |
|---|
| Pinhole size |
| Pinhole settings (current vs measurement) |

| *Pellicle* |
|---|
| Pellicle position (if installed) |
| Pellicle settings (current vs measurement) |
| Enable reflective alignment measurements |

| *Custom Axis* |
|---|
| Parameter sweeps (via additional scan axis) |
| Optimising parameters for illumination, adaptive optics & more |
| Testing parameters for alignment procedures |
| Stage scanning (manual tiling) |

| *Laser Live Dialogs* |
|---|
| Direct laser power output (Full laser power level before attenuation) |
| Laser control mode (power/current) |
| Laser sync mode |
| Laser ON/OFF, shutdown, initialisation |

| *Measurement Setup* |
|---|
| Scan axis order |
| Sync mode settings (frame, line, none) |
| Bidirectional scan |
| Permanent Adding |

| *Line Steps* |
|---|
| Line accumulations: Advanced control |
| Line step sequences (sequences of channels) |
| Line step creation & patterns |
| Interleaving |

| *Pixel Steps* |
|---|
| Pixel step creation & patterns |
| Pixel step duration (advanced dwell time) |
| Rise times |

| *DM Settings (Deformable Mirror)* |
|---|
| Adaptive optics (all beams & detection) |
| Depth-dependent aberration compensation |
| Custom aberration compensation |
| Global aberration compensation (e.g., objective imperfections) |
| Deformable Mirror alignment |

| *SLM (spatial light modulator)* |
|---|
| Phase pattern selection (2D/3D-STED, custom) |
| Phase pattern options (Phase delay, radius, rotation) |
| Phase pattern adjustment (STED donut position & shape) |
| Adaptive optics (STED beams only) |
| Upload custom phase patterns |

| *MRS* |
|---|
| Manual control of 2D/3D STED ratio |

*Table 2.7-2: List of live dialogs and their functions (part 2)*

Click on live dialog headings in the table to jump to the relevant manual section.

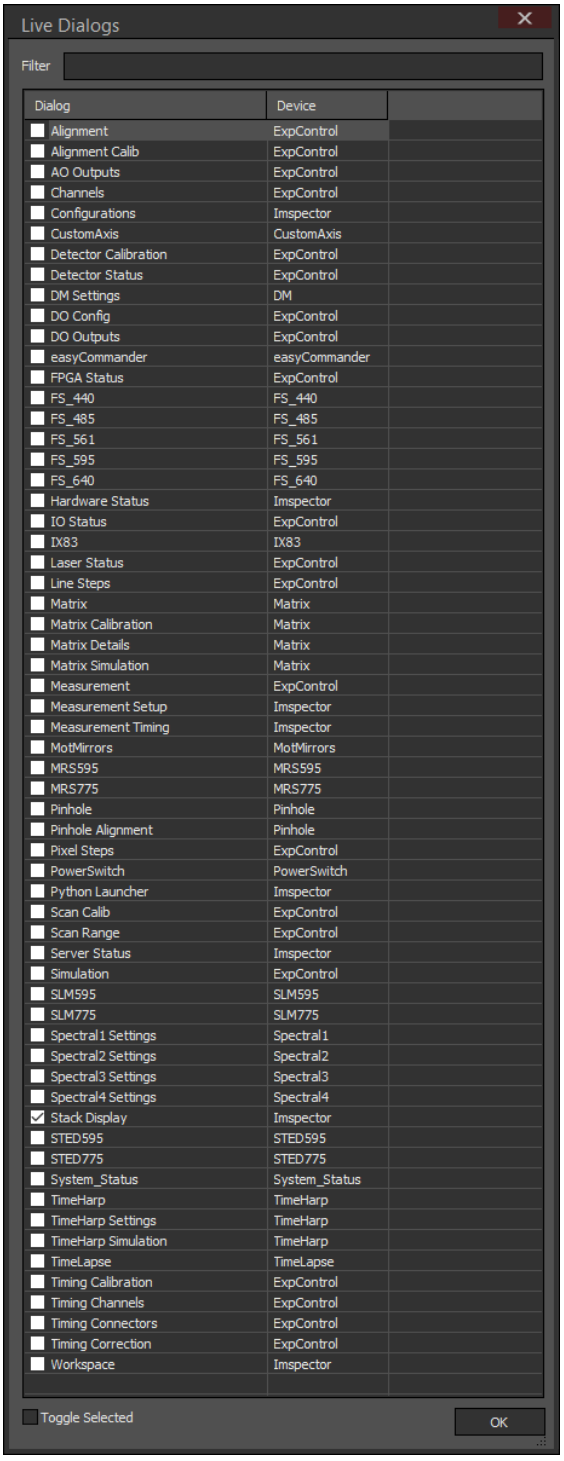



*Figure 2.19: 'Live Dialogs' list*

**Advanced dialogs for maintenance only**

There is many live dialogs that offer access to core settings and calibrations if the user has the relevant permissions (see section Imspector Multiuser Management System). These dialogs should only be used by maintenance staff or users with the relevant training. These live dialogs are listed in TABLE 2.7-3.

| | | |
|---|---|---|
| Alignment Calib | IO Status | Scan Calib |
| AO-Outputs | MATRIX Details | Simulation |
| Detector Calibration | Matrix Calibration | Timing Calibration |
| DO Config | Matrix Simulation | Timing Channels |
| DO-Outputs | MotMirror | Timing Connectors |
| FPGA-Status | Pinhole Alignment | Timing Correction |

*Table 2.7-3: Live Dialogs for support and trained staff only*

### 2.7.1 Stack Display live dialog

The Stack Display live dialog is the main hub controlling the visualization of data (images, graphs…). Unlike other live dialogs, this dialog cannot be closed. A detailed description of the Stack Display live dialog can be found in the visualization section (*2.10.4*).

### 2.7.2 Configurations Live Dialog

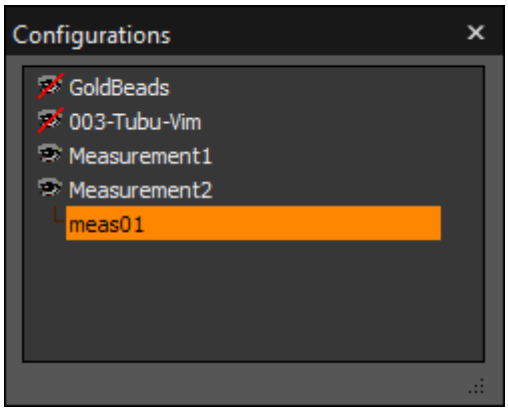



*Figure 2.20: 'Configurations' live dialog*

The 'Configurations' live dialog shows all measurements including measurement windows (sub-windows that are currently open). It allows users to hide/unhide measurements (via clicking the eye symbol) in order to comfortably arrange their workspace and desktop.

When saving a file, the corresponding top-level item (measurement) in the list will adopt the file name. The file will include all sub-windows (measurement windows with one or more data stacks).

For navigation purposes, the user can click on a sub-item in the list to bring the corresponding measurement window to the front. Clicking on a sub-item in the list allows the user to rename sub-items.

Individual data stacks in each measurement window are not listed here. Their names can be found in the respective measurement window (tabs towards the top) and renamed via a double click (see *IMSPECTOR MEASUREMENTS*). Renaming individual data stacks can be helpful when importing Imspector data files e.g., via the BIOFORMATS importer, as these stack names are automatically appended to the resulting files.

### 2.7.3 Microscope body Live Dialogs

Imspector can communicate with the microscope stands of several vendors and their motorized components. It is able to drive most functions of the microscopy body and autofocus units (see hardware notes on *OLYMPUS FOCUS CONTROL UNIT (MCZ)*, *OLYMPUS TPC (TOUCH PANEL CONTROL)*, *OLYMPUS ZDC: FOCUS SEARCH / CONTINUOUS AUTOFOCUS*).

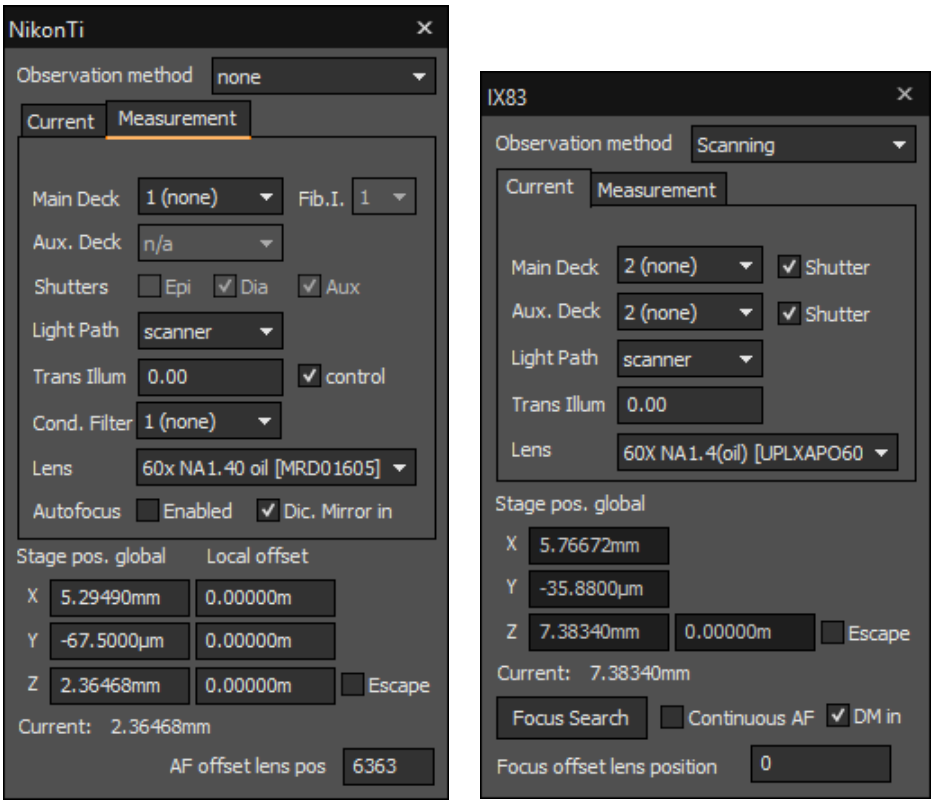



*Figure 2.21: 'Microscope Body' live dialogs for Nikon and Olympus*

The available microscope stand functions include:

**'Current / measurement':** Select these tabs to show either the current settings or the measurement-specific settings. Changing settings in the tab “Current” has immediate effect. Settings changed in the tab “Measurement” only become active once the respective measurement is started. The settings in the tab “Measurement” are tied to individual measurement windows and can be saved with those.

**'Observation method':** The observation method contains presets for several of the individual settings below, optimized for modes such as epifluorescence or scanning. These presets help switching between modes more easily, e.g., ensuring that the microscope is in a laser-safe state to allow confocal laser scanning (otherwise prevented by laser safety features).

**'Light path' / 'Main Deck' / 'Aux. Deck':** This allows the selection of the beam path (Light path) and its mirrors/splitters (Aux. Deck) or filter sets (Main Deck) in the microscope stand (if motorized).

**'Lens':** This allows the selection of the objective lens if a motorized objective revolver is installed. Before rotating the revolver, the software will ask for confirmation.

**'Escape':** Sets the nosepiece to the escape position, e.g., for exchanging samples or sample holders.

**Z-position:** Read-out of the nosepiece position (adjustment see *SCAN RANGE LIVE DIALOG* and *THE SMART WAY TO START: EASYCOMMANDER*).

**Stage position:** Read-out of the stage X/Y position (adjustment see *SCAN RANGE LIVE DIALOG* and *THE SMART WAY TO START: EASYCOMMANDER*). Only available if a motorized stage is installed.

**Trans Illum.:** Brightness of the tungsten lamp in the transillumination path (e.g., for brightfield observation)

**Shutter:** Open/close shutters for epifluorescence illumination and other light paths.

**Dichr. Mirror in / DM in:** Change the autofocus dichroic mirror (DM) position (in/out of beam path)

***Important note for STED microscopes with autofocus***
*If a STED microscope is equipped with an autofocus device (Olympus ZDC or Nikon PFS), the autofocus dichroic mirror should always be in the beam path (tick the checkbox "DM in"), even when the autofocus device is not in use. All beams are optimized for the presence of the dichroic mirror in the beam path. Removing the dichroic mirror from the beam path reduces performance, especially in STED mode.*

***Important note for hybrid STED/MINFLUX microscopes with autofocus***
*If a hybrid STED/MINFLUX microscope is equipped with an autofocus device (Olympus ZDC or Nikon PFS), proper performance in STED mode requires the autofocus dichroic mirror positioned in the beam path (tick the checkbox "DM in") while proper MINFLUX performance requires the dichroic mirror positioned out the beam path (untick the checkbox "DM in").*

**Focus search:** If an Olympus autofocus unit (ZDC) is installed, this option runs a focus search to find the interface of the coverslip and the sample medium. If successful, you will hear one beep and the new focal position should be near the coverslip. For more details and troubleshooting see hardware quick guide section *OLYMPUS ZDC: FOCUS SEARCH / CONTINUOUS AUTOFOCUS*.

**Continuous Autofocus / Autofocus enabled**: If an autofocus unit (Olympus ZDC or Nikon PFS) is installed, this activates/deactivates the continuous autofocus mode (focus lock). The continuous autofocus will compensate drift over longer periods of time while allowing acquisition of Z-stacks. This is especially helpful for live-cell imaging. In continuous autofocus mode, the zero position of the Z axis (as shown e.g., in easyCommander) now refers to the detected position of the coverslip surface.

***Notes for continuous autofocus mode***
*Depending on objectives and samples, the ZDC autofocus has a working range of up to 35 µm and an axial resolution down to 100 nm. This can become a resolution bottleneck in measurements with high-resolution 3D-STED. More details on autofocus operation and trouble-shooting can be found in the hardware quick guide section* **OLYMPUS ZDC: FOCUS SEARCH / CONTINUOUS AUTOFOCUS**.

***Important note for setups with Z-piezo and autofocus***
*If a Z-piezo is installed, it needs to be deactivated to allow proper function of the continuous autofocus mode (Z-piezo is the preferred focusing device of Imspector but will be counteracted by the autofocus). For Z-piezo deactivation see section Experimental Control – Scanning. Note that the Z-position of the sample may change ~100 µm upon deactivation. After finishing your session, please re-activate the Z-piezo for other users.*

### 2.7.4 Alignment Live Dialog

For optimal alignment and performance of the microscope, Imspector offers an 'Alignment' live dialogue. This dialog contains controls for motorized optical elements in the system as to manually fine-tune the overlay of different lasers in the beam path, the position of the pinhole or the alignment of the Matrix detector. Besides these manual alignment controls, the live dialog can be used to start and control an auto alignment function (optional upgrade).

**Manual alignment:**

This is a complex procedure. Please refer to the microscope operation section.

**Auto alignment:**

The auto alignment function allows the automatic alignment of excitation and STED laser lateral positions, the optimization of SLM settings for 2D and 3D STED and the alignment of the lateral pinhole position. This provides the user with an optimally aligned microscope for best performance.

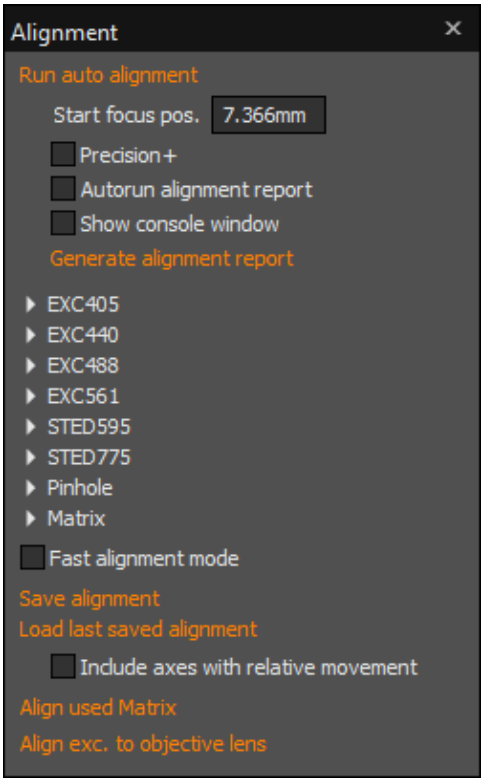



*Figure 2.22: 'Alignment' live dialog.*

The interval to run the auto alignment is usually once a week (depending on stability of the room temperature).

The auto alignment is performed on a dedicated "auto alignment sample" provided at microscope installation (fresh replacement samples available from abberior). The sample contains a custom mix of fluorescent beads and should be **stored at 4°C** and **protected from light**.

Before auto-alignment, the microscope should be switched on for at least 20 minutes (better 30 minutes) to allow warming up. Best practice is to switch on the microscope (power on), mount the 'auto alignment sample' in the sample holder and let the system and sample settle to working temperature and conditions for 30 minutes.

Auto alignment is always performed with the same default objective. For nearly all microscopes, this is the 60x oil objective (e.g., UPLXAPO60XO), only in some cases a 100x oil objective (these microscopes usually do not have a 60x oil objective).

Upon installation of a new objective lens (e.g., 60x water immersion), alignment is optimized for the default objective lens and for the new objective lens. Any alignment drift affecting the default objective will affect the new objective in a similar, predictable way. Auto alignment performed with the default objective therefore also optimizes the STED alignment for the new objective.

Mount the auto alignment sample with default objective (see above) using immersion oil appropriate for this objective. Take care to avoid bubbles in the immersion oil (jiggling the stage laterally via the joystick in coarse mode can help dispelling any remaining small or invisible bubbles).

The autoalignment process can now be started using the "Run auto alignment" option.

> ***Note:*** *If a Z-piezo is installed, it needs to be activated for auto alignment (see* EXPERIMENTAL CONTROL – SCANNING*). If an Olympus autofocus (ZDC) is installed, continuous autofocus should be deactivated.*

The following advanced options for auto alignment are available:

**Run auto alignment:** This starts the auto alignment process with the selected set of parameters (as seen below).

**Start focus pos.:** The start focus position describes the z-position where the auto alignment will start to search for the bead layer. If you switch sample holder (e.g., from slide to dish holder) or if you generally change the position of the sample holder by removing the z-piezo or adding a live cell chamber this starting focus position should be updated. To do this, ensure continuous autofocus is OFF, start a measurement, manually bring the bead layer into focus, note the focus position displayed in the live dialog easyCommander or Scan Range and update the "Start focus pos." value accordingly.

**Precision +:** The precision plus mode re-iterates part of the auto alignment routines to minimize the effect of drift between alignment steps. All devices are aligned at least two times to reduce residual deviation of beam positioning accuracy. This increases the required time but can be useful in environments with high thermal instability.

**Autorun alignment Report:** If this option is activated, the auto alignment will generate and automatically save an alignment report after finishing the alignment process. This enables the user to judge the quality of the alignment.

**Generate Alignment Report:** This immediately generates an alignment report with the current alignment status of the system without performing an auto alignment beforehand.

**Show console window:** Activation of this checkbox enables the console window providing additional information during auto alignment. After alignment is done the console needs to be selected and the end of the alignment procedure needs to be confirmed. As the "show console" checkbox is a global Imspector setting it should be deactivated once the auto alignment was performed.

**Auto alignment Log**: An auto alignment log file is written during every auto alignment.

**Auto alignment reports**: The auto alignment reports are saved in the following folder "C:\Users\User Name\Documents\Abberior Instruments GmbH\Auto Alignment\date-time"

**Align exc. to objective lens:** As described above, autoalignment is always run using the same default objective lens providing concomitant STED alignment for all other calibrated STED-compatible objectives. However, on some newer microscope models running the latest Imspector version, this option can automatically optimize the excitation laser alignment when switching objectives. As different objectives may show different chromatic aberrations, this option can improve performance. This option does not require the autoalignment sample and can be run with any sample. Requires initial calibration of the objective lens by support staff.

**MATRIX alignment:** The MATRIX alignment option of the alignment dialogue is a way of manually triggering the automated MATRIX alignment for optimum performance in the current measurement. This process aligns the MATRIX detector module for the current parameters of the spectral detection ensuring centered illumination of the MATRIX array. It is recommended to run MATRIX alignment before a MATRIX measurement if any spectral detection windows (of any of the detectors) have been changed since the last MATRIX measurement.

### 2.7.5 Scan Range Live Dialog

In the 'Scan Range' live dialog (*FIGURE 2.23*) the basic scan parameters can be controlled in more detail than in the simplified easyCommander live dialog (see *THE SMART WAY TO START: EASYCOMMANDER*).

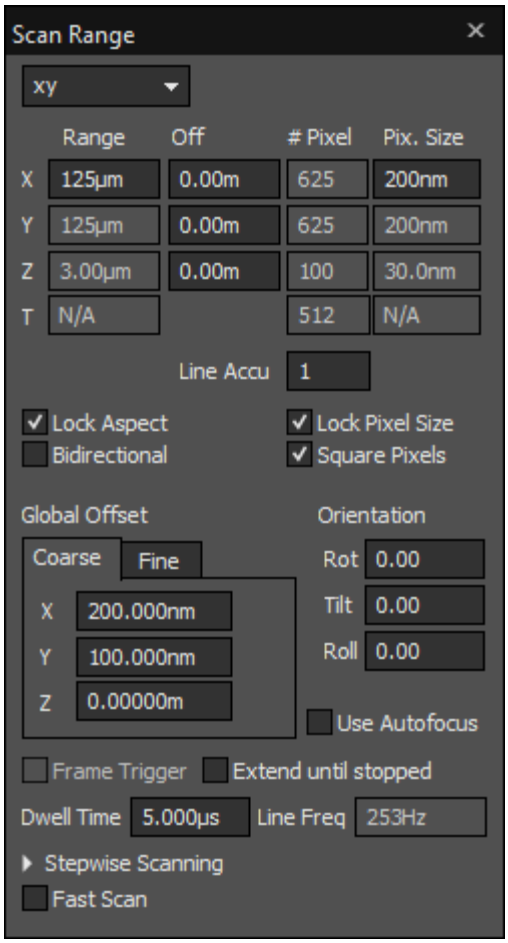



*Figure 2.23: 'Scan Range' live dialog*

**Offset classes**

In the Scan Range live dialog, we need to distinguish between local (measurement) offsets and global offsets. Local (measurement) offsets are tied to particular measurements. Activating a different measurement window will automatically replace these offsets with those from the new measurement. Local offsets can also be saved with the measurement. In contrast, global offsets will stay in place, even when a new measurement is opened or activated. Global offsets are not saved with the measurement. Local and global offsets are summed to produce the actual parameter used during acquisition.

The following parameters can be controlled:

**Scan axis:** The user can either expand the drop-down menu (top-left corner) to select from a list of common scan axis combinations (xy, yx, yz, xz, xt, xyt, xyz ...) or type an axis combination in the intended order. The following axis are directly available from this dialog:

x, y, z, t, l (= lambda for spectral scans).

There are additional axis types that are available only from additional live dialogs, e.g., the custom axis or timelapse axis which will both be displayed as a "?" axis (see ***CUSTOM AXIS LIVE DIALOG*** and ***TIMELAPSE LIVE DIALOG***).

There is currently a maximum number of four axis. The axis will be scanned in the order displayed here. Generally, it is recommended to scan the fastest axis first, e.g., XZ instead of ZX. The exact timing (syncing) of all scan axis can be modified in the ***MEASUREMENT SETUP LIVE DIALOG***.

> ***Note:*** *The "t" axis here only produces consecutive time steps with the length of the current dwell time. This allows extremely fast time series suitable e.g., for FCS measurements or movies at maximum frame rate. If you intend to produce time steps independent of the dwell time with a defined break/delay in between (e.g., for movies with a defined frame rate), please activate a timelapse axis via the* **TIMELAPSE LIVE DIALOG**.
>
> ***Note:*** *The z axis here always refers to the same Z-positioning device as the fine global offset (see fine global offset)*

**Range:** The Range defines the size of the acquired image or measurement for each axis in the units displayed. Drawing and applying ROIs, the range is automatically inserted. The maximum range of the x and y axis is defined by the current objective (e.g., 133 µm x 133 µm FOV for a standard 60x objective).

**Off:** The offset parameters at the top section of the live dialog are local offsets. They are typically used to define the lateral position of a ROI (x, y) within the field of view (FOV) while the global stage position remains fixed (global parameter). Drawing a ROI automatically generates these local offsets. Local offsets can also be used to define measurements that are acquired with an axial offset relative to other open measurements. Those relative axial offsets can be re-loaded with the measurement file.

**#Pixel:** By default, the number of pixels cannot be adjusted as it is automatically calculated from the range and pixel size (step size). This is the recommended setting. If the user prefers to set the number of pixels, the checkbox "Lock Pixel Size needs to be unticked (see below).

**Pixel Size:** The pixel size can be set by the user for every axis independently. The pixel size is kept constant even when the image range or scan speed are changed. For axis other than x or y, this becomes a more general step size (e.g., time steps or step size between image planes).

**Lock Aspect:** This checkbox can be used to link or unlink the range of the x and y axis. When locked, the image aspect ratio is fixed, i.e., when the X axis is changed, the Y axis is automatically adjusted by the same factor.

**Lock Pixel Size:** If this checkbox is ticked (default), the range and pixel size (step size) are adjustable producing a number of pixels that cannot be changed. This relates to the concept that the pixel size and dwell time should be kept constant (independent from image size and scan speed) to make images quantitatively comparable. If the user prefers to adjust the number of pixels, the checkbox "Lock Pixel Size needs to be unticked.

> ***Note:*** *The range may be rounded when a change is incompatible with pixel size and range.*

**Bidirectional:** Usually, at the end of one line, all lasers are blanked, and the mirrors drives quickly back to its starting position. In Bidirectional scan mode, the image is also acquired during this flyback. This increases the scan speed.

> ***Note:*** *A slight lateral displacement between two consecutive lines may occur. The size of the displacement depends on the scan settings (range, line frequency) and may be more visible in STED images due to the high resolution.*

**Square Pixels:** Ticking this checkbox (default) ensures square pixels along the x and y axis. If the user prefers non-square pixels with independent x and y dimensions, this checkbox needs to be unticked.

**Line Accu:** Just as in easyCommander, this field is a master switch controlling how many times a line of pixels in the image is scanned.

> ***Note:*** *In contrast to most other control software, Imspector calculates the signal sum, not an average signal, for repeated line scans. The idea is to preferably provide the user with raw signal (e.g., raw photon counts from APDs), without hidden adjustments or factors.*

If the measurement has been set up via easyCommander (without changes in the live dialog Line Steps, see ***2.7.9***), "Line Accu" simply sets the number of consecutive scans of the same line for each channel before moving on to acquisition of the next channel. For each channel, the signal is accumulated (summed) over the respective number of consecutive line scans. For all STED channels, the actual number of line accumulations is 3x the value displayed in the Scan Range live dialog. Hence, setting the line accumulations to 3 in easyCommander will generate a confocal image with 3-line accumulations and a STED image with 9-line accumulations. This usually generates more comparable signal levels in confocal and STED channels as STED switches off many fluorophores to generate a smaller scanning spot.

It is important to note that Imspector allows the creation of far more complex line scan patterns via additional live dialogs. For a complex line scan pattern, the easyCommander field "Line Accu." only sets the number of repetitions of this complex line pattern. For detailed information on the organization of channels and line scan patterns, please see sections ***Channels and Measurement Live Dialog*** and ***Line Steps Live Dialog***.

**Global Offsets:** As described above, these offsets are global, apply to all measurements and are not changed or saved by the measurement. The global offsets found here concern the X, Y and Z axis. The user switch between the coarse and fine global offsets via the tabs:

**Coarse global offsets** are generally the X, Y, Z position of the motorized stage and the nosepiece (unless custom hardware is used).

**Fine global offsets X, Y** are usually set by the galvo scanner (scanning mirrors).

**Fine global offset Z** can refer to the Z-piezo position if a Z-piezo if installed and connected/activated. When the continuous autofocus is activated, the fine global offset Z can also refer to the current stabilized Z position (distance from the detected coverslip position). As the Z-piezo allows much faster Z-positioning than most other devices, it is the preferred device for the fine offset Z when multiple devices are available. For this reason, if you would like to make use of the continuous autofocus, please disconnect/deactivate the Z-piezo in the software (see ***Microscope body Live Dialogs*** and ***Experimental Control – Scanning***).

**Use autofocus:** This checkbox has the same function as the checkbox “Continuous Autofocus / Autofocus enabled” in the Microscope body Live Dialogs. If an autofocus unit (Olympus ZDC or Nikon PFS) is installed, this activates/deactivates the continuous autofocus mode (focus lock). The continuous autofocus will compensate drift over longer periods of time while allowing acquisition of Z-stacks. This is especially helpful for live-cell imaging. In continuous autofocus mode, the zero position of the Z axis (as shown e.g., in easyCommander) now refers to the detected position of the coverslip surface. For more details on the autofocus, see *MICROSCOPE BODY LIVE DIALOGS* and *OLYMPUS ZDC: FOCUS SEARCH / CONTINUOUS AUTOFOCUS*).

> ***Note:*** *if a Z-piezo is installed, the continuous autofocus can only be controlled by the software, if the Z-piezo is deactivated (see fine global offset above). For information on deactivation of the Z-piezo see* EXPERIMENTAL CONTROL – SCANNING.

**Dwell Time:** This is defined as the integration time per pixel during the laser scanning process. See below for continuous and stepwise scanning processes.

**Line Frequency:** The Line Frequency cannot be adjusted directly as it is a result of the settings for the axis range, the pixel size and the dwell time. To change the line frequency indirectly, adjust axis range, pixel size and/or dwell time. In addition, the line frequency can be affected by factors such as scan mode (see Fast Scan), uni/bidirectional scanning, stepwise scanning, syncing options and flyback options.

**Stepwise scanning:** Ticking this checkbox will force stepwise scanning of an axis, with steps either between lines or pixels. Per default, the axis will be scanned in a continuous movement whenever possible (checkbox unticked). This is the fastest method, especially for devices with inertia (e.g., galvo scanners) but not all devices are capable of this (e.g., autofocus devices). A different option of altering the scan progress is the *MEASUREMENT SETUP LIVE DIALOG* which allows to modify the syncing options (when is the scan stopped and the position checked) without introducing steps.

**Orientation:** These parameters allow rotation and tilting of the scan field in all three dimensions.

- Rot: This option rotates the scan field around the Z axis.
- Tilt: This option rotates the scan field around the X axis.
- Roll: This option rotates the scan field around the Y axis.

**Fast Scan:** Tick this checkbox to switch the galvo scanner from the default sawtooth scan mode to a sinusoidal scan mode (Fast Scan mode). The duty cycle value represents the percentage of the scan curve actively read out for the image. The higher the duty cycle value, the higher is the line frequency. Values between 60% and 95 % are recommended, depending on the ROI size.

> ***Note:*** *The sinusoidal Fast Scan is not always faster than the default sawtooth scan. Generally, for scan ranges <10µm, the Fast Scan mode results in a higher line frequency. For larger images, the sinusoidal Fast Scan mode may also lead to lateral shifts, distortions.*

**Frame Trigger:** This option may be utilized by expert users to wait for a frame trigger signal in the acquisition process.

**Extend until stopped**: Acquisition option only for specialized applications.

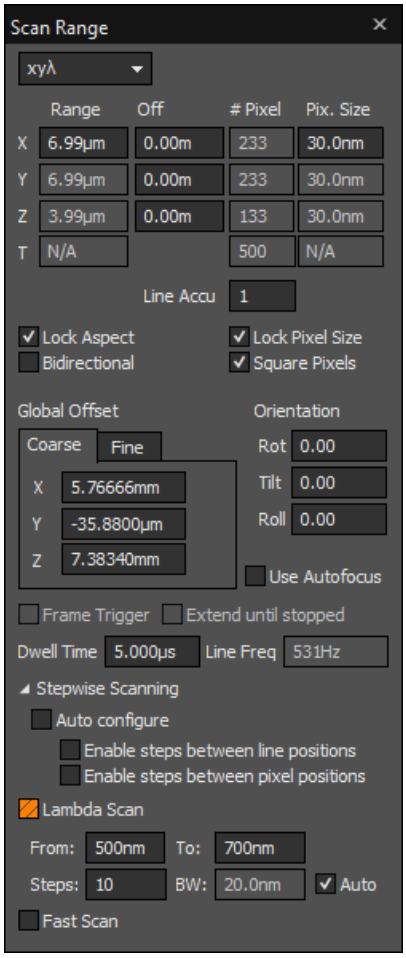



*Figure 2.24: 'Scan Range' live dialog with additional options for Lambda Scan and Stepwise Scanning*

**Lambda Scans (spectral scans)**

A lambda scan will set the current spectral detector to a narrow spectral detection window with a fixed width (bandwidth = BW) and will move this spectral window along the spectral axis in a defined range.

A Lambda Scan can be setup by clicking on the axis drop-down menu, typing an axis combination including "l" for lambda, e.g., xyl. (Later, the "l" will automatically be displayed as a "λ"). It is recommended to set the spectral axis as the last axis, as this is the commonly the slowest axis.

Once the scan includes a spectral axis, additional options become available in the Scan Range live dialog (see ***FIGURE*** 2.24). The user can determine the spectral range to be covered by the lambda scan and set the number of steps for the lambda scan. By default, the bandwidth (BW) is adjusted automatically. The user can choose to manually adjust the bandwidth (see checkbox) to allow more or less signal intensity or to adjust the spectral resolution of the Lambda Scan.

> ***Note:*** *Lambda Scans can be performed only if a single channel and single detector with spectral detection is active in the measurement.*

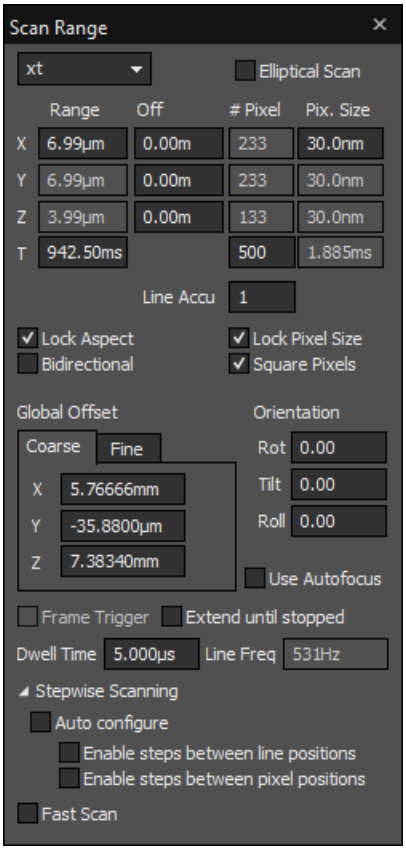



*Figure 2.25: 'Scan Range' live dialog with line and elliptical scan options*

**Fast time series for FCS measurements**

The "t" axis in the Scan Range live dialog produces consecutive time steps with the length of the current dwell time. This allows extremely fast time series suitable for FCS measurements.

A point scan FCS measurement can be set up by clicking on the axis drop-down menu and typing a single "t". Pease use the local offsets x, y, or the ROI functions to determine the position of the point scan. The length (and integration time) of each time step can be adjusted via the field "Dwell Time". Adjust the number of pixels or the t axis range to determine the length of the FCS measurement.

Similarly, a line scanning FCS measurement can be set up by clicking on the axis drop-down menu and typing single "xt" (see *FIGURE* 2.25). Pease use the local offsets x, y, or the ROI functions to determine the position and length of the line scan. The length of the line scan can be adjusted via the x axis range. If the user prefers ROI handling transferring only the ROI center position (not the range), this can be achieved by holding the keyboard short "c" while transferring ROIs via drag&drop.

The user can change a line scanning FCS measurement into an elliptical scanning FCS measurement via the checkbox "Elliptical Scan". Now the size of the elliptical scan (or in some cases circular) can be adjusted via the x and y axis range. Please note that the aspect ratio of the ellipse is fixed as long as the checkbox "Lock Aspect" is ticked. Using ROI drag&drop, the position and size of the elliptical scan can be transferred (holding keyboard shortcut "R") or only the center position of the elliptical scan is transferred (holding keyboard shortcut "C").

The number of line accumulations should be set to one for FCS measurements.

### 2.7.6 Measurement Setup Live Dialog

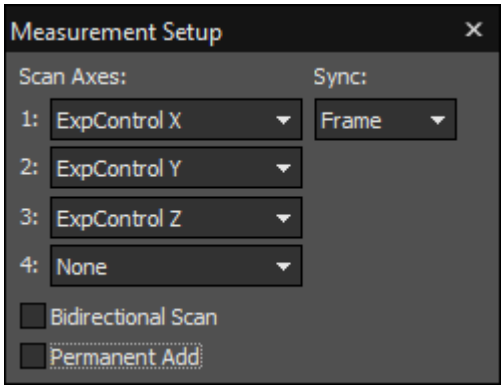



*Figure 2.26: 'Measurement Setup' live dialog.*

Alternatively, to the 'Experimental Control' live dialog, the 'Scan Axes' can be selected in the Measurement Setup live dialog (***FIGURE 2.26***). Note that here in addition to the three axes that are available in the 'Experimental Control' live dialog a fourth axis may be defined. The additional axis might be the *time lapse* (***2.7.7***) or the *custom* axis (***2.7.16***).

The user can choose the order in which the axis are scanned. Please note that selecting a slow axis as the first axis will slow the measurement down considerably (e.g., spectral scan axis or Z-axis without a fast Z-piezo).

The type of syncing between axis can also be changed. This determines, if the microscope will wait and check before proceeding e.g., after each single line or after a full frame. Frame syncing allows faster continuous scans (good for X/Y/Z) while slower line syncing may be required for slower hardware components (e.g., vertical XZ scan with continuous autofocus).

> ***Note:*** *These settings are automatically set when using the easyCommander (**2.4.1**) live dialog.*

### 2.7.7 TimeLapse Live Dialog

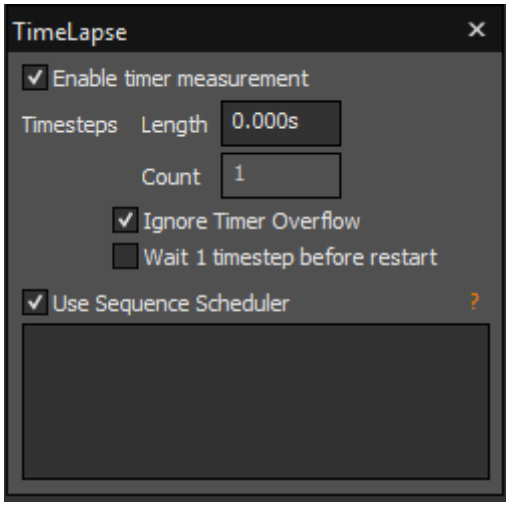



*Figure 2.27: 'Time Lapse' live dialog.*

In the 'TimeLapse' live dialog (***FIGURE 2.27***), time lapse measurements with waiting times in between the individual steps can be devised. More advanced settings with unequal steps can be further devised using the 'Sequence Scheduler'. The length parameter defines the time in between the starting point of two consecutive scans. If the time needed to perform one scan is longer than the length parameter a warning would occur during the measurement. If this warning can be ignored, activate the check box *"Ignore Timer Overflow"*. In this case, if the scan duration is longer than the wait time, the 2 consecutive scans will be done without a waiting time (time lapse runs as fast as possible).

> ***Please note*** *that when "Ignore Timer Overflow" is activated, there is no notification if the actual timestep length differs from the originally selected timestep length.*
>
> ***Please note*** *the tooltips on the usage of the sequence scheduler which can be opened using the question mark at the right side of the live dialog.*

## 2.7.8 *Channels* and *Measurement* Live Dialog

The easiest way to set up a measurement is via the easyCommander which provides all the required standard settings based on a dye selection. For more experienced users, live dialogs offer more advanced measurement designs, settings beyond the standard parameters and direct control of individual parameters.

The live dialogs “Channels” (*FIGURE 2.28*) and “Measurement” *(FIGURE 2.29*) are central to organizing and modifying a measurement. Both dialogs are used to define the number of channels used in the measurement, the corresponding detectors and which lasers are switched to a given power in each channel. Furthermore, the imaging modes 'Line Steps' (*2.7.9*) and ‘Pixel Steps’ (*2.7.10*) can be activated.

In most cases, it is recommended to start with parameters generated via easyCommander followed by manual changes via these live dialogs. For more information on easyCommander and how it handles manual (sometimes conflicting) changes, see section *2.4.1*.

After creating and defining channels via the “Channels” and “Measurement” live dialogs, users often proceed to the “Line Steps” live dialog to arrange the order and timing of the channels (see *2.7.9*).




*Figure 2.28: ‘Channels’ live dialog*

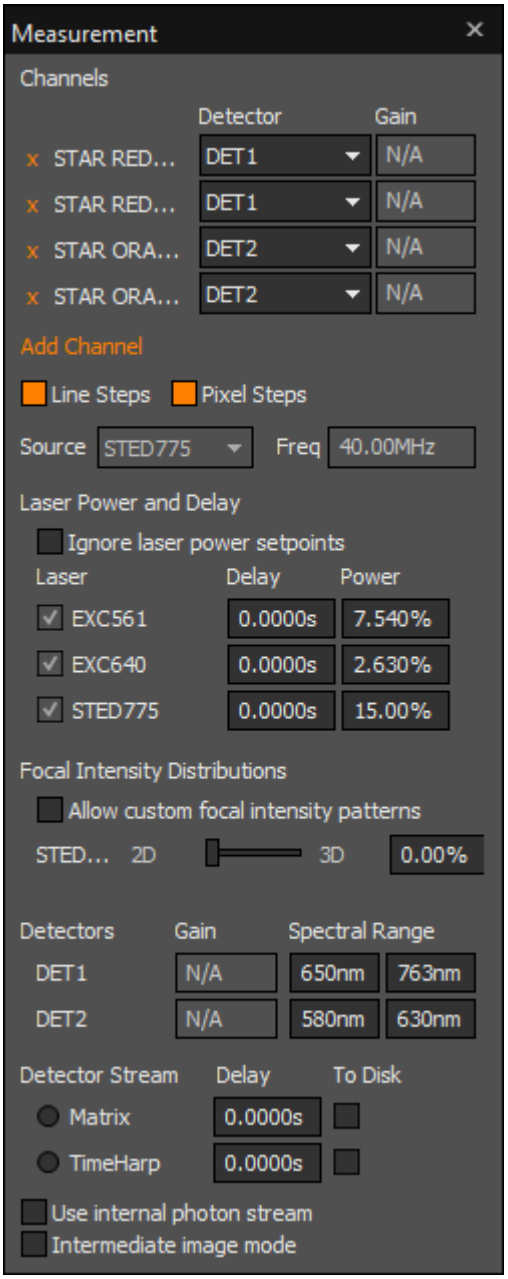



*Figure 2.29 'Measurement' live dialog*

### 2.7.8.1 Channels

Detection channels can be navigated using the tabs at the top of the Channels live dialog. Each channel can be assigned exactly one detection source and a range of settings, e.g., one or multiple active lasers, laser powers, detection modes and more.

Detection channels can be added or removed in the *Channels* dialog via the “+” and “remove” signs. Per default, newly added channels are named *Ch1, Ch1, Ch3, Ch4*. If the measurement has been set up via the *easyCommander* live dialog, each channel automatically receives a name according to the dye and imaging mode (confocal, STED…). The user can also assign channel names manually.

The temporal arrangement of channels (simultaneous, line sequential etc.) can be set up in the *Line Steps* (**2.7.9**) and *Pixel Steps* (**2.7.10**) live dialogs.

> ***Note:*** *Up to 16 logical (detection) channels can be used.*

### 2.7.8.2 Detectors

In the *Detector* section of the *Channels* live dialog, the different detectors (APDs/PMTs/MATRIX…) can be selected and assigned to the (logical) detection channels.

If PMT detectors are selected, the gain of the PMT may be adjusted. For APDs a gain is not applied.

If a MATRIX detector is selected, its various imaging mode can be adjusted (see section **2.7.8.7**).

The field *Spectrum* in the *Channels* live dialog allows adjustment of the spectral detection window if the selected detector is connected to a spectral detection module. If not, the field is greyed out and fixed values for the respective filter set are shown.

In setups with multiple spectral detection modules, the order of detectors (and their connected spectral detection modules) is crucial as these are usually connected in series (see **FIGURE** 2.30). Each spectral detection module diverts the intended part of the spectrum to its detector. Only light of wavelengths shorter than the selected detection window is transferred to the next detector while the longer wavelength portion is lost. As a result, the first detector should be selected for the detection window with the longest wavelength, followed by the next detectors in numerical order, each set to detection windows with increasingly shorter wavelength.

> ***Note:*** *A detector in the series may be skipped by setting its spectral range to 780 nm – 780 nm for bypassing. This can easily be achieved by adding an otherwise unused detection channel (without active lasers), selecting the respective detector set to 780 nm – 780 nm.*

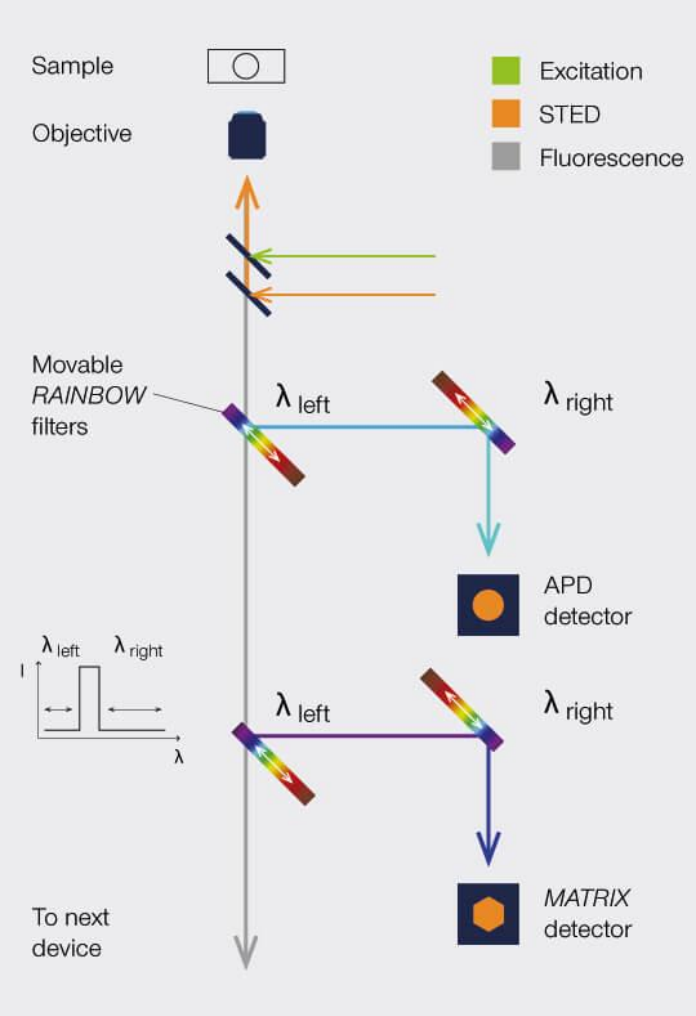



*Figure 2.30: Rainbow spectral detection concept*

### *2.7.8.3 Lasers*

In the *Lasers* section of the *Channels* live dialog, one can select which lasers are switched on and at what power. Power values are given in % of full power (full power level may be adjusted separately for some lasers, e.g., STED lasers, in a separate laser live dialog).

Multiple lasers may be switched on for one channel, e.g., an excitation laser and a STED laser (see figure). Each laser may be active in multiple channels and can be set to a different power in each channel, as long as those channels are not active simultaneously (see *Line Steps* and *Pixel Steps* live dialogs).

> ***Note:*** *Switching laser power between channels may require a more recent FPGA hardware version.*

If no STED lasers are active in any of the channels, the user can adjust the repetition rate for the excitation lasers by changing the value in the *Freq* field of the *Measurement* live dialog. However, it is not recommended to change the frequency from the default value of 40 MHz. If a STED laser is active in any of the channels, the repetition rate for all lasers is fixed to the repetition rate of the STED laser (generally 40 MHz).

### 2.7.8.4 Gating

The APD and MATRIX detection channels built in the microscope are all capable of time-gated detection. In short, detection of photons can be limited to a defined time span that starts with a delay after the excitation laser pulse (see *FIGURE 2.31*). Amongst other functions, this enables time-gated STED microscopy. In short, detection can be limited to those photons that are emitted after the STED laser pulse has completed depleting the dyes at the periphery of the scanning spot. This rejects earlier "low resolution" photons providing increased resolution.

See the following references for details on STED and time-gated detection: G. Vicidomini, et al., *Nat. Meth.* **8**, 571 – 573 (2011) and F. Göttfert, et al., *Biophys. J.* **105**, L01 – L03 (2013).

The time-gating settings are defined in the *Gating* section of the *Channels* live dialog. By default, *easyCommander* will set the delay of the time gate for STED to 750 ps. Manually setting a shorter delay may recover some signal intensity for dyes with a short fluorescence lifetime but may also reduce resolution or introduce hazy background. The default width of the time gate for STED images is 8 ns. For dyes with a very long fluorescence lifetime, a longer time gate may provide additional fluorescence signal.

In confocal images, *easyCommander* will deactivate time gating by default. If confocal images have a strong background from unwanted fluorescence with a short fluorescence lifetime (e.g., auto fluorescence), the user can manually activate time-gating for confocal images in the *Channels* live dialog.

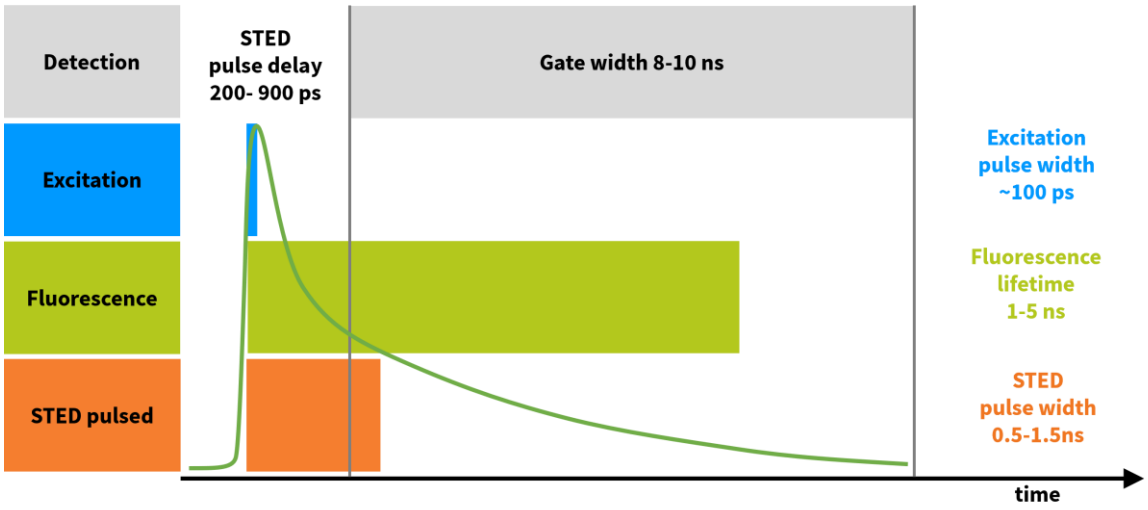



*Figure 2.31: Time-gated detection and laser pulses*

### 2.7.8.5 FLIM (APD with TCSPC electronics)

The 'Channels' live dialog allows control of FLIM (fluorescence lifetime imaging microscopy) measurements.

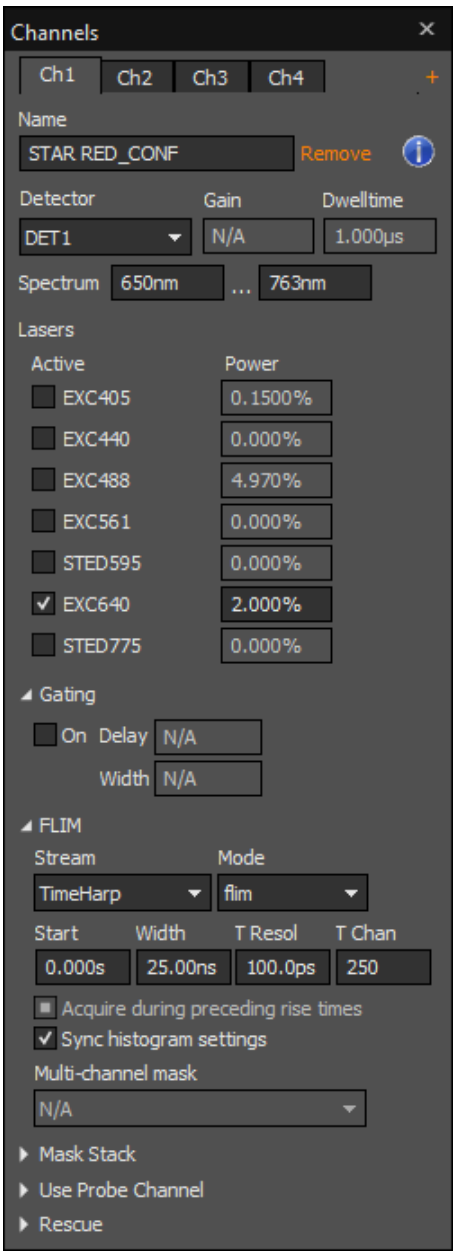



*Figure 2.32: 'Channels' live dialog (FLIM controls)*

### Hardware

The standard APD (avalanche photodiode) detectors in abberior microscopes readily allow time-gating (see section *2.7.8.4*) via the inbuilt (internal) fast electronics. While time-gating uses a single time-gate to select or reject detected photons based on their arrival time after the laser pulse, FLIM can create a full histogram of the arrival times of all photons (and other advanced time-correlated functions). FLIM mode via the APD detectors requires dedicated FLIM hardware (optional upgrade), i.e., TCSPC (time-correlated single photon counting) electronics. The TCSPC electronics may be supplied by different manufacturers, e.g., Becker&Hickl or PicoQuant and are integrated by our production team. Depending on the model, one or several APD detectors may be connected to the TCSPC electronics. Generally, APD detector signal is split and simultaneously sent to the inbuilt internal electronics (for time-gating) and the TCSPC electronics (for FLIM).

Alternatively, FLIM can be performed with our MATRIX detector without further hardware upgrades (see section *2.7.8.6*).

### FLIM controls

The 'Channels' live dialog section can be expanded (see *FIGURE* 2.32) to reveal FLIM controls. The functions of the following parameters are also illustrated in *FIGURE* 2.33.

**'Stream':** The drop-down menu 'Stream' allows selection of the 'internal' electronics (time-gating only) or a TCSPC module (if installed) for FLIM options.

**'Mode':** For the streams of TCSPC modules, multiple modes are available. The 'counter' mode handles the same way as the internal streams and allows only time-gating. However, the internal stream is usually preferred for simple time-gating functions as many TCSPC models saturate at lower photon count rates. The 'flim' mode enables FLIM measurements and reveals additional options for the resulting photon arrival time histogram. The 'plim/pat' mode enables phosphorescence measurements which are time-resolved at a much longer time-scale (described elsewhere).

**'Start':** The 'Start' parameter shifts the start point of the detection time window in which the FLIM measurement is performed, i.e., defines the start point for the histogram of photon arrival times. The shift occurs relative to the laser pulse triggers. Before delivery, timing has usually been adjusted so that a 'Start' value of zero will include detection of the laser pulse shortly after the start histogram.

**'Width':** This adjusts the length of the detection time window in which the FLIM measurement is performed, i.e., width of the histogram of photon arrival times. As our microscopes mostly run at laser repetition rates of 40 MHz (one pulse every 25 nanoseconds), a maximum width of 25 ns is recommended. The histogram will contain arrival times of photons from a large number of laser cycles as the pixel dwell time is usually much longer than the FLIM histogram width.

**'T Resol':** Photons detected via the TCSPC module are sorted according to their arrival time into a large number of time bins to produce a histogram. The parameter 'T Resol' sets the size of each time bin.

**'T Chan':** This parameter defines the number of time bins into which the detected photons will be sorted according to their arrival time.

> ***Note:*** *The parameters 'Width', 'T Resol' and 'T Chan' are related, as they set the range, pixel size and number of pixels for the FLIM time axis. When adjusting one parameter, another parameter will automatically be calculated accordingly.*

**'Sync histogram settings' checkbox:** To simplify FLIM control in multi-channel measurements, the histogram settings for multiple channels/detectors can be synced by checking this box.

> ***Note:*** *To avoid artifacts and saturation effects of the APD detectors or TCSPC electronics, please adjust excitation laser power to reach sufficiently low count rates. Increase pixel dwell time and/or line accumulations to provide sufficient signal (e.g., >50 µs).*

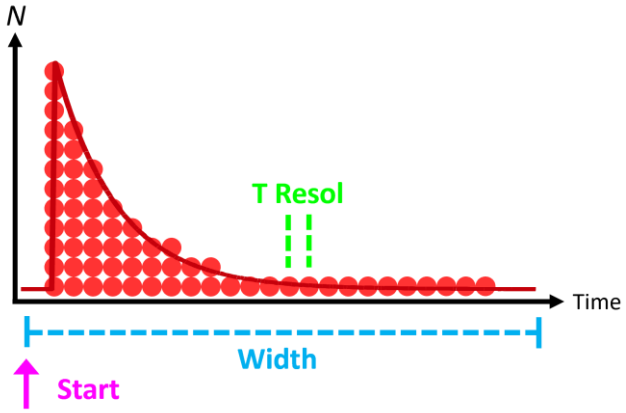



*Figure 2.33: Channels Live dialog (FLIM parameters)*

Once the FLIM measurement has been started, the histogram can be displayed by right-clicking on the image and selecting 'T Profile'. To generate a fluorescence lifetime image, fractional lifetime analysis can be performed via the 'Stack Display' life dialog (see section **2.10.4**) or the or via 'Menu bar>Analysis>Fractional lifetime analysis'.

> ***Note:*** *For more advanced TCSPC features, the raw photon streams containing the individual photon arrival times can be written to the disk. This mode can be activated by ticking the checkbox 'To disk' behind the individual streams in the 'Measurement' live dialog (see* FIGURE 2.29*). The location and format of the photon stream files can be defined in the 'Hardware Settings' menu (see section* 2.8.3*).*

### *2.7.8.6 FLIM (MATRIX detector)*

The MATRIX detector comprises more than twenty individual APD detector pixels which allow time-resolved detection (see section **2.7.8.7**). In contrast to the conventional APD detectors, the MATRIX detector does not require additional hardware and is ready for FLIM. The MATRIX detector has a superior time resolution when compared to the standard APD detectors in our microscopes (upgrades to specialized APD detectors with higher time-resolution available). With the MATRIX detector, saturation artefacts may occur at higher photon count rates than with other APD-based detectors as the signal is distributed onto multiple individual detector pixels.

FLIM measurements are performed similar to the conventional APD detectors, and the controls are mostly identical. However, the user can choose whether to perform FLIM for each detector pixel individually or for the total signal summed over all detector pixels.

> ***Note:*** *Performing FLIM individually for each MATRIX detector pixel will reduce the signal available for each FLIM analysis.*

> ***Note:*** *Performing FLIM individually for each MATRIX detector pixel will add another axis to the measurement (in addition to the conventional FLIM axis). This can limit the measurement options as a maximum of four axis can be selected per measurement.*

### 2.7.8.7 Pixel array (MATRIX detector)

The MATRIX detector comprises an array of more than twenty individual APD detector pixels. This allows the analysis of the point spread function (PSF) of the detected light and the discrimination of photons based on their spatial information. In practice, this allows the discrimination of in focus versus out-of-focus light which can dramatically reduce the background signal in STED images. Traditionally this level of background suppression requires the use of a very small pinhole which reduces the overall signal strongly. In contrast, for MATRIX detection the pinhole can be opened widely (recommended above 3 AU) for maximum signal collection. The In addition, it allows improved Z-sectioning and improved lateral resolution (image scanning microscopy).

Apart from spatial discrimination, the MATRIX detector also allows time-resolved detection enabling time-gating (see *2.7.8.4*) and FLIM (see *2.7.8.5*).

The image analysis features necessary for the beneficial features of the MATRIX detector are implemented in our Lightbox software interface. However, the Imspector software allows the user to record MATRIX detector raw data in several different array modes for custom analysis by the user. In the 'Channels' live dialog, the pixel array mode can be changed in the 'FLIM' section (see *FIGURE 2.34*):

**'Multi channel mask':** Here the user can select which detector pixel of the MATRIX array detector are active and how the signal is recorded.

- **'Raw':** Individual image per detector pixel. This introduces an additional measurement axis for the detector pixel order.
- **'Center map 2px':** Similar to 'raw', individual images are recorded per detector pixel. Only the central pixel and the first ring of pixels around it are included.
- **'Center map 3px':** Similar to 'raw', individual images are recorded per detector pixel. Only the central pixel and the first two rings of pixels around it are included.
- **'Sum':** The signal from all detector pixels is summed. Only a single image is generated (no additional measurement axis).
- **'Center sum 2px':** Similar to 'Sum', individual images are recorded per detector pixel. Only the central pixel and the first ring of pixels around it are included.
- **'Center sum 3px':** Similar to 'Sum', individual images are recorded per detector pixel. Only the central pixel and the first two rings of pixels around it are included.

When using the MATRIX detector, several other functions may be essential:

**Spectral detection:** The MATRIX detector is commonly installed as the third detector in a block of spectral detection modules. To transfer detection light of the intended spectrum to the MATRIX detector, the first two detectors need to be skipped and the spectral detection range needs to be adjusted for the third detector (see 'Channels' live dialog section *DETECTORS*).

**Alignment:** The detection PSF should be centered onto the central detector pixel for optimum performance. This can be achieved automatically for the current settings via the 'Alignment' live dialog (see *2.7.4*). The alignment can be checked manually in the live dialog 'MATRIX'.

**Overload:** Should the MATRIX detector become overloaded due to extremely high signal intensity, there is the option to reset and reactivate it via the 'Clear overload' button in the 'MATRIX' live dialog.

> ***Note:*** *Acquiring data individually for each MATRIX detector pixel will add another axis to the measurement. This can limit the measurement options as a maximum of four axis can be selected per measurement.*

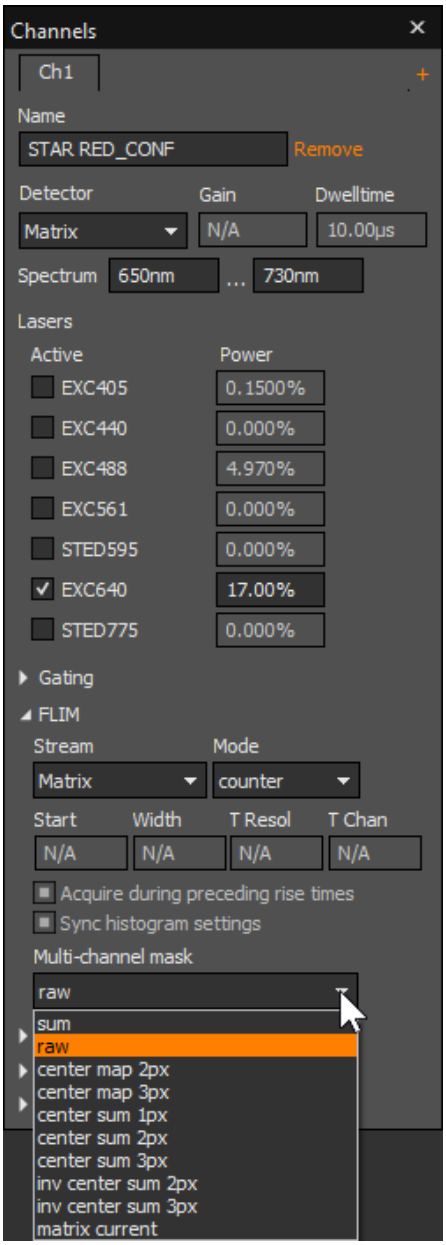



*Figure 2.34: 'Channels' live dialog (MATRIX pixel options)*

#### 2.7.8.8 Adaptive illumination

Adaptive Illumination is best controlled via easyCommander (see **2.4.3**) as most parameters in the Channels live dialog have a more intuitive counterpart in easyCommander.

## 2.7.9 Line Steps Live Dialog

For measurements with more than one active channel, the acquisition order or pattern of these channels can be designed *(see* **FIGURE** 2.35) using the acquisition modes *Line Steps* or *Pixel Steps. Line Steps* and/or *Pixel Steps* can be activated in the *Measurement* live dialog (**2.7.8**).

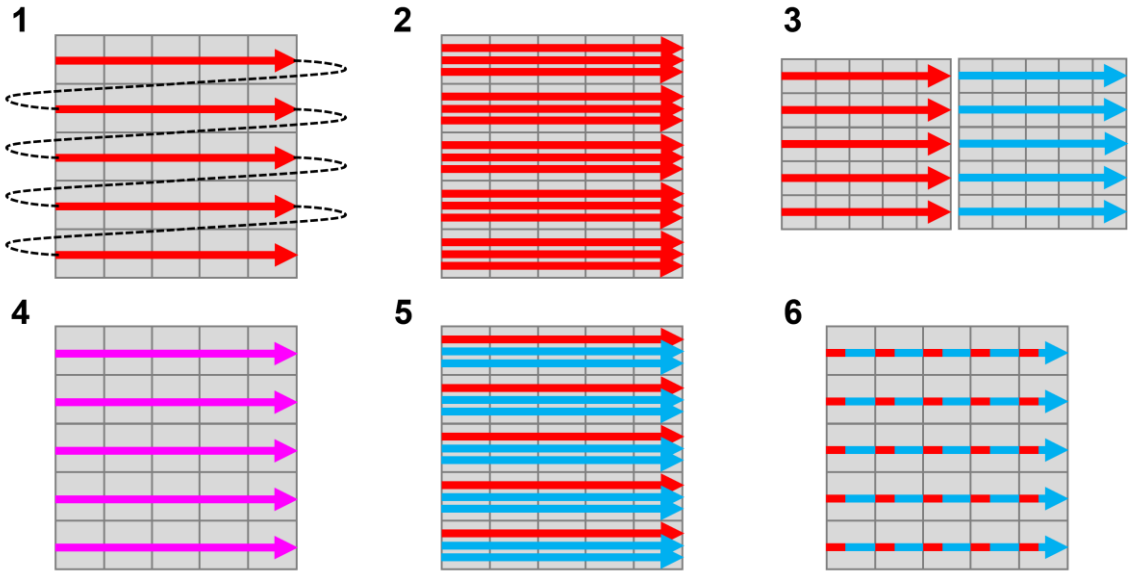



*Figure 2.35: Acquisition modes for single and multi-channel imaging*

**1.** Single-channel image scan (red) with cartoon representations of pixels (grey boxes) and scanner accelerations / fly-back (black dashed line). **2.** Single-channel scan with three-line accumulations (red) per line of pixels. **3.** Frame-sequential scan of two channels (red, blue). **4.** Simultaneous scan of two channels (magenta) can cause higher channel crosstalk. **5.** Line-sequential scan of two channels (red, blue) with different numbers of line accumulations per channel. **6.** Pixel-sequential scan of two channels (red, blue).

With *Line Steps*, the channels are recorded line-sequentially. The user can choose to scan a line repeatedly to allow separate line scans for each channel or choose to scan two or more channels simultaneously within the same line scan. Compared to frame-sequential scanning, line-sequential scanning offers much shorter delay times when switching between channels. Line-sequential scanning therefore enables multi-channel colocalization analysis when imaging live or fast-moving samples.

**Line Step concept**

A *Line Step* is used to define a line scan, its number of repetitions, its position in a sequence of line scans and its active channels and lasers.

### Line Step control

In the *Line Steps* live dialog (*FIGURE* 2.36), we can activate multiple sequential *Line Steps* by ticking the boxes *S1* to *S8*. Each *Line Step* is then graphically represented by a column of boxes. We can assign one or more channels to a *Line Step* column, by clicking the boxes in the row of the desired channels, highlighting active channels in orange. Below the active channels (rows), we can check which lasers (rows) are activated in each *Line Step* (columns). Lasers are automatically activated according to the settings of the active channels (see *Channels* live dialog *2.7.8*).

> ***Note:*** *By default, easyCommander will activate only one channel per Line Step. A line is scanned repeatedly with channels active one at a time until all channels are completed. This mode minimizes crosstalk between channels in multi-color imaging as only the correct excitation lasers and detection channels are active when imaging each dye.*

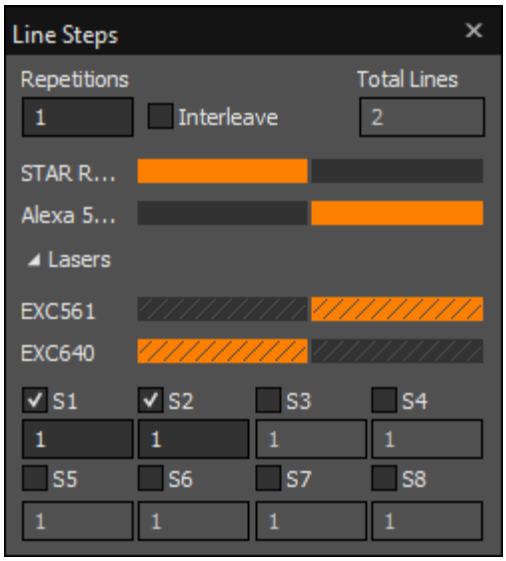



*Figure 2.36: 'Line Steps' live dialog*

For optimization of signal intensity in each channel, line accumulations can be applied. Here, a line is repeatedly scanned for the same channel(s) with summation of the signal intensities. In the Line Steps live dialog, the number of accumulations (repeats) can be set for each Line Step individually in the field below the tick box of the respective line step.

> ***Note:*** *EasyCommander will automatically set a single line accumulation for confocal channels and three-line accumulations for STED channels to promote comparable intensities.*

The displayed pattern of Line Steps and line accumulations can be repeated multiple times. In the Line Steps live dialog, the number of pattern repetitions can be set in the field Repetitions.

> ***Note:*** *The overall number of accumulations of a Line Step is the product of pattern Repetitions and the individual line accumulations per Line Step shown in the pattern.*

> ***Note:*** *The complete number of line accumulations of a channel is the sum of accumulations of all line steps featuring this channel (see above). This is due to the fact that a channel may be assigned to more than one Line Step.*

> ***Note:*** *The number of line accumulations set in the simple easyCommander live dialog is mirrored in the Repetitions field in the Line Steps live dialog. Please multiply this value as described above to find the overall number of line accumulations for a given channel.*

Repetitions of the displayed line scan pattern can be performed in two ways. When activating the Interleave tick box, the line scan pattern is completed as displayed and then repeated a given number of times as a whole block. Without Interleave mode, the first line step is accumulated as shown in the individual line step accumulations. Then, for the same line step, the individual line accumulations are repeated as defined in the Repetitions field. Only after all line accumulations and Repetitions are completed for the first line step, the second line step is tarted. Generally, most measurements work best without Interleave mode.

> ***Note:*** *Interleave mode should only be activated when absolutely required. It is incompatible with some other imaging modes, e.g., Adaptive Illumination (2.4.3).*

**Examples**

The examples in FIGURE 2.37 illustrate the large range of possible *Line Step* patterns and adjustments. Here, two channels are arranged: One channel for abberior STAR RED with active 640 nm excitation laser and one channel for Alexa Fluor 594 with active 561 nm excitation laser.

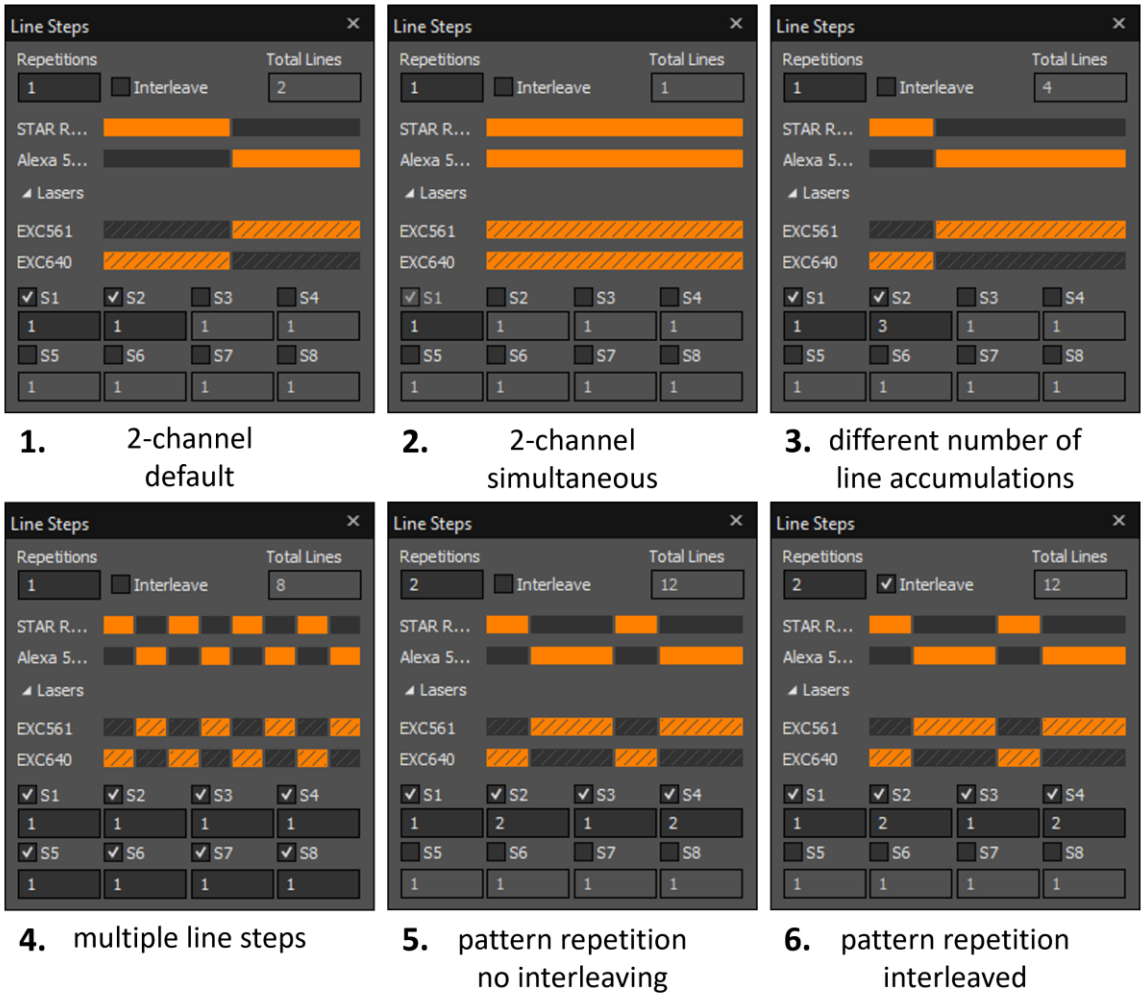



*Figure 2.37: Examples of Line Step patterns*

**1.** Per default, the easyCommander live dialog will arrange two channels line-sequentially, with each channel active in a separate line step for lowest channel crosstalk. Note that in each line step only one excitation laser is active.
**2.** The user can choose to record two channels simultaneously in one *Line Step* for increased speed. However, both excitation lasers are active simultaneously, increasing crosstalk.
**3.** The user can also adjust the number of line accumulations for each *Line Step* individually.
**4.** Activation of additional *Line Steps* allows custom patterns and complex assignments of the channels.
**5.** The number of Repetitions of a pattern can be chosen. Without interleaving, the *Line Steps* will be recorded as follows: 2x1 accumulation of *Line Step* 1, 2x2 accumulations of *Line Step* 2, 2x1 accumulations of *Line Step* 3, 2x2 accumulations of *Line Step* 4. The STAR RED channel signal is accumulated 2x1 + 2x1 = 4x while the Alexa Fluor 594 channel signal is accumulated 2x2 + 2x2 = 8x.
**6.** With interleaved two-fold pattern repletion, the line steps will be recorded as follows: 1x1 accumulation *Line Step* 1, 1x2 accumulations *Line Step* 2, 1x1 accumulation *Line Step* 3, 1x2 accumulations *Line Step* 4 and again 1x1 accumulation *Line Step* 1, 1x2 accumulations *Line Step* 2, 1x1 accumulation *Line Step* 3, 1x2 accumulations *Line Step* 4. The STAR RED channel signal is accumulated 1x1 + 1x1 + 1x1 + 1x1 = 4x while the Alexa Fluor 594 channel signal is accumulated 1x2 + 1x2 + 1x2 +1x2 = 8x.

**Combination of *Line Steps* with *Pixel Steps***

Analogous to *Line Steps*, the channels can be organized in *Pixel Steps* (*2.7.10*). For simplicity, it is recommended to use either multiple *Line Steps* or multiple *Pixel Steps*. A combination of *Line Steps* and *Pixel Steps* is possible but quickly creates highly complex patterns. (For combinations, a given channel is only active if it is active in both the current *Line Step* and the current *Pixel Step*).

## 2.7.10 Pixel Steps Live Dialog

Pixel steps and *Line Steps* (*2.7.9*) work very similarly. For *Line Steps*, active channels and lasers are switched line sequentially. For *Pixel Steps*, active channels and lasers are switched pixel-sequentially.

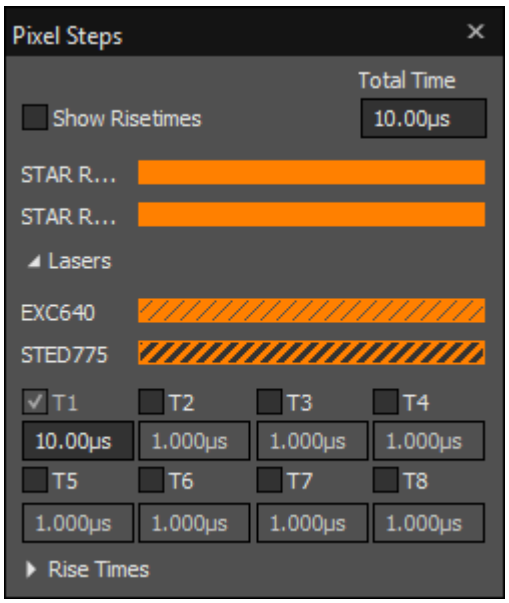



*Figure 2.38: 'Pixel Steps' live dialog*

abberior microscopes allow very fast adjustment of laser powers (sub-microsecond rise-times) which enables *Adaptive Illumination* (*2.4.3*) and also pixel-sequential scanning. Pixel-sequential scanning is required for other super-resolution techniques such as RESOLFT, where dyes need to be illuminated with a sequence of lasers during the acquisition of a single pixel. On the other hand, pixel-sequential scanning is rarely needed for STED imaging. Compared to frame-sequential and line-sequential scanning, pixel-sequential scanning offers even shorter delays between the acquisition of multiple channels. However, for most live specimen, line-sequential scanning is more than sufficient and therefore the default mode when using *easyCommander*.

**Pixel Step concept**

The user can choose to split the original pixel dwell-time (the time period the confocal scanning spot stays on a given pixel) into a sequence of several *Pixel Steps*. Each *Pixel Step* can be defined with an individual duration (dwell time), its position in a sequence of pixel steps and its active channels and lasers.

### Pixel Step control

The *Pixel Steps* live dialog (*FIGURE* 2.38), is organized analogous to the Line Steps live dialog.

In the *Pixel Steps* live dialog, we can activate multiple sequential *Pixel Steps* by ticking the boxes *S1* to *S8*. Each *Pixel Step* is graphically represented by a column of boxes. We can assign one or more channels to a *Pixel Step* column, by clicking the boxes in the row of the desired channels, highlighting active channels in orange. Below the active channels (rows), we can check which lasers (rows) are activated in each *Pixel Step* (columns). Lasers are automatically activated according to the settings of the active channels (see *Channels* live dialog **2.7.8**).

For optimization of signal intensity in each channel, the *Pixel Step* duration (dwell time) can be adjusted (analogous to line accumulations for Line Steps). In the *Pixel Steps* live dialog, the duration can be set for each *Pixel Step* individually in the field below the tick box of the respective *Pixel Step*. The box *Total Time* shows the sum of the durations of all *Pixel Steps*.

> ***Note:*** *The overall dwell-time for a channel is the sum of the dwell-times of all Pixel Steps featuring this channel. This is because a channel may be assigned to more than one Pixel Step. If line accumulations are also active, we need to further multiply the channel dwell-time with the number of line accumulations (see Line Steps* **2.7.9***) to get the final dwell-time of a channel.*
>
> ***Note:*** *The easyCommander live dialog mirrors the field Total Time in the Pixel Steps live dialog.*

### Rise Times

As described above, abberior microscopes can adjust laser powers within sub-microsecond time periods. Still, pixel-sequential scanning requires very short breaks between *Pixel Steps* to allow completion of these fast laser powers adjustments. *Pixel Steps* are therefore most efficient for longer dwell-times (Pixel Step durations above around 2 µs). To visualize the rise times in the *Pixel Steps* live dialog, please activate the tick box *Show Rise Times*.

### 2.7.11 SLM Live Dialog (spatial light modulator)

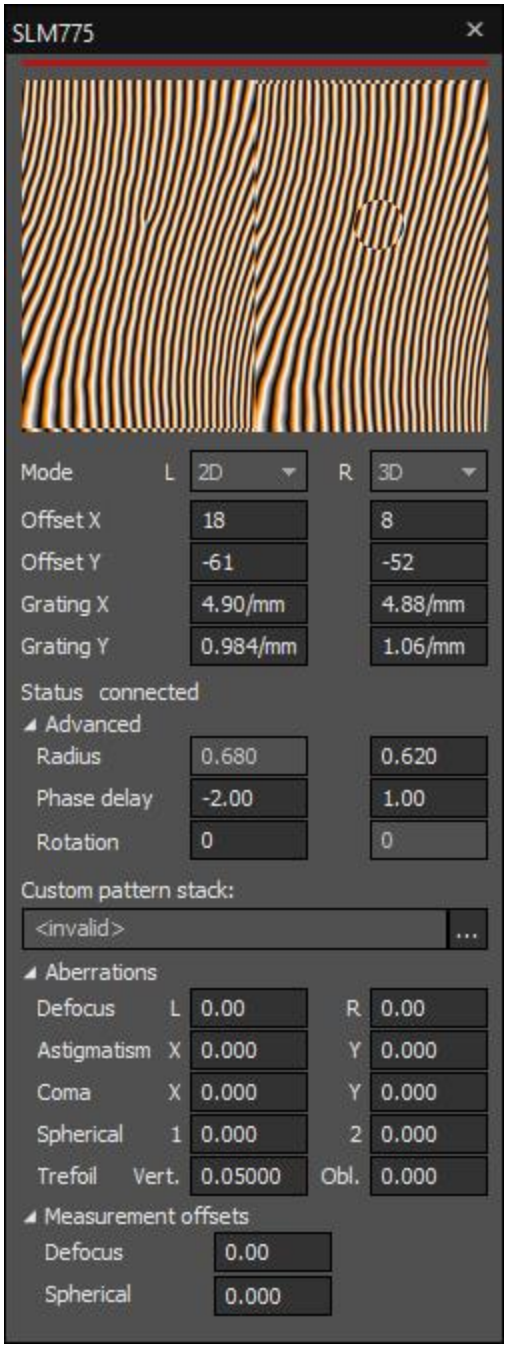



Figure 2.39: SLM live dialog – controlling position and shape of the STED beam

### SLM concept

STED super-resolution microscopy requires the STED laser to form a special pattern when focused into the sample, e.g., forming the well-known donut shape for 2D-STED or more complex patterns for 3D-STED. Generally, the intended shape at the focal point is generated by imprinting a phase pattern onto the STED laser beam. The phase patterns change the laser interference pattern at the focal point in the sample finally giving rise to the intended shapes of STED laser light (see *FIGURE* 2.40).

The easy3D module of abberior microscopes features a spatial light modulator (SLM) which can display and adapt different phase pattern as needed. In contrast to conventional fixed optical elements for phase pattern generation, the SLM offers several advantages:

- 3D-STED with a wider range of objectives than in conventional STED microscopes.
- High stability: The single-beam layout minimizes drift between 2D-STED and 3D-STED PSFs.
- Adaptive optics: Compensation of aberrations provides brighter and clearer STED images. (The SLM provides Adaptive Optics for the STED beam only. Adaptive optics via deformable mirror provide simultaneous improvement of all beams: Excitation, STED and detection)
- Fast and reproducible alignment: Phase patterns are generated, optimized and aligned without any moving parts. This improves manual alignment as well as auto-alignment (optional upgrade).

### SLM live dialog

Each STED laser can be equipped with an easy3D module featuring an SLM. A separate live dialog is available for each STED laser and its respective SLM. This allows separate control of STED PSFs for each STED laser. In addition, the SLM live dialog values are specific for each objective lens switching automatically when selecting a measurement using a different objective.

### Phase pattern display

The upper half of the live dialog (see *FIGURE 2.39*) shows the image that is currently displayed on the spatial light modulator. In short, gray values from 0 to 255 correspond to phase shifts from 0 to 2π. The vertical stripes represent an additional grating pattern used for clean-up of the STED beam. The frequency of the grating can be tuned to reposition the STED beam laterally to optimize the overlay of the STED and excitation beams.

### Modes

The SLM displays two phase patterns side-by-side. In most cases, the left pattern generates the 2D-STED shape while the right pattern generates the 3D-STED shape at the focal point.

The easy3D module can adjust how much the left and right phase patterns are imprinted onto the STED beam. This allows a power distribution between 2D-STED and 3D-STED mode which can be controlled by the user can via the 2D/3D slider in the easyCommander live dialog (see *2.4.1*) or the Measurement live dialog (see *2.7.8*). Physically, the MRS sets the ratio of left/right pattern imprinting (see *2.7.12*).

The 'Mode' pull-down menus allow custom selection of the phase pattern type on each half of the SLM. This selection may need to be enabled in the Measurement live dialog (see *2.7.8*). Common options are '2D' for the vortex phase pattern required for 2D-STED, '3D' for the ring phase pattern required for 3D-STED and 'None' for a normal, unchanged (Gaussian) beam (see *FIGURE* 2.40).

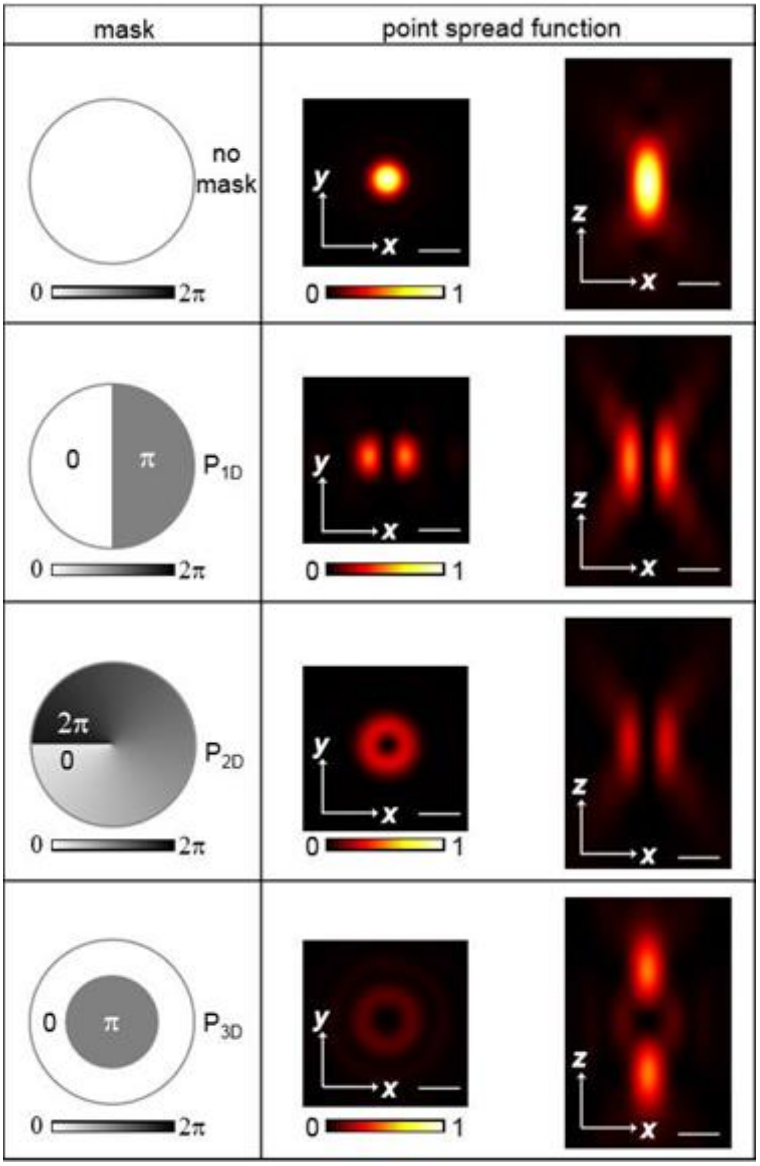



*Figure 2.40: Phase patterns and the resulting shapes at the focal point*

STED laser PSFs have been calculated for a 760 nm STED laser and an objective numerical aperture of 1.4. Scale bars 500 nm.

### 'Offsets': Manual STED alignment

The 'Offset X' and 'Offset Y' editing fields permit shifting of the phase patterns as displayed on the SLM resulting in a shift of the pattern in the back focal plane of the objective lens. As a result, the STED beam shape (e.g., the donut) at the focal point in the sample will tilt and also slightly shift.

> ***Note:*** *Offset values should only be adjusted by support staff or specifically trained users as they affect performance for all users.*
>
> ***Note:*** *Generally, it is more convenient to adjust the STED beam shape via the Alignment live dialog (see 2.7.4). The alignment live dialog will perform this adjustment for all objectives, while the SLM live dialog is objective specific (objective selected for the active measurement).*
>
> ***Note:*** *In most cases, Offset X will provide tilt visible in the YZ plane while Offset Y provides tilt visible in the XZ plane.*

### 'Grating': Manual STED alignment

The 'Grating X' and 'Grating Y' fields permit lateral shifting of the STED shape position at the focal point in the sample (see above). This may be used to manually optimize the overlay of STED shapes versus excitation focal points in the sample or to optimize the overlay of 2D-STED and 3D-STED PSFs.

> ***Note:*** *Grating values should only be adjusted by support staff or specifically trained users as they affect performance for all users.*
>
> ***Note:*** *Generally, it is more convenient to adjust the STED beam position via the Alignment live dialog (see 2.7.4). The alignment live dialog will perform this adjustment for all objectives, while the SLM live dialog is objective specific (objective selected for the active measurement).*
>
> ***Note:*** *In most cases, Grating X reposition the beam focal point along the Y axis, while Grating Y will reposition along the X axis.*

### 'Advanced' settings

The 'Radius' value can be used to optimize 3D-STED for new objective lenses. Different objectives can feature different back aperture sizes preventing their use for 3D-STED in conventional STED microscopes. The SLM can adjust the ring size (radius) of the 3D-STED phase pattern to match the objective aperture size.

> ***Note:*** *Before installing a new objective for STED, please check compatibility with our support or sales team. Our support team can also help with installation and optimization of SLM values.*
>
> ***Note:*** *Radiik are optimized during installation and values should only be adjusted by support staff or specifically trained users as they affect performance for all users.*
>
> ***Note:*** *'Phase delay' and 'Rotation' values should only be adjusted by maintenance staff, as their appropriate values strongly depend on the particular beam geometry.*

### 'Custom pattern stack'

Here, custom phase patterns can be uploaded and displayed on the SLM. For details and format requirements, please contact our support team.

**'Aberrations'**

The SLM can, similar to the deformable mirror (DM, see ***2.7.13***), perform adaptive optics functions such as compensation of aberrations. Compared to conventional STED microscopes, the SLM can improve STED performance providing clearest and brightest STED images. However, the adaptive optics functions of the SLM are limited to the STED beam while the DM can improve all beam paths and imaging modes simultaneously (excitation lasers, STED lasers, detected light). This makes the DM the preferred tool during imaging (***2.7.13***), if installed (optional upgrade).

Nevertheless, the SLM aberration functions are crucial for optimum STED performance, even in microscopes with both modules (SLM and DM) as some compensations need to be performed specifically for the STED beam.

The parameters in the 'Aberrations' section of the live dialog are mainly used to compensate aberrations originating from imperfections in the optical components of the microscope (see ***2.7.13.2*** for more detail on sources and types of aberrations). For this reason, these parameters are global parameters and will affect all measurements.

> ***Note:*** *Aberration values should only be adjusted by support staff or specifically trained users as they affect performance for all users. These parameters have been optimized during installation and rarely require adjustment.*

The SLM aberration functions change the STED beam shape at the focal point in the sample in a fashion similar to the DM aberration functions (see ***2.7.13*** for more details on aberrations and procedures):

- **'Defocus':** The STED PSF can be repositioned vertically using this parameter. Generally, defocus is adjusted to match the position of the PSFs of the STED and excitation lasers along the Z axis.
- **'Astigmatism':** This parameter can introduce/compensate astigmatism for the STED laser. Typically, astigmatism can cause the fluorescence PSF in 2D-STED mode to become asymmetric (oval-shaped in XY) and may cause intensity losses. Astigmatism can be controlled separately for the X and Y axis, though the effects may be noticeable at slightly different angles.
- **'Coma':** This parameter can introduce/compensate coma for the STED laser. Typically, coma can cause the 2D-STED fluorescence PSF to appear bent when observed in XZ or YZ slices (often referred to as “banana” shape). Coma can be controlled separately for the X and Y axis, though the effects may be noticeable at slightly different angles.
- **'Trefoil':** This parameter can introduce/compensate trefoil for the STED laser. Typically, trefoil can cause the donut-shaped PSF of the STED laser in 2D-STED mode to become asymmetric, causing distortions or intensity losses of the resulting fluorescence PSF in 2D-STED mode (more complex effects than for astigmatism). Trefoil can be controlled separately for the X and Y axis, though the effects may be noticeable at slightly different angles.
- **'Spherical':** This parameter can introduce/compensate spherical aberrations for the STED laser. Typically, the **'**Spherical**'** parameter is used to compensate spherical aberrations caused by non-ideal (but stable) room temperature or non-optimum adjustment of the objective correction collar. Some users may prefer adjusting the “Spherical 3rd" parameter instead of fine-tuning the correction collar of water or silicone objectives, as some correction collars may show hysteresis effects requiring further alignment adjustments.

**'Measurement corrections' for local spherical aberration compensation**

Sample-specific spherical aberrations are often caused by a mismatch between the refractive index (RI) of the objective immersion medium (oil, silicone oil, water, air…) and the RI of the sample medium (see *2.7.13.2* for more details and *TABLE 2.7-4* for a list of objectives and sample media). While negligible near the coverslip, the spherical aberrations increase with distance from the coverslip causing progressively stronger image deterioration and distortion in deeper sample areas. Especially 3D-STED images are affected by spherical aberrations. If available, spherical aberrations are best compensated via the DM (optional upgrade), as the DM acts on all beams simultaneously (excitation, STED, detection) and can automatically adjust the compensation to the level required at the current sample position (see *2.7.13.3*).

> ***Note:*** *Before attempting aberration compensation, it is recommended to select an objective with a good refractive index match between the immersion liquid and the sample medium to minimize aberration from the start.*

In absence of a DM module, the SLM can be used to compensate local spherical aberrations. However, the SLM compensates spherical aberrations only for the STED laser, recovering less image quality and requiring additional adjustments to overlay the (compensated) STED focal shape and the (uncompensated) confocal PSF. The 'Measurement corrections' parameters at the bottom of the SLM live dialog allow measurement-specific adjustments, i.e., the compensations are only valid for the current measurement window (and do not affect other users):

- **'Spherical':** This parameter compensates the spherical aberrations for the STED laser. Typically, this is used for 3D-STED imaging. If the refractive index of the sample medium is lower than the refractive index of the correct objective immersion liquid, the user may test positive compensation values (e.g., 0.15, 0.3. 0.5 …) until STED image quality is improved sufficiently. Conversely, if the refractive index of the sample medium is higher than the refractive index of the correct objective immersion liquid, the user typically tests negative compensation values to improve STED image quality.
- **'Defocus':** Whenever spherical aberrations are compensated for the STED beam via the SLM, the confocal PSF (excitation and detection) remains distorted by the spherical aberrations. This distortion shifts the confocal PSF vertically, while the STED shape remains in the (nearly) correct position. This necessitates repositioning of the STED focal shape along the Z axis via the 'Defocus' parameter. Typically, a good starting value for 'Defocus' is half of the 'Spherical' value.

## 2.7.12 MRS live dialog

In setups with easy3D module, the user can seamlessly adjust the power distribution between pure 2D-STED mode and 3D-STED mode. In short, this adjust whether STED increases resolution purely in the XY plane (2D-STED), in a near-isotropic fashion in the X, Y and Z planes (3D-STED) or as a variable mixture between both modes.

The user will typically adjust the power distribution via a simple slider (0-100% 3D-STED)in the easyCommander (see section ***2.4.1***) or in the Measurement live dialog (see ***2.7.8***).

The hardware component achieving power distribution can be manually adjusted in the MRS dialog. Here the user can adjust the degree of rotation of the hardware which translates into a given power distribution (% 3D-STED). For easier access, this translation has been analyzed and calibrated to allow the user simple control via the 2D-3D sliders.

For a description of the STED laser patterns for 2D-STED and 3D-STED generated by the SLM device, please see section ***SLM LIVE DIALOG (SPATIAL LIGHT MODULATOR)***.

> ***Note:*** *There is a MRS live dialog for each installed easy3D module (optional for each installed STED laser).*

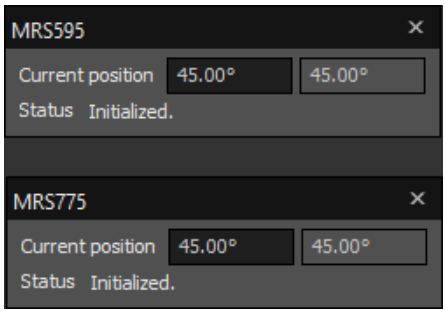



*Figure 2.41: MRS live dialog (power distribution)*

## 2.7.13 DM (Deformable Mirror for Adaptive Optics)

### *2.7.13.1 DM concept*

Optical aberrations can affect image quality in confocal and other optical microscopes. High-end confocal microscopes employ objective lenses with a high numerical aperture (NA) to improve confocal image resolution and these are also best suited for STED microscopy. However, high NA lenses are more susceptible to optical aberrations. STED super-resolution image quality is similarly affected by aberrations and 3D-STED shows an even higher susceptibility to aberrations.

Adaptive Optics can compensate the relevant aberrations to restore image quality. This is why abberior introduced the Deformable Mirror (DM), a versatile Adaptive Optics module compensating aberrations for confocal and STED optics simultaneously. This allows quick and easy improvement of image quality for confocal and STED mode, without the need for offset adjustments between confocal and STED beams. In contrast, our easy3D module (see ***SLM LIVE DIALOG (SPATIAL LIGHT MODULATOR)***) provides Adaptive Optics for the STED laser beam only.

The DM hardware can be described as a miniaturized version of deformable mirrors found in optical telescopes for astronomy which can compensate for aberrations caused by the atmosphere. In essence, the topology of the mirror surface can be shaped. Different topology patterns can imprint aberrations onto a beam reflected off the DM. If an aberration affects the microscope, the DM can imprint the opposite aberration onto the main beam compensating the existing aberration and restoring perfect focus quality in the sample. In our microscopes, all beams (excitation lasers, STED lasers, detected light) are combined and pass the DM together before entering the scanner and objective. In this way, the DM can compensate different aberration types (originating from the sample or optical components) simultaneously for STED and confocal mode.

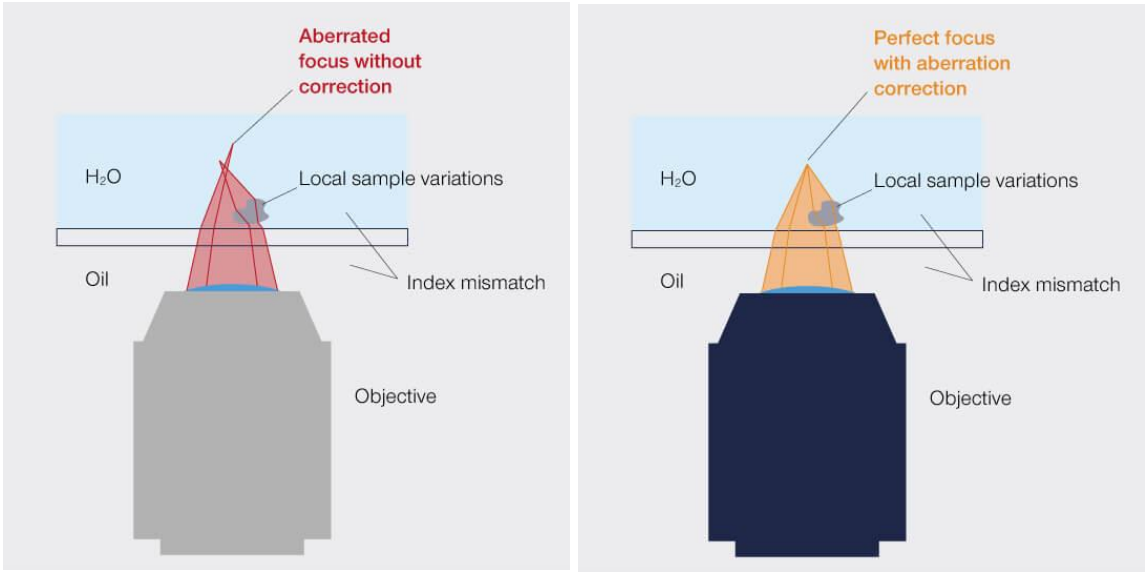



*Figure 2.42: Compensation of optical aberrations affecting focus*

### 2.7.13.2 Sources of optical aberrations

| Objective type | RI immersion liquid | Sample type | RI sample medium | RI match |
|---|---|---|---|---|
| **Oil** | 1.518 | Fixed samples in methyl salicylate | 1.53-1.54 | +++ |
| | | Fixed samples in ProLong Glass, TDE-O (e.g., from abberior) | ~1.52 | ++++ |
| | | Fixed samples in abberior Mount Solid AF, Mowiol+Dabco, ProLong Gold/Diamond | 1.38-1.49 * | ++ |
| | | Cultured live cells (use only near coverslip) | 1.33-1.38 | - |
| **Glycerol** | 1.46-1.47 | Fixed samples in glycerol | 1.45-1.47 | ++++ |
| | | Fixed samples in abberior Mount Solid AF, Mowiol+Dabco, ProLong Gold/Diamond | 1.38-1.49 * | +++ |
| **Silicone** | 1.4 | TDE-S (e.g., from abberior) | 1.4 | ++++ |
| | | Dense tissue in aqueous buffer. Fixed samples in abberior Mount Liquid Solid AF or fresh Mount Solid AF (no curing) | ~1.38 | +++ |
| | | Cultured live cells | 1.33-1.38 | +++ |
| **Water** | 1.33 | Cultured live cells | 1.33-1.38 | +++ |
| | | Dense tissue in aqueous buffer Fixed samples in abberior Mount Liquid AF or fresh Mount Solid AF (no curing) | ~1.38 | ++ |
| | | Expansion microscopy samples (gel/water) | ~1.33 | ++++ |
| **Air** | 1 | Mostly non-biological samples | >1 | varies |

*Table 2.7-4: Refractive indices for common objectives and sample media*

* refractive index increases with curing time (water evaporation)

Aberrations can originate from multiple sources, including imperfections in the objective lenses and other optical elements. Here, the DM offers new opportunities to boost microscope performance and robustness, e.g., for temperature changes or non-perfect objectives.

More relevant to the microscope user, aberrations can be caused by the sample. An extreme example are unsuitable coverslips which introduce extremely strong aberrations that can barely be compensated. Most objectives delivered with our microscopes require glass coverslips with thickness #1.5 (0.17 mm). If you are unsure about more specialized objectives, e.g., dipping objectives, please check with our support team.

In contrast, spherical aberrations caused by a sample medium with a non-ideal refractive index (RI) can be compensated very comfortably and with a large degree of automation (see below). While these aberrations are negligible near the coverslip, their compensation can greatly improve image quality deeper in the sample. Compensation enables the particularly sensitive 3D-STED mode at previously unreachable sample depths. Larger RI mismatches will limit the imaging depth more strongly as aberrations increase more quickly with depth. *TABLE 2.7-4* lists objective types, the refractive indices of their immersion liquids and typical sample media.

### 2.7.13.3 Depth-dependent aberration compensation

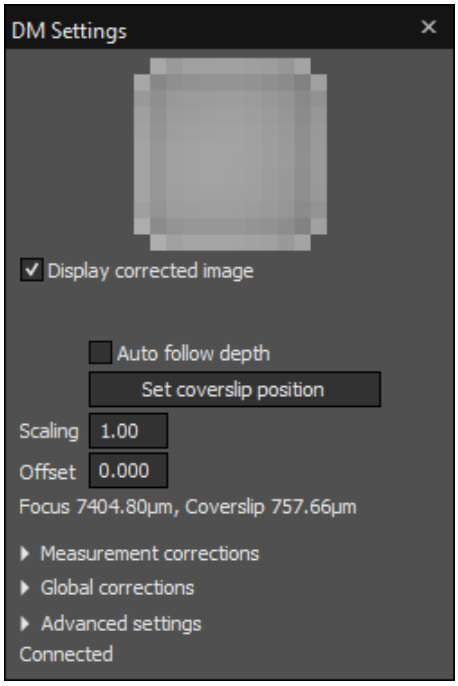



*Figure 2.43: DM Settings live dialog (collapsed)*

**Introduction**

A mismatch between the RI of the objective immersion medium (oil, silicone oil, water, air…) and the RI of the sample medium can cause spherical aberrations. While negligible near the coverslip, the spherical aberrations increase with distance from the coverslip causing progressively stronger image deterioration and distortion in deeper sample areas. The DM can not only compensate the spherical aberrations, but it can also automatically adjust the compensation to the level required at the current sample position (depth).

> ***Note:*** *Before attempting aberration compensation, it is recommended to select an objective with a good refractive index match between the immersion liquid and the sample medium to minimize aberration from the start.*

The DM Settings live dialog (*FIGURE* 2.43) allows the user to calibrate and control the automated compensation process. For this purpose, the user needs to define the coverslip position (zero compensation) and the required compensation level at a position deep in the sample before activating automated compensation. The relevant options in the live dialog are listed below:

**“Set coverslip position” (step 1):** In a first step, the user needs to identify the coverslip level (interface between coverslip and sample medium). This can easily be done via an XY scan in live mode (auto repetition). Once the coverslip surface is in focus, click the “Set coverslip position” button. Now the live dialog should display the coverslip position as well as the current focus. The coverslip position is a global parameter and applies not only to the current measurement but also to any other measurement involving the DM.

> ***Notes on finding the coverslip surface:*** *Often, the end of most fluorescent structures marks the coverslip position. Should there be a gap between coverslip and larger fluorescent structures, fluorescent background staining may reveal the coverslip location. If all else fails, the user can directly exploit laser reflection at the interface of coverslip and sample medium to detect the coverslip position via reflected light signal. This can be done either by setting the spectral detection window to the excitation laser wavelength (requires spectral detection, see Detectors) or by selecting the reflective PMT for detection (requires reflective PMT, see* PELLICLE LIVE DIALOG *and* DETECTORS*). For the latter option, ensure the PMT is deactivated, and the pellicle returns to UP position for subsequent standard fluorescence imaging.*

**Checkbox: “Auto follow depth” (step 2):** Once the coverslip position has been found, the user should create and activate a vertical scan measurement (XZ or YZ slice) for the following steps. The vertical slice should span the coverslip surface as well as the deeper regions that the user would like to image. Now activate the automatic, depth-dependent compensation of spherical aberrations. This is necessary to test, optimize and use the parameters below.

**“Scaling” (step 3):** The software can automatically adjust spherical aberration compensation for any sample position by increasing the compensation level as a function of imaging depth (distance from the coverslip level). The “Scaling” parameter defines the slope of this function and needs to be adjusted by the user to allow accurate automated compensation.

The user can access and adjust the “Scaling” parameter when a vertical scan is selected (see above). While running the vertical scan in auto repetition mode, the user can optimize the “Scaling” parameter and see the effects live in the image. Adjusting the “Scaling” mostly changes the image quality in the deeper regions. We recommend initially adjusting the “Scaling” value via vertical scans in confocal mode until satisfied with image quality at all depth levels. If the user intends to perform 3D-STED imaging (see ***SLM LIVE DIALOG (SPATIAL LIGHT MODULATOR)*** and ***THE SMART WAY TO START: EASYCOMMANDER***), the “Scaling” value may need to be fine-tuned in 3D-STED mode which is much more sensitive to spherical aberrations. Please note that low STED laser powers are recommended as 3D-STED mode may induce bleaching during repeated scanning.

> ***Notes on “Scaling” value range:*** *For a sample with perfectly matched refractive indices, the “Scaling” factor should be approximately zero. If the sample medium has a lower refractive index (RI) than the immersion liquid, a positive “Scaling” value is required for compensation. If the sample medium has a higher RI than the immersion liquid, a negative “Scaling” value is required. Larger RI mismatches will require larger “Scaling” values (positive or negative). However, overcompensation with exceedingly large “Scaling” values will negatively affect image quality in the deeper sample regions. Generally, we recommend testing a few values between zero and (+/-) one before moving to larger values if required. Please note that the DM may not be able to fully compensate the most severe aberrations in the deepest sample regions when reaching the DM maximum stroke (indicated by red pixels in DM topology pattern displayed at the top of the live dialog).*

**“Offset” (step 4):** This parameter usually does not require adjustment if the coverslip Z position has been set correctly (see above). In some cases, spherical aberrations may affect the image even at the coverslip surface. This could be caused by slight deviations in coverslip thickness, non-optimum temperatures or non-optimum adjustment of the objective correction collar. In these cases, it may help to adjust the apparent position of the coverslip to provide an appropriate compensation gradient throughout the sample. However, instead of using the “Offset” parameter, it is often easier to identify the general aberrations affecting the system and to compensate those via custom aberration compensation (see section *2.7.13.4*) before setting up depth-dependent aberration compensation.

**Applying optimized compensation settings to measurements (step 5):** Once the “Scaling” value has been optimized, the user can modify the vertical slice measurement into any measurement type to acquire the intended images. If the user prefers applying the “Scaling” factor to another existing measurement, select the existing measurement (click on the respective measurement window) and go to the DM Settings live dialog to update “Scale” with the optimized value. For some measurement types (e.g., XY scans), the “Scaling” parameter is hidden. In this case, temporarily change the scan axis of the measurement to XZ, update the “Scaling” parameter and then change back to the original scan axis selection (all ranges and pixel sizes should automatically restore).

> ***Note on Z-shifts:*** *As compensation of spherical aberrations removes images distortion along the Z axis, the Z position of the current image may not appear the same as before compensation. Please update the image Z position as required or use a larger margin along the Z axis for volume imaging.*

### *2.7.13.4 Custom aberration compensation (measurement-specific)*

Some aberrations may affect the whole measurement (i.e., all areas of the image) or are more complex, exceeding what can be compensated via depth-dependent compensation (see *2.7.13.3*). One example is a fixed spherical aberration caused by slight deviations in coverslip thickness, non-optimum temperatures or non-optimum adjustment of the objective correction collar. Other examples include coma aberrations caused by positioning the coverslip at an angle. In these cases, we can apply custom aberration compensations via the DM live dialog section “measurement corrections” (see *FIGURE 2.44*) that are valid for the current measurement only.

> ***Note:*** *The DM Settings live dialog section “Measurement corrections” uses local parameters that apply to the current measurement only. This is important to avoid accidentally applying general aberration compensations to all measurements which would affect all microscope users. For the user it is important to remember that, if the measurement corrections should be applied to a newly created measurement, they need to be entered manually again.*

The following parameters can be adjusted to control various aberrations in the measurement. Both, positive or negative values may be used to either apply or compensate a specific aberration.

**“Defocus”:** This parameter shifts the focal spot of all beams up or down. Please note that strong shifts can result in additional aberrations affecting the point spread function (PSF).

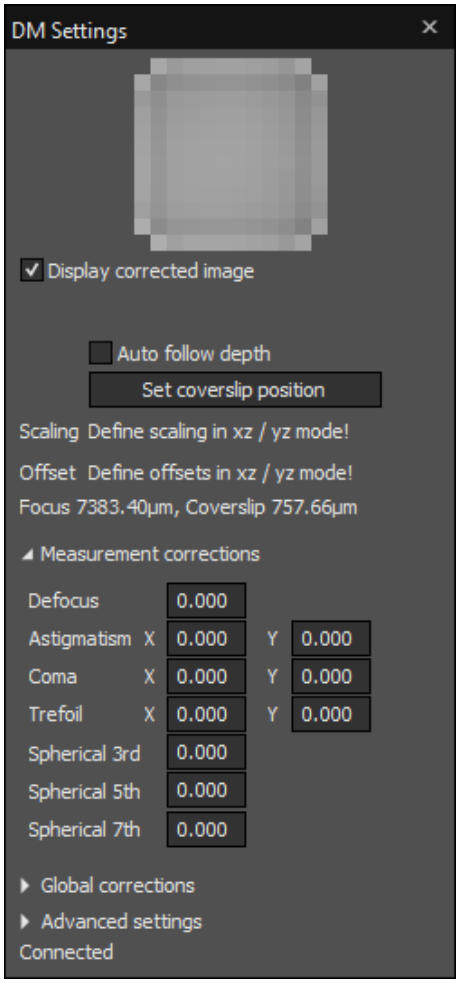



*Figure 2.44: DM Settings live dialog (Measurement corrections)*

**"Astigmatism":** This parameter can introduce/compensate astigmatism. Typically, astigmatism can cause the fluorescence PSF in 2D-STED mode to become asymmetric (oval-shaped in XY) and may cause intensity losses. Astigmatism can be controlled separately for the X and Y axis, though the effects may be noticeable at slightly different angles. Astigmatism may be caused by small imperfections of the objective lens or internal optics.

**"Coma":** This parameter can introduce/compensate coma. Typically, coma can cause the confocal fluorescence PSF to appear bent when observed in XZ or YZ slices (often referred to as "banana" shape). This can similarly affect the STED PSFs. Coma can be controlled separately for the X and Y axis, though the effects may be noticeable at slightly different angles. Coma is often caused by minor imperfections of the objective lens or imperfect sample positioning (e.g., angled coverslips).

**"Trefoil":** This parameter can introduce/compensate trefoil. Typically, trefoil can cause the donut-shaped PSF of the STED laser in 2D-STED mode to become asymmetric, causing distortions or intensity losses of the resulting fluorescence PSF in 2D-STED mode (more complex effects than for astigmatism). Trefoil can be controlled separately for the X and Y axis, though the effects may be noticeable at slightly different angles. Trefoil is often caused by minor imperfections of the objective lens or internal optics.

**“Spherical 3rd”:** This parameter can introduce/compensate spherical aberrations of the 3rd order (Zernike polynomials). This aberration type is also automatically compensated during depth-dependent aberration compensation (see *2.7.13.3*). In contrast to depth-dependent compensation (intended to help with sample medium refractive index mismatches), the “Spherical 3rd” parameter can be used to apply fixed compensation of depth-independent effects (e.g., from slight deviations in coverslip thickness, non-optimum temperatures or non-optimum adjustment of the objective correction collar). Some users may prefer adjusting the “Spherical 3rd" parameter instead of fine-tuning the correction collar of water or silicone objectives, as some correction collars may show hysteresis effects requiring further alignment adjustments.

The required fixed compensation can best be identified in a vertical scan (XZ or YZ) close the coverslip. Depth-dependent compensation can be combined with fixed compensation of spherical aberrations with “Spherical 3rd" parameter acting as an offset value.

Alternatively, the “Spherical 3rd” parameter can be used for fixed compensation of spherical aberrations in images with a small Z-axis range (low variation of spherical aberrations).

**“Spherical 5th” / “Spherical 7th”:** These parameters can introduce/compensate spherical aberrations of the 5th and 7th order (Zernike polynomials). This parameter also influences spherical aberration compensation. However. It is only recommended for expert users who would like to experiment with the specific effects of higher-order polynomials. For all other users, the parameter “Spherical 3rd" typically suffices.

### *2.7.13.5 Custom aberration compensation (global)*

Some aberrations may affect not only the whole measurement (i.e., all areas of the image) but also all other measurements. Typical examples are fixed aberrations caused by slight imperfections of the objective or internal optics. As these aberrations will not change much over time, they may be compensated via the global parameters available in the DM Settings live dialog section “Global corrections” (see *FIGURE 2.45*). This has the advantage, that the intended compensations are applied by default for all users and for all measurements.

> ***Important note:*** *Using the global parameters in the section “Global corrections” poses the risk of affecting all users and all future measurements negatively if done incorrectly. To limit access to these sensitive parameters to specific users, the multi-user management system (MMS) may be employed (see section 2.9).*
>
> ***Note:*** *“Global corrections” are objective specific, i.e., they apply only to the objective selected in the currently active measurement. This is important to allow fine-tuning the alignment for each objective.*
>
> ***Note:*** *When closing software, a dialog appears asking whether the user would like to save changes to hardware settings. Selecting “Yes” will make “Global corrections” permanent, while “No” ensures potentially remaining “Global corrections” are reset. Affects all other changes to hardware settings.*

Apart from global relevance instead of measurement-specific relevance, the DM Settings live dialog section “Global corrections” contains the same parameters as the section “Measurement corrections” (see section *2.7.13.4*).

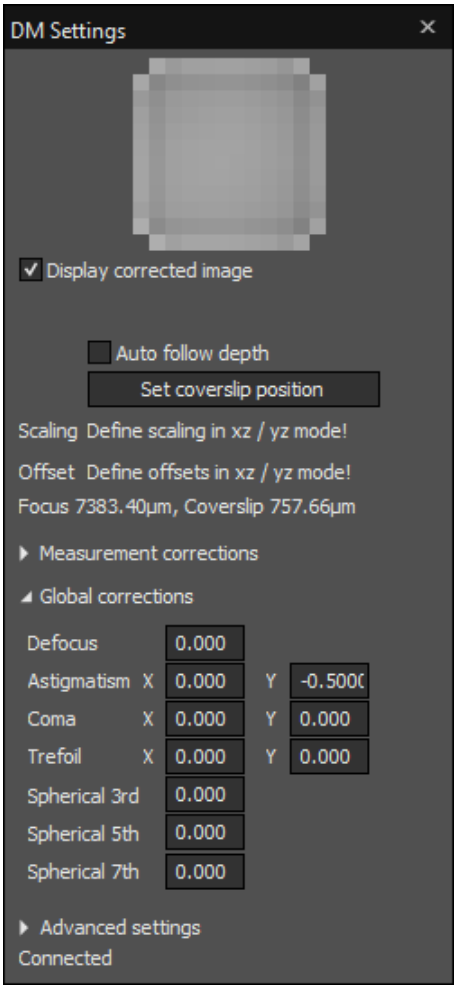



*Figure 2.45: DM Settings live dialog (Global corrections)*

### *2.7.13.6 Application examples: Global vs measurement-specific corrections*

In the following, we consider selected applications and the advantages/disadvantages of using global corrections.

**Objective correction collar adjustment:** For many microscopes, the correction collar of water or silicone immersion objectives is adjusted once during installation and subsequently kept in the same position to avoid adjustment hysteresis effects that may affect alignment. Here, fine-tuning via the parameter “Spherical 3rd" is a great alternative. “Global corrections” should be used for general fine tuning. For temporary effects, e.g., when using the objective at a different temperature, “Global corrections” are comfortable (automatically applied to all measurements) but risky, as great care needs to be taken to return to the original settings for other users. Measurement corrections are recommended for such temporary effects. The custom axis (see section ***2.7.16***) offers a great way of checking the current correction collar position by running a parameter sweep of the “measurement correction” parameter “Spherical 3rd".

**Refractive index mismatch of sample medium:** Aberrations are only relevant if imaging deeper into the sample. If aberrations are observed, first check whether other objectives may match the refractive index of the sample better than the current objective. Use depth-dependent aberration compensation if imaging and multiple different depth-levels. Use “Spherical 3rd" in “Measurement corrections” if only within a small area along the Z-axis. Do not use “Global corrections”.

**Complex local aberrations in samples:** Only use “Measurement corrections”, as these parameters apply only to one measurement / one sample position.

**Room temperature changes:** A change in room temperature may cause a visible spherical aberration. Adjusting “Spherical 3rd" parameter may improve image quality, but adjusting global parameters is only recommended if the temperature change is expected to be long term. For significant long term room temperature changes, it is recommended to contact support to check whether other settings and alignments require changes as well.

**Objective damage:** Slight damage to the objective may introduce aberrations. These may be counteracted via the global corrections as the damage will most likely persist long term. However, it is recommended to contact our support team for best results as STED alignment may need additional changes for this objective.

> ***Note:*** *Please clean the objective thoroughly with isopropanol and lens wipes to exclude other sources affecting performance.*

**Tilted sample:** If the sample is tilted, aberrations such as coma may occur. It is recommended to either improve sample positioning or to use measurement corrections only, as this is a temporary effect.

### *2.7.13.7 DM alignment*

For optimum performance, the topology pattern of the DM needs to be laterally centered on the position of the laser and detection beams. The DM alignment has been optimized during installation but can be fine-tuned again if significant alignment drift occurs.

> **Note:** DM alignment affects image quality for all users. It should only be performed by our support staff or specifically trained users. If you would like to prevent most users from changing the DM alignment, refer to the section Imspector Multiuser Management System.

The DM Settings section “Advanced Settings” (see ***Figure*** 2.46) offers an opportunity to laterally align the DM without physically moving the DM or any mirrors. Instead, lateral offsets can be set that change the position of the topology pattern generated by the DM:

**Offset X:** This moves the center of the DM topology pattern along the X axis. Depending on the hardware layout, the effects can be observed in images mostly along the Y axis

**Offset Y:** This moves the center of the DM topology pattern along the Y axis. Depending on the hardware layout, the effects can be observed in images mostly along the X axis

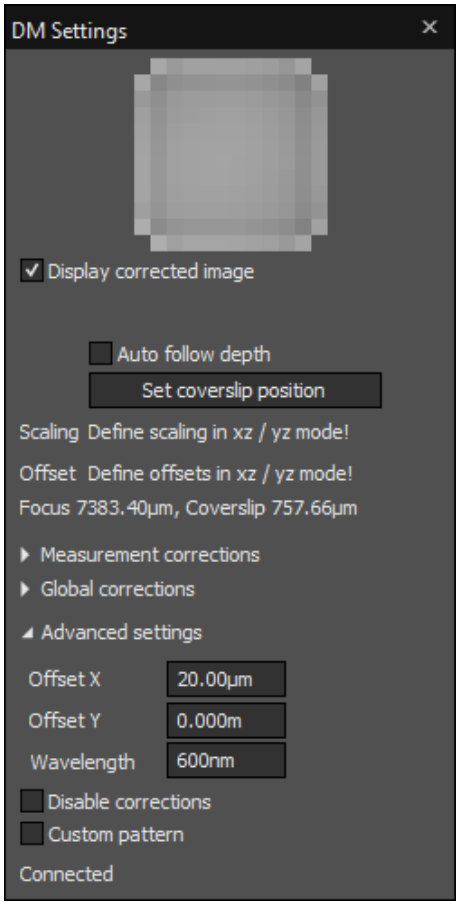



*Figure 2.46: DM Settings live dialog (advanced)*

## 2.7.14 Pinhole Live Dialog

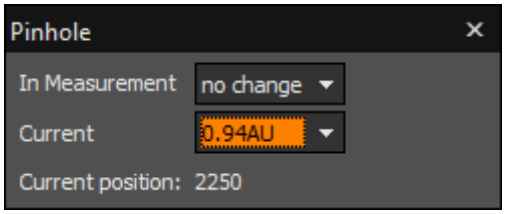



*Figure 2.47: 'Pinhole' live dialog.*

abberior Expert Line, INFINITY and FACILITY microscopes have a motorized pinhole module with 16 different physical pinhole sizes installed. The different pinholes sizes can be selected in the 'Pinhole' live dialog (see ***FIGURE 2.47***) or selected via the easyCommander (see ***2.4.1***). The pinhole size is displayed in airy units (AU) calculated from the physical pinhole size, magnification, and the current objective properties.

The pinhole size can be made part of the measurement settings. Selecting a pinhole size in the drop-down dialog 'In Measurement' will cause the pinhole position to be set upon starting a measurement (waiting until the pinhole position is reached before actually starting the scan). If 'In Measurement' is set to “no change”, then the current pinhole position is kept when starting the measurement. The pinhole may even be changed during a scan (causing a short black stripe in the image during the switch).

When no measurement is running, the current pinhole size can be adjusted immediately via the 'Current' parameter. Please keep in mind, that starting a measurement will reset the pinhole as defined in the measurement settings.

## 2.7.15 Pellicle Live Dialog




*Figure 2.48: 'Pellicle' live dialog.*

Reflected light measurements can be helpful, e.g., for measuring and analyzing PSFs using reflecting nanoparticles. Facility microscopes require an additional PMT maintenance tool for reflective measurements while most Infinity microscopes have an inbuilt PMT. To detect reflected light signal on these Infinity microscopes via the PMT, the motorized pellicle has to be set to the 'down' position using the 'Pellicle' live dialog (*FIGURE 2.48*).

The Pellicle position can be changed live without running a measurement via the *"Flip"* button. In the moment a measurement starts the pellicle will change to the position set in the 'Measurement' tab.

> ***Note:*** *Avoid setting the pellicle to the "down" position in regular fluorescence measurements as the pellicle removes a significant amount of excitation laser light as well as fluorescence signal. STED measurements can be affected strongly, as the STED laser polarization may be altered leading to very dim STED images.*
>
> ***Note:*** *Alternatively, some reflected light may also be detected by matching the spectral detection window to the wavelength of an active excitation laser (see* CHANNELS AND MEASUREMENT LIVE DIALOG*). However, the signal strength may vary strongly from weak to strong detector saturation.*

## 2.7.16 Custom Axis live dialog

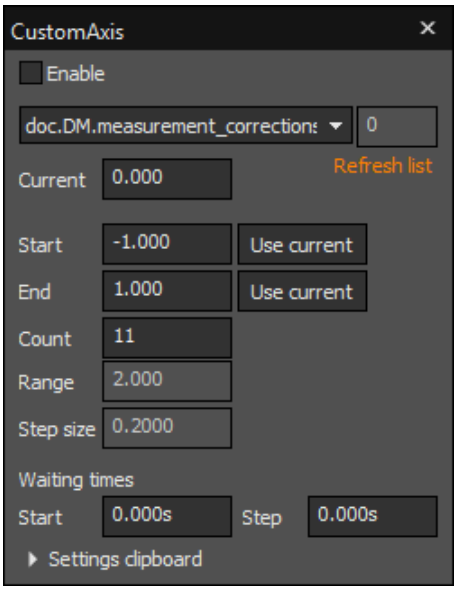



*Figure 2.49: 'Custom Axis' live dialog*

### Concept

The Custom Axis live dialog (see *FIGURE* 2.49) is an extremely powerful tool to perform parameter sweeps without the need of scripting (e.g., via the Python interface). The custom axis acts like any additional axis (e.g., time or Z position), creating multidimensional measurements (e.g., extending a simple XY scan to a timelapse movies or volume image). The custom axis allows the user to select one measurement/instrument parameter to be scanned like a conventional axis. The resulting data stack can be viewed like a movie (showing e.g., one image slice per parameter step) or analysed further to characterise the effect of the parameter variation. If required, a second custom axis can be installed in the software to allow the sweep of two parameter dimensions in one measurement.

> ***Note:*** *Currently a maximum number of four scan axis can be acquired per measurement (including all standard and custom axis).*

### Applications

A very popular application for the custom axis is parameter sweeps of deformable mirror (DM) parameters for Adaptive Optics. One example is an XZ scan with depth-dependent aberration compensation running a custom axis sweep of the 'Scaling' parameter. This allows quickly identifying the optimum 'Scaling' parameter by scrolling through the resulting image stack. Another example is running an XZ scan near the coverslip with a sweep of the DM measurement correction 'Spherical 3rd' (e.g., -1 to +1) to optimise the position of the objective correction collar. For the ideal correction collar position, scrolling through the resulting image stack should show the best image where 'Spherical 3rd' reaches zero.

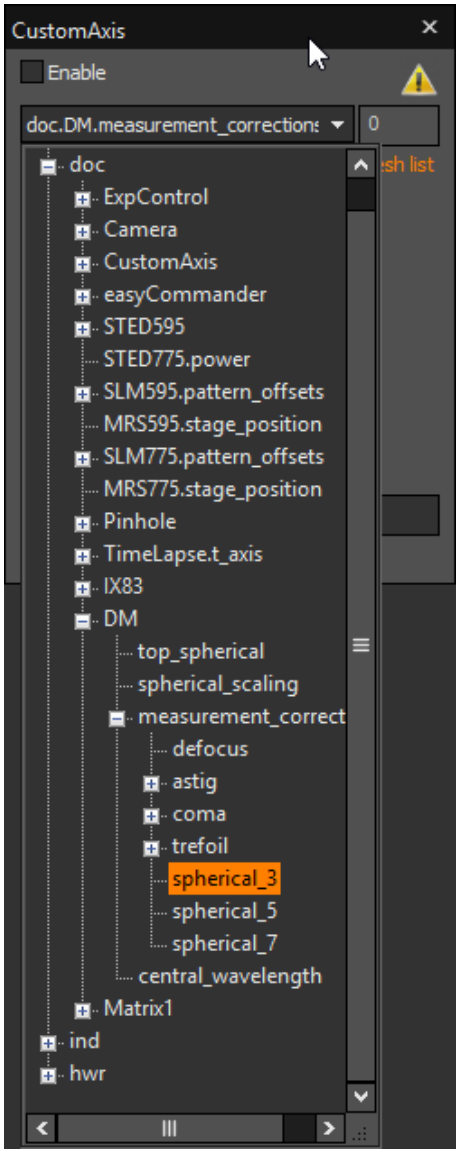



*Figure 2.50: 'Custom Axis' live dialog (selection of parameters)*

The following options are available in the Custom Axis live dialog:

**Drop-down menu:** Here, the user can select the parameter that will be scanned for the custom axis. The available parameters are presented in an extensive tree view, like the metadata shown in the workspace live dialog. For example, the measurement parameter “Spherical 3rd" (spherical aberration compensation via the deformable mirror) can be selected as shown in FIGURE 2.50.

> ***Note:*** *To simplify parameter selection, we can identify parameters in their usual live dialog. For this purpose, go to the appropriate live dialog and press 'SHIFT+F2' while hovering the mouse over the parameter of interest. This reveals additional details about the parameter, including how to select it for the custom axis or how to call it in Python.*
>
> ***Note:*** *Not all parameters can be used for the custom axis. The parameter must be capable of being stepped in equidistant steps. Some parameters, such as the pinhole, can only be adjusted in irregular steps and are therefore unavailable. Other parameters may be unavailable, as they are already in use and therefore locked.*

**Parameter instance:** Right next right next to the drop-down menu, a number field becomes available

if the parameter is active for multiple channels. In this case, the correct instance / channel needs to be selected.

**'Refresh list':** When the measurement or parameters are changed, it is advised to refresh the list of available parameters and instances.

**'Start' / 'End':** Like a conventional scan axis, the scan range of the parameter range needs to be defined for the custom axis. Instead of defining the range directly, the 'Start' and 'End' of the range must be selected (the range is automatically calculated and displayed below).

**'Use current':** To make the selection of the start and end of the custom axis range more convenient, the user can simply insert the current parameter value (e.g., the current focal plane or stage position) by clicking the 'use current' button.

**'Count':** Like the number of pixels of a conventional axis, the number of steps for the parameter sweep can be adjusted (the step size is calculated automatically and displayed below).

**'Range':** Automatically calculated from 'Start' and 'End'.

**'Step size':** Automatically calculated from 'Start', 'End' and 'Count'.

**Waiting time 'Start':** Defines how long the software should allow for initial parameter initialisation before starting the measurement. Increase for parameters relying on slow hardware.

**Waiting time 'Step':** Defines how long the software should allow for parameter initialisation between each parameter scan step. Increase for parameters relying on slow hardware. To increase speed, it is often advisable to use the custom axis as the last axis in the measurement.

**Checkbox 'Enable':** After adjusting all parameters of the custom axis, the axis needs to be added to the measurement by ticking the 'Enable' checkbox. Only then will the custom axis be scanned in the current measurement.

**'Settings Clipboard':** The user expands this section to store the custom axis settings to the clipboard or recall previously stored settings via the respective buttons. The checkbox 'Range only' can be ticked to recall the range only while keeping the current position.

### 2.7.17 Laser Live Dialogs

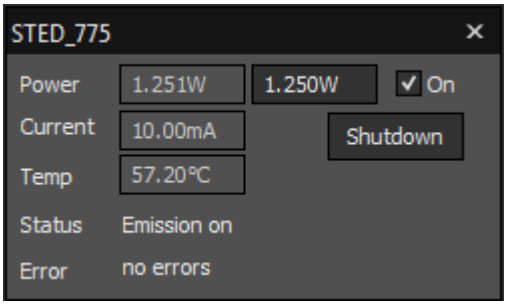



*Figure 2.51: 'STED_775' live dialog.*

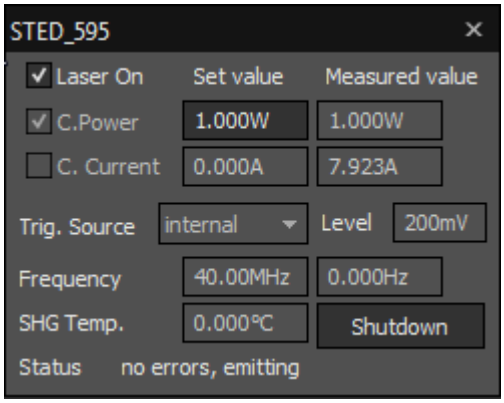



*Figure 2.52: 'STED_595' live dialog.*

In the laser live dialogs (***Figure 2.52*** and ***Figure 2.51***, the current settings of the lasers e.g., power, trigger source etc. can be set. All necessary settings will be done prior to setting up the microscope in your lab. The laser power shown here is the direct output of the laser. Other live dialogs can be used to set the percentage of laser light coupled into the beam path (see sections ***Channels and Measurement Live Dialog*** and ***The Smart Way to Start: easyCommander***)

## 2.7.18 Workspace Live Dialog

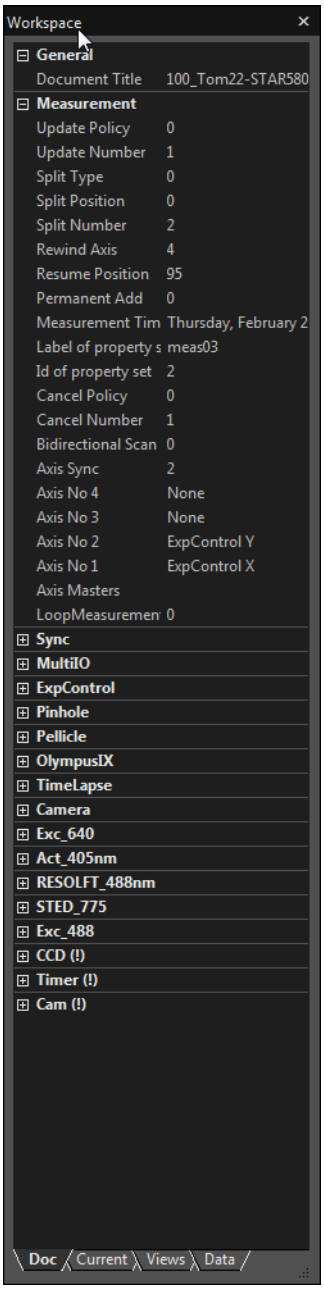



*Figure 2.53: 'Workspace' live dialog.*

Many instrument parameters and settings can be found in detail in the 'Workspace' live dialog. The ‘Workspace’ live dialog is available in the Imspector version at the setup as well as in Imspector Base versions.

Many of the instrument and measurement parameters found here can be used as a variable parameter for the ‘Custom Axis‘ live dialog (see ***2.7.16***) To find the exact location of a parameter in this tree view, users can hover the mouse over the relevant field in a live dialog and press Shift+F2 to display the parameter path.

## 2.7.19 Camera and Camera Settings Live Dialog

The Camera that is mounted at the microscope is operated using the Camera live dialog (Figure 2.54). In this dialog, the 'Exposure time' and 'Gain' of the camera can be set. Further the live video mode can be started and stopped.

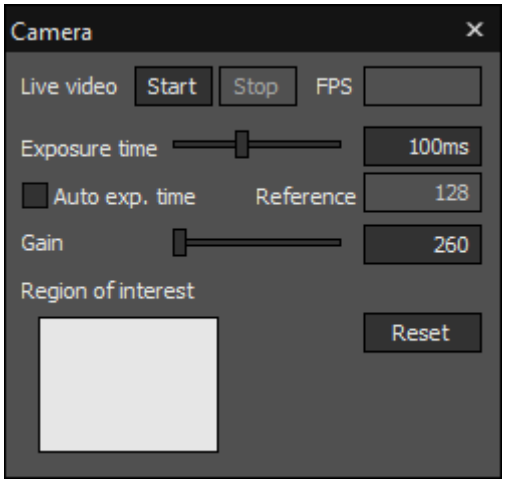



*Figure 2.54: Camera live dialog*

In addition to the camera live dialog the Camera Settings live dialog is used to determine the measurement parameters of the Camera. Here further settings like bit depth, binning, frame rate and others may be set.

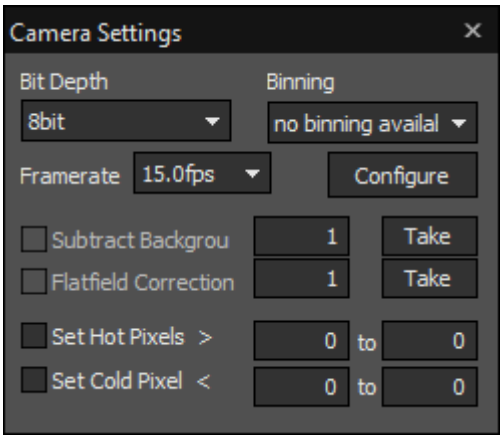



*Figure 2.55: Camera settings live dialog*

## 2.8 Hardware Settings menu

The Hardware Settings menu mostly contains hardware settings that are defined during installation of the microscope, during maintenance or after system modifications. A selection of hardware settings that may be useful during everyday imaging is described in the following.

The Hardware Settings menu can be opened from the menu bar (“Hardware > Configure…”).

Many settings changes become active only after clicking “Initialize” in the Hardware Settings menu (e.g., changes to the z-piezo). Altered settings can be saved directly in the Hardware Settings menu. In addition, if settings have been altered, Imspector will display a dialog when closing the software offering to save the new settings.

> ***Note:*** *Please be aware that any changes made in the Hardware Settings menu are global changes. If the altered settings are saved, these will affect all users (if using the same configuration file). For this reason, not all users are permitted to make changes in the Hardware Settings menu (see* IMSPECTOR MULTIUSER MANAGEMENT SYSTEM*).*

### 2.8.1 Experimental Control – Scanning

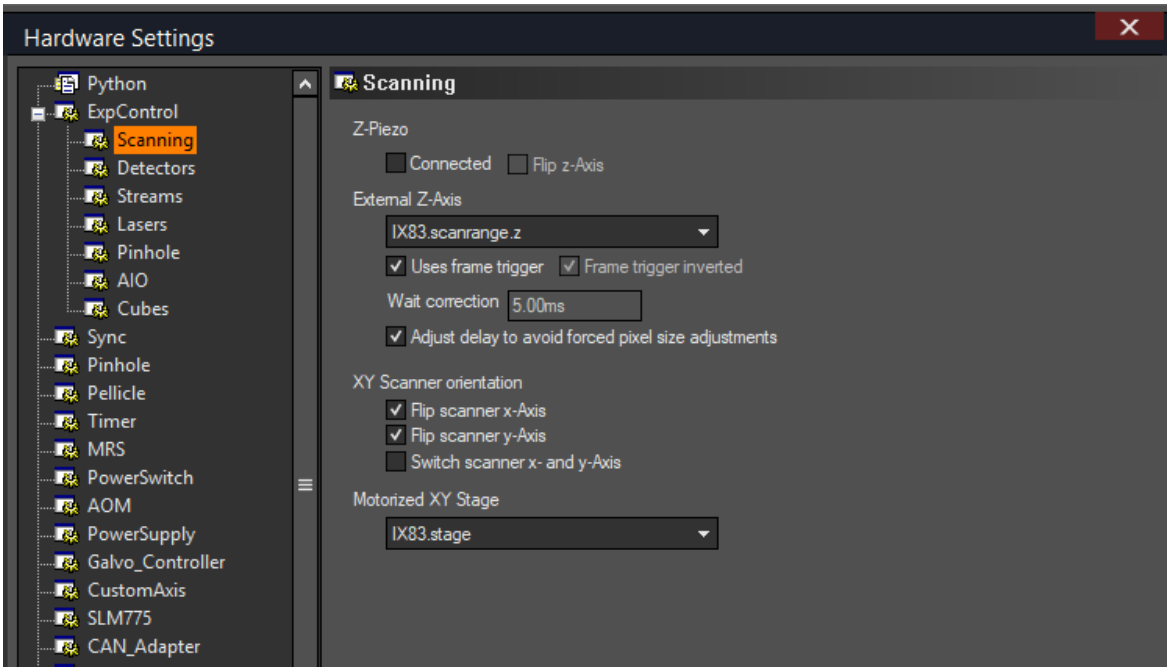



*Figure 2.56: Hardware Settings – ExpControl - Scanning*

#### Z-piezo

Here, the Z-piezo can be activated/deactivated using the tick box “Connected”. Changes become active only after clicking “Initialize”. For most purposes, the Z-piezo should remain connected, as it offers much faster and more accurate Z-positioning than the microscopes nosepiece. However, there are situations, where the user may want to deactivate the Z-piezo: Most Z-piezos can cover Z ranges up to 220 µm within a single scan, above that the nosepiece will offer larger ranges. If the continuous autofocus (see OLYMPUS ZDC: FOCUS SEARCH / CONTINUOUS AUTOFOCUS) is active, the software allows adjustments of the Z position only if the Z-piezo is deactivated (see MICROSCOPE BODY LIVE DIALOGS).

**Nosepiece**

Here, the option “Adjust delay to avoid forced pixel size adjustments” allows more freedom in the choice of Z step sizes for vertical scanning. When deactivated, the limited minimum and maximum Z positioning speed will cause limitations to the possible Z step sizes in vertical scans.

### 2.8.2 Experimental Control – Lasers

Here, detailed settings for each laser can be found.

> ***Note:*** *All settings have already been optimized by production staff before installation. Only our support staff or specifically trained users may make changes if required.*

**Laser blanking during flyback:** In most cases, it is a good idea to blank the laser during flyback in a unidirectional scan as this reduces sample illumination. However, under certain conditions some lasers with slower power adjustment capabilities may perform better for Adaptive Illumination without blanking.

**Fast power changes:** Fast power changes allow line-wise power adjustment of a laser beam , which is great to allow independent power control of the same laser for two channels. It also allows pixel-wise blanking of the laser during Adaptive Illumination measurements. However, under certain conditions Adaptive Illumination may provide better images if fast power changes are disabled for some excitation lasers with slower power adjustment capabilities (e.g., 561 nm).

**Laser calibration files:** A different calibration file can be chosen for the power control of a laser. However, to guarantee a linear power response, it is recommended to use the calibration file provided at delivery or to contact support for recalibration.

### 2.8.3 FLIM settings

If optional dedicated FLIM hardware is installed (e.g., from PicoQuant or Becker&Hickl), it will be listed in the tree view of the Hardware Settings menu (either as TCSPC or with under a specific brand name).

**'Streaming Directory':** Here, the user can set the folder where photon stream files are written whenever the option “to disk” is enabled in a measurement. These streams are not required for FLIM analysis within the Imspector software but allow analysis in third-party software.

**'Stream Format':** Here, the user can select the format of the photon stream output to better suit third-party software for analysis.

### 2.8.4 Microscope body

The microscope body can be selected from the tree view in the Hardware Settings dialog (see ***Figure 2.57***). Here, several settings of the microscope body (e.g., Olympus IX83, Nikon Ti etc.) can be adjusted. Most parameters have been optimized before delivery of the microscope.

**Autofocus search range:** If a compatible autofocus system (e.g., Olympus ZDC, Nikon PFS) is installed, the user can adjust the focus search range here. This concerns the focus search range when triggered from the software (see section ***2.7.3***). When changing the sample holder or using different sample types, the search range may need adjusting to allow the autofocus to reach the sample position. In addition, the focus limits should also match the current sample holder and sample type (adjustable in the menu of the Olympus touch screen panel, see ***Olympus TPC (touch panel control***).

**Acquire AF calibration:** If a compatible autofocus system (e.g., Olympus ZDC) is installed, the user can calibrate the Z-axis positions for the continuous autofocus mode (focus lock, see sections ***MICROSCOPE BODY LIVE DIALOGS*** and ***OLYMPUS ZDC: FOCUS SEARCH / CONTINUOUS AUTOFOCUS***). The calibration enables Imspector to use the autofocus feedback signal to adjust the focus position accurately while still compensating for Z-drift.

> ***Note:*** *In continuous autofocus mode, the AF calibration is required to adjust the Z-position via Imspector and to acquire volume images. Without any calibration, continuous autofocus mode completely fixes the Z-position to the current position.*

Generally, the AF calibration has already been acquired by abberior during production for all compatible objective lenses. A new calibration may be required if the original calibrations are lost or inaccurate. In this case, please contact abberior support (***1.4***) or try reacquisition of the AF calibration with a suitable sample (e.g., aqueous sample medium with a thickness #1.5 glass coverslip and immersion medium without any bubbles). First run a focus-search (see above), then activate continuous autofocus (focus lock) and acquire the AF calibration via the Hardware Settings dialog. If successful, click “Initialize” and “Save” in the Hardware Settings dialog.

If a new compatible objective has been installed (please check autofocus compatibility with Olympus or abberior support), calibrations need to be acquired via the Olympus touch screen panel (***3.3***) and in Imspector (see above). As this process is more complex, please contact abberior for support (***1.4***).

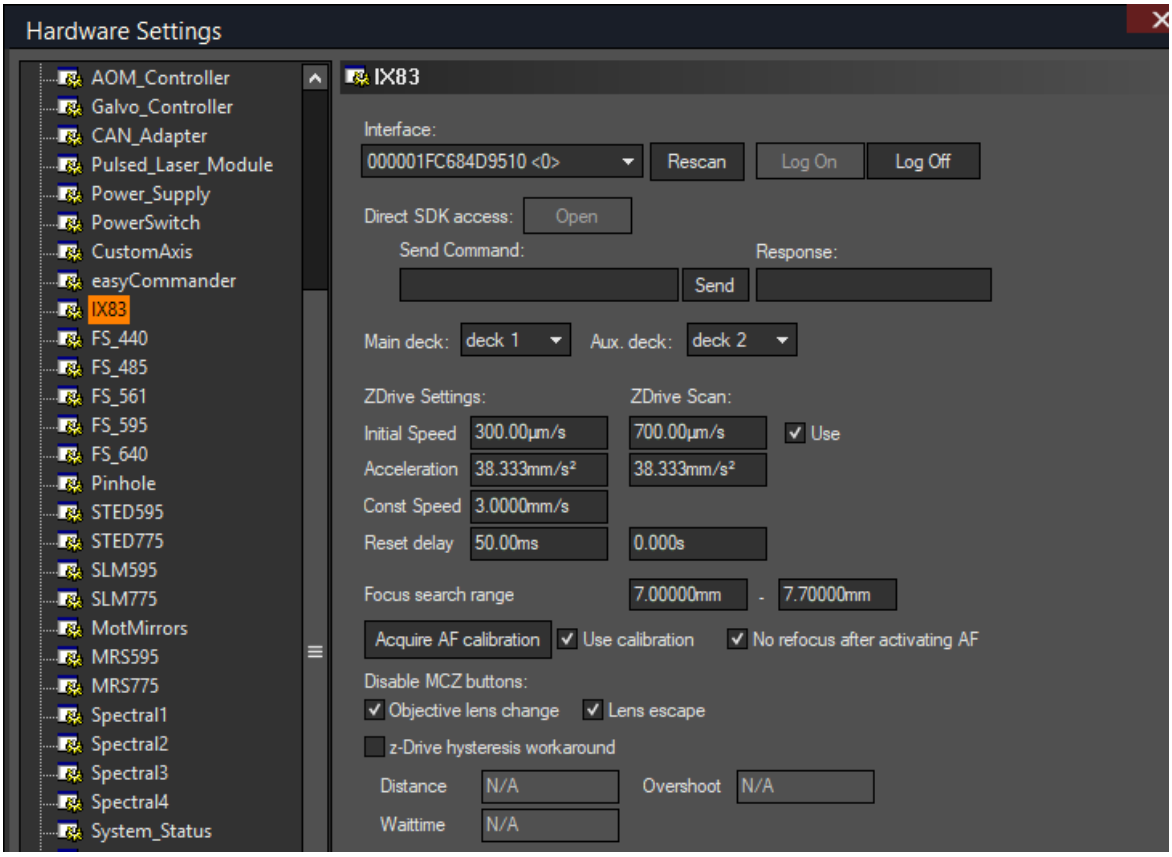



*Figure 2.57: Hardware Settings – Microscope Body*

## 2.9 Imspector Multiuser Management System

Imspector offers functionalities to handle multiple users, e.g., in a core facility environment. It further allows restricting access to sensitive software parameters (e.g., critical hardware settings that affect all users) to advanced users only.

### 2.9.1 Setting up users

The administrator of the Imspector software can define multiple users. This relates to the fact, that individual users can have their own configuration file (this personalized configuration file saves e. g. the user specific measurement templates etc.). However, it is recommended to use a common configuration file to guarantee smooth operation with a common state of microscope alignment and performance.

The Imspector Admin Interface (*FIGURE 2.58*) is a separate program and allows the administrator to configure individual users (i.e., individual user's config files). It is located in the Imspector folder on the hard drive of the setup PC.

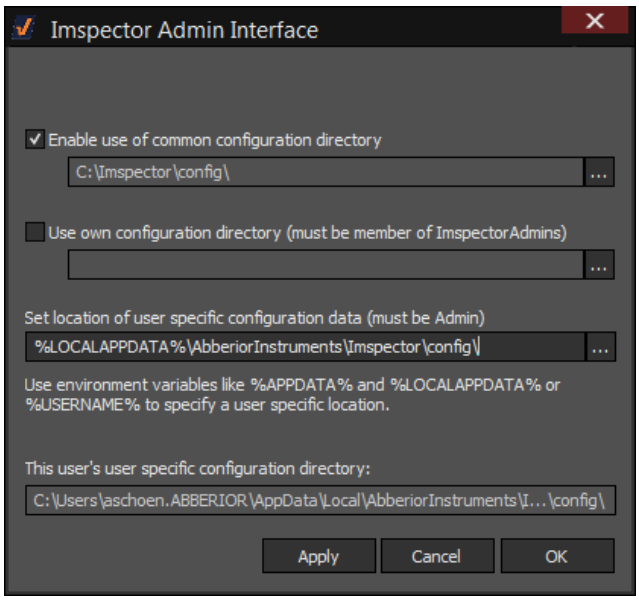



*Figure 2.58: 'Imspector Admin Interface'*

## 2.9.2 Setting up user levels

The Imspector multiuser management system (MMS) allows to define restrictions of software functionality for each user individually: Different restrictions apply to different user levels: Default, advanced or expert level.

At default level, the user has no access to system-critical hardware settings. The advanced user has access to all parameters of the microscope, including some critical hardware settings. The expert user has full control over all devices, parameters and settings that can be addressed in the software – even critical settings such as timing corrections or scanner settings.

In the Imspector software, the user level can be set to default, advanced or expert (*FIGURE 2.59*) in the preferences menu (Edit > Preferences… > General). However, the available user level options may be restricted for different users of Microsoft Windows. To allow a windows user selection of 'advanced' or 'expert' user levels in Imspector, the windows user must be included in the Imspector Admin user group in Windows. Only Windows users with admin rights can add windows users to the Imspector Admin group (see Microsoft Windows help). Further, the user must be elevated to the Imspector Admins user level via the 'ImspectorAdmin.exe' program.

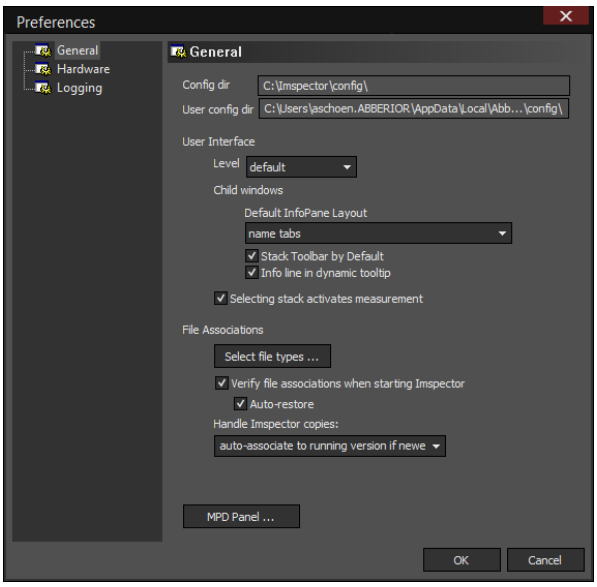



*Figure 2.59: 'Preferences' dialog showing the user interface level.*

The Imspector multiuser management system (MMS) allows the microscope system to be operated by multiple users with their individual measurement templates and workspaces saved.

Each user can save and load his own measurement templates ('user templates', see *2.5*). Each user template can be additionally commented with a description of the template. Alternatively, any user can use globally defined measurement templates ('common templates'). These templates are of general interest, can be opened and used by any user but cannot be overwritten by the (default) user. The common templates are created and modified by the administrator only.

Additionally, users can define the arrangement of the live dialogs needed to perform their measurements individually. Such a 'workspace' contains the information which live dialogs are displayed and where they are positioned on the screen (*FIGURE 2.60*). The settings and values within the live dialogs are not included in workspaces, only the positions of windows and live dialogs. Loading a workspace reopens the respective live dialogs in the correct position.

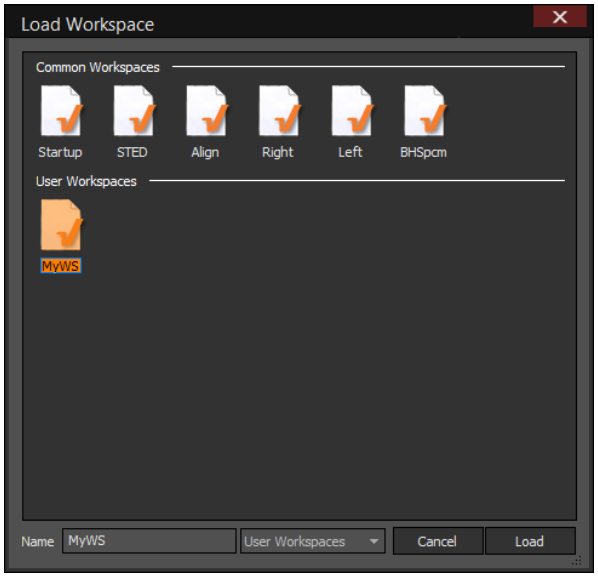



*Figure 2.60: 'Load Workspace' dialog*

## 2.10 Image visualization, processing and analysis

The Imspector software offers a wide range of functionalities for data visualization, image processing and data analysis. Many functions can be applied in real time allowing instant image analysis that updates throughout image acquisition.

Most functions for quick visualization (e.g., LUT, projections) or quick analysis (e.g., line plots, lifetime) are available via toolbars, right-click menus, or the visualization hub of Imspector: The 'Stack Display' live dialog (*2.10.4*). More complex analysis and processing functions are generally available via the top menu bar ('Data', 'Analysis').

Among others, the user can choose from the following functionality: Line profiles, add values in any ROI, interpolation, resizing of data, histograms of any ROI, two- and three-dimensional function generation (e.g., Gaussian or Lorentzian functions for model PSFs), different deconvolution algorithms (e.g., linear deconvolution with arbitrary PSFs defined by the user).

### 2.10.1 Colormaps / LUTs

The active colormap or look-up table (LUT) is displayed at the bottom of the measurement window. In Imspector, the colormap is typically 'Fire' ( Figure 2.61).

*Figure 2.61: Colormap example 'Fire'*




Below the left and right end of the colormap, the minimum and maximum gray value is given. The minimum and maximum intensity value of the colormap determine the look of the displayed image (the raw data remains unchanged). The minimum value typically determines to which level we suppress the display of low intensity values. The maximum determines the saturation point of the colormap (not of the raw data). In addition, some colormaps feature underglow and overglow colors that mark pixel with intensities below or above the minimum and maximum values.

Properties of the colormap can be changed via the right-click context menu of the colormap bar. To access this menu the active colormap has to be right clicked.

#### Selecting colormaps

A large range of alternative colormaps can be selected. For this purpose, first select the intended image via left-click, then select a colormap from the colormap bar on the bottom of the measurement window (right-click onto colormap bar > 'Colormaps').

Popular colormaps for single image channels are 'Grayscale' for high contrast and 'Fire' for a high dynamic range. For most colormaps, Imspector offers standard versions (best for overlaid images) and hot versions (best for high dynamic range in single channel images). For overlaid image channels, popular colormaps are red/green/blue or cyan/magenta/yellow.

### Adjusting the colormap values

The minimum and maximum intensity values of a colormap can be adjusted by different means:

- The minimum and maximum values can be accessed and changed directly via double-clicking the values located below the colormap ends.
- The colormap minimum/maximum values can be adjusted to the dimmest/brightest pixels found in the entire image or a ROI selection of the image via the colormap right-click context menu.
- There are keyboard shortcuts to adjust the colormap minimum/maximum values to the dimmest/brightest pixels found in the entire image or a ROI selection of the image.
  Simply click on an image or ROI selection and press F9 orF10 to adjust the minimum or maximum values.
  Pressing Shift+F9 or Shift F10 will adjust minimum or maximum values for all images in the current measurement window simultaneously.

  ***Note:*** *The colormap must be visible on screen to use shortcuts for adjustment.*

### Locking the values of a colormap

The minimum/maximum values of a colormap can be fixed via the colormap right-click context menu (right-click onto colormap bar > 'Lock').

> ***Note:*** *This option may be particularly helpful during image acquisition when monitoring intensity changes in a continuous acquisition.*

### Copying colormaps

Colormaps can be copied between measurements using standard Windows shortcuts.

The colormap has to be selected via left-click on the colormap bar and is copied using <Ctrl> + C. Then insert the copied colormap into the new image using <Ctrl> + V.

Alternatively, the colormap can be copied using drag and drop. To this end, the active colormap bar has to be selected via <Ctrl> + left-click and transferred to the new image via drag and drop.

### Linear and Logarithmic Representation of the Colormap

In Imspector gray values can be represented in a linear or logarithmic mode. Typically, the colormap is set to a linear mode. To change the colormap mode to logarithmic scale, please enter colormap right-click context menu and select 'Logarithmic'. If the logarithmic mode is active, a tick is shown in front of this entry.

### Creating and editing custom colormaps

New colormaps can be created via the colormap right-click context menu (right-click onto colormap bar > 'New') opening the 'Colormap editor' dialog. Once a new colormap has been created, it can be edited (default colormaps cannot be modified), saved and loaded via the colormap right-click context menu.

When editing a colormap in the 'Colormap editor', the colormap gradient is displayed via 256 boxes. The color of each box can be modified by selecting a box via left-click followed by clicking the 'Modify' button. Importantly, the user can change individual boxes into nodes, which allow the creation of smooth gradients between nodes. Interestingly, the entire gradient can be inverted via the 'Flip' button.

## 2.10.2 Zoom toolbar and zoom shortcuts

The 'Zoom' toolbar (see *FIGURE* 2.62) allows the user to adjust the image display size and to zoom in/out of images. The toolbar can be found at the edge of the Imspector main window and/or at the edge of each measurement window. The following buttons are available:




*Figure 2.62: The 'Zoom' toolbar.*

1. Magnification button 1 allows to **zoom to selection**.
2. Magnification button 2 allows to **zoom in**.
3. Magnification button 3 allows to **zoom out**.
4. The 'Reset Zoom' button zooms the data to a scale where **one screen pixel equals one pixel in the measurement**.
5. The 'Fit to Window' button adjusts the image size to fit the measurement window size. The raw data remains unchanged.
6. The 'Resize Window to Data' button adjusts the measurement window to match the size of the data shown within the window.
7. The 'Lock' button **locks the dimensions** of shown data.

Alternatively, the user can also zoom in to specific image regions or zoom out via the <Ctrl> + mouse wheel selecting the area of interest via the mouse cursor.

## 2.10.3 Display toolbar: Orientation, projections, overlays and scalebars

The 'Display' toolbar ( Figure 2.63) allows to quickly change the way the data is displayed. This toolbar can be found at the edge of the Imspector main window and/or at the edge of each measurement window.




*Figure 2.63: The 'Display' toolbar*

1. **Orthogonal views:** The 'XY view' button rotates the view of a 3-dimensional data stack to XY.
2. **Orthogonal views:** The 'XZ view' button rotates the view of a 3-dimensional data stack to XZ.
3. **Orthogonal views:** The 'YZ view' button rotates the view of a 3-dimensional data stack to YZ.
4. **Viewing orientation:** The 'Tilt' button exchanges the displayed X and Y axis and thereby tilts the data axes.
5. **Slice view:** The 'Single Slice' button switches the visualization to viewing of single slices. Each slice is an image displaying the first and second viewing axis (this depends on the order

of axis in the current visualization). The user can move along the next viewing axis via the mouse wheel effectively scrolling through the slices. This function is mainly relevant for data with 3 or 4 axes.

6. **Axis add-up:** The 'Add Up' button will add up all data points along the fourth axis in the current visualization See section ***2.10.4.3*** or more detailed options available via the 'Stack Display' live dialog. This function is helpful for displaying data with four axes, allowing 3-dimensional navigation after summation of one axis.

   ***Note:*** *This generates only a visualization and the raw data remains unaffected. The operation can be applied to the raw data as follows: Right-click on the image > 'Manage and Manipulate Data' > 'Delete Hidden Axis'.*

7. **Maximum projection:** The 'Maximum Projection' button will create a maximum projection of all data points along the fourth axis in the current visualization See section ***2.10.4.3*** for more detailed options available via the 'Stack Display' live dialog. This function is helpful for displaying data with four axes, allowing 3-dimensional navigation after summation of one axis.

   ***Note:*** *This generates only a visualization and the raw data remains unaffected. The operation can be applied to the raw data as follows: Right-click on the image > 'Manage and Manipulate Data' > 'Delete Hidden Axis'.*

8. **Overlays:** The 'Gallery' button changes the data representation from overlaid to side-by-side. As this button will overlay all images in a window, users may want to create a new empty window (see 'File' toolbar in section ***2.6.1***) and transfer a selection of images (see section ***2.10.6***) into the new window for a custom overlay.
   Per default, overlaid images are produced by adding up the RGB values (colormap not raw intensities) of all overlaid image channels. If needed, the RGB values can instead be multiplied (e.g., to combine intensity and lifetime information via a grayscale and a rainbow colourmap). This option can be selected from the colormap right-click context menu ((right-click onto colormap bar > 'Channels' > 'Multiply RGB values').
9. **Preset colormaps:** The 'RGBize' button applies preset LUTs (color tables) to the different data channels with one click, e.g., red/green/blue for the first/second/third channel. The default LUTs can be changed in the preferences menu ('Edit' > 'Preferences…' > 'Display Settings'). Channel #4 or higher will not be changed.
10. **Scalebar:** The 'Scalebar' button shows hides the scalebar.
11. The **'Info pane'** button shows/hides the information and comment pane on top of the measurement view.
12. The **'Change Size'** button shows information on the data stack dimensions allowing adjustment of these dimensions (see section ***2.10.7***).

    ***Note:*** *For orthogonal views, XYZ refers to the first/second/third scan axis in the image acquisition process. In most cases scan axis #1/#2/#3 are indeed the physical X/Y/Z axis. However, Imspector allows measurements with different orders of acquisition for the individual scan axis, e.g., XZY. In this case, the visualization axis X/Y/Z refer to scan axis 1/2/3 or X/Z/Y. While this can be confusing, the correct order can be checked via the dialog Stack Display (see* **2.10.4***).*

### 2.10.4 Stack Display live dialog: The visualization hub

Depending on the type of data shown in Imspector the user has different display options. The 'Stack Display' dialog, the visualization hub of Imspector, collects all display options in one place and automatically adjusts to the currently selected data type.

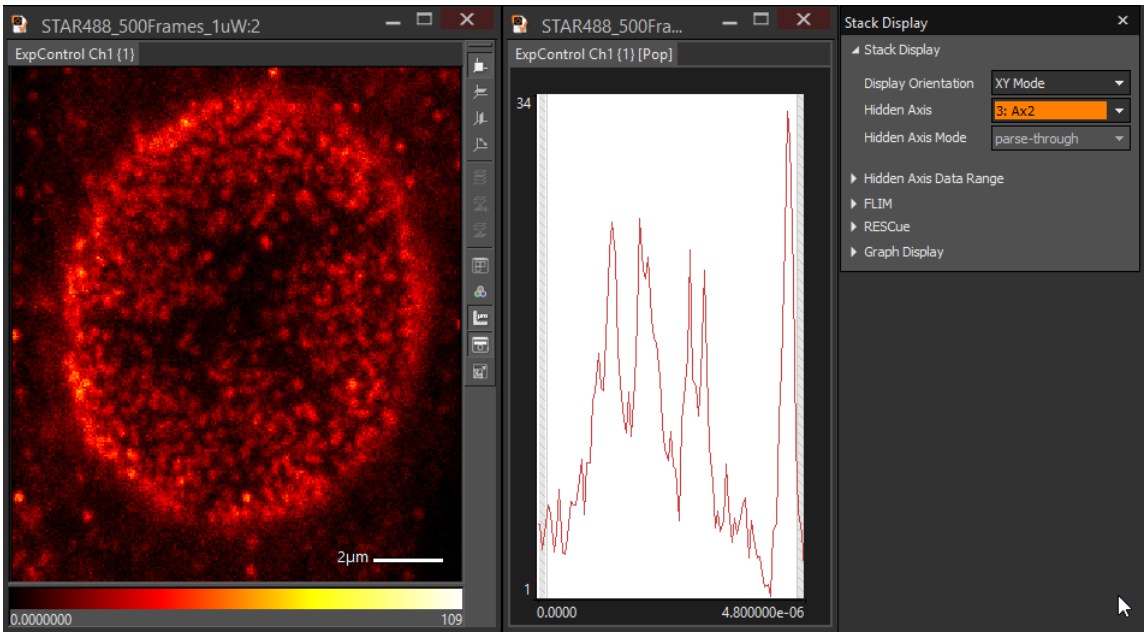



*Figure 2.64: Stack Display dialog appearance for image data*

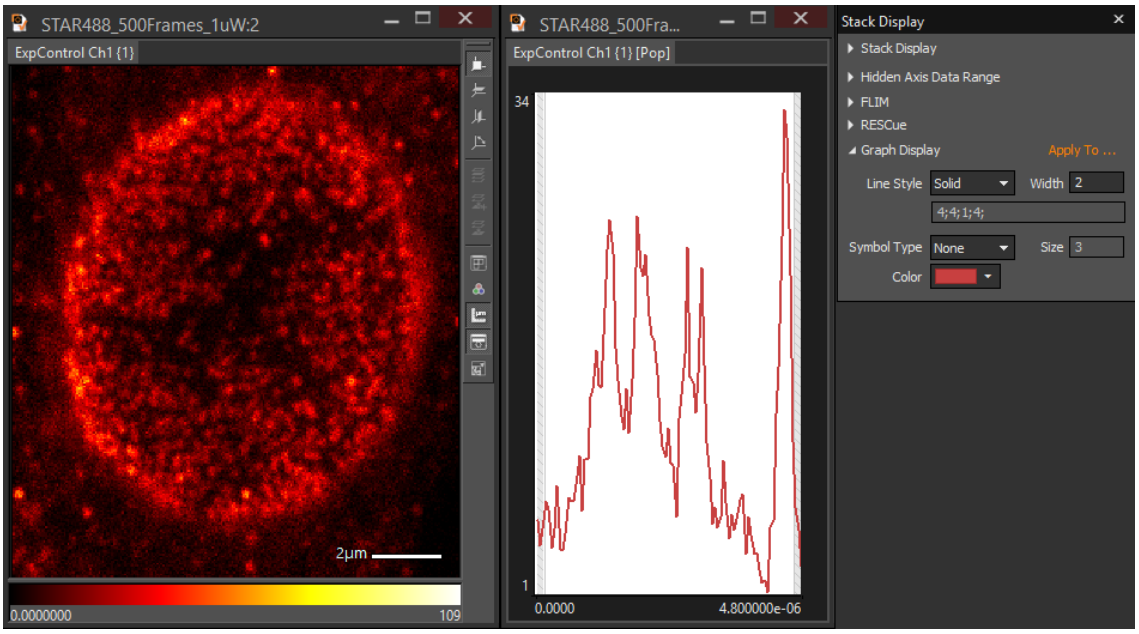



*Figure 2.65: Stack Display dialog appearance for plot data*

### 2.10.4.1 Displaying images vs plots

The examples in FIGURE 2.64 and FIGURE 2.65 show a scenario with an image (left) and a corresponding graph (center) of a line profile. The appearance of Stack Display dialog (right) changes depending on which data type is selected.

When image data is selected (via left click), the 'Stack Display' live dialog offers options for the axis orientation of the displayed image as well as projection options.

When graph data is selected, the 'Stack Display' live dialog shows plot options. In overlaid plots, the plot options apply to the currently selected graph (left-click).

- Line 'Style' (solid, dashed) and line 'Width'
- Symbol 'Type' (none, circle…), 'Size' and 'Color'

### 2.10.4.2 'Display Orientation'

For multi-dimensional data, the user can quickly select the 'Display Orientation' in the 'Stack Display' live dialog. In this drop-down menu, the user can select two axes to be displayed as a two-dimensional image on the screen. FIGURE 2.66 shows different orientations (XY and XZ) for a volume image (XYZ).

> ***Note:*** *The 'Display orientation' drop-down menu lets the user select combinations of the axis X, Y and Z. For simplicity, the X, Y and Z labels refer to the first, second and third axis available for displaying. Depending which axis have been recorded, here the labels X, Y and Z could refer to many different axis types, e.g., X, Y, Z, timelapse, FLIM, lambda and more. If unsure, users can check the actual axis order and the corresponding physical axis in the drop-down menu 'Hidden axis'*

The remaining axis are not visible on screen directly but may be scrolled though (via mouse wheel on the image window) or analyzed in various ways (e.g., via projections, see below). As an example, 4-dimensional data could be displayed showing two axes as an image on screen, using a maximum projection for the third axis and allowing scrolling through the fourth axis via mouse wheel.

### 2.10.4.3 'Hidden Axis' concept: Projections and analysis

The third axis (or fourth axis if available) can be selected for visualization operations via the 'Hidden Axis' drop-down menu in the 'Stack Display' live dialog (see FIGURE 2.66). As this dropdown menu always lists four axes, an empty axis may be listed as generic 'Ax1' or 'Ax2'. Once a (non-empty) axis has been selected as 'Hidden Axis', the type of visualization operation can be selected from the 'Hidden Axis Mode' drop-down menu. The available analysis types depend on the axis type:

- **Maximum projection:** Performs a maximum intensity projection of data points along the hidden axis.
- **Add-up:** Adds up data points along the hidden axis.
- **FLIM:** Ratiometric analysis of lifetime data axis.
- **Parse through:** Shows a single slice at a given position along the hidden axis.

As a result, the 2-dimensional image displayed on-screen in the measurement window, includes information generated from a third axis (unless selecting 'parse-through' mode).

By default, the full range of the 'Hidden Axis' is analyzed/visualized. However, the user can limit the visualized range via the 'Hidden Axis Data Range' drop-down menu (see FIGURE 2.67).

### 2.10.4.4 Display and projection of volume images

For 3-dimensional volume data, the user can quickly select the 'Display Orientation' in the 'Stack Display' live dialog. In this drop-down menu, the user can select two axes for display as a 2-dimensional image on the screen. FIGURE 2.66 shows orientations XY and XZ are for a volume image (XYZ) recorded in 3D-STED mode.

> ***Note:*** *For data recorded with the usual axis order XYZ, the axis labels in the 'Display orientation' drop-down menu are correct. For data recorded in a different order, the XYZ labels may refer to physical XZY axis. If unsure, users can check the actual axis order and the corresponding physical axis in the drop-down menu 'Hidden axis'.*

The third axis will not be visible on screen directly but can be scrolled though (mouse wheel) or analyzed

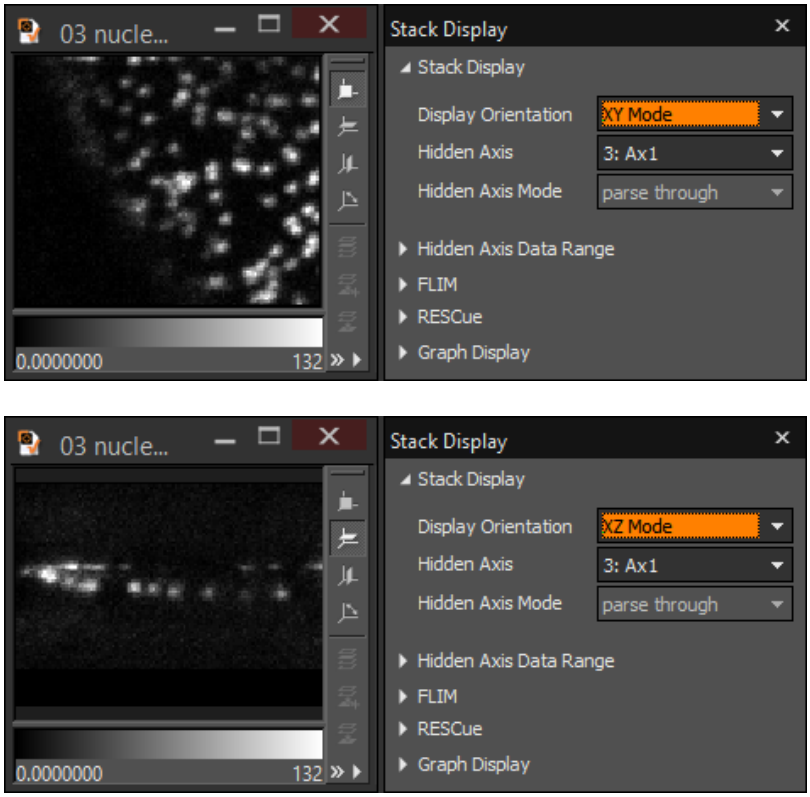



*Figure 2.66: 'Stack Display' live dialog – XY and XZ display orientation of volume images*

(e.g., maximum projection). For a simple volume image, the third axis can be the Z-axis (if in XY display orientation), the Y axis (if in XZ display orientation) or the X axis (if in YZ display orientation).

The third axis (or fourth axis if available) can be selected for visualization operations via the 'Hidden Axis' drop-down menu in the 'Stack Display' live dialog (see FIGURE 2.66). The type of visualization operation can be selected from the 'Hidden Axis Mode' drop-down menu. Common 'Axis Mode' selections for volume images are:

- **Maximum projection:** Performs a maximum intensity projection of data points along the hidden axis (maximum Z-projection, maximum Y-projection or maximum X-projection).
- **Add-up:** Adds up data points along the hidden axis. While creating bright images, often this option also creates a very high background (see FIGURE 2.67).

By default, the full range of the 'Hidden Axis' is projected or added up. However, the user can limit the

visualized range via the 'Hidden Axis Data Range' drop-down menu (see *FIGURE* 2.67). Often, this function can reveal the desired axis section without being obscured by signal and/or background from other sections.

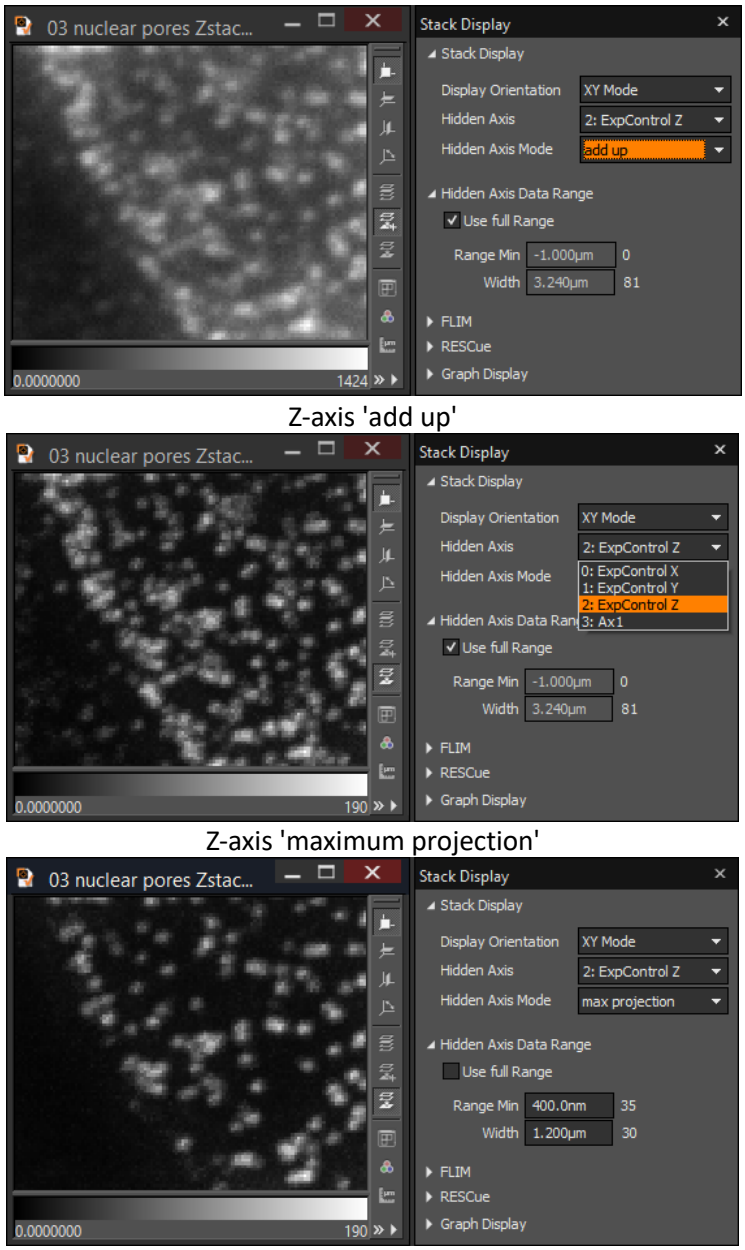



*Figure 2.67: 'Stack Display' live dialog – 'Hidden Axis Mode' and 'Range' in volume images*

### 2.10.4.5 FLIM: Ratiometric lifetime analysis

Instructions following soon.

### 2.10.4.6 Adaptive Illumination: Displaying thresholds

Impector can display the thresholds and illumination reduction for images recorded with Adaptive Illumination (see *2.4.3*). These Adaptive Illumination parameters are shown in the 'Stack Display' section 'RESCue' (see *FIGURE* 2.68). Users can set default colors for Adaptive Illumination thresholds in the preferences menu ('Edit' > 'Preferences…' > 'Display Settings').

***Note:*** *Adaptive Illumination requires optional software upgrades.*

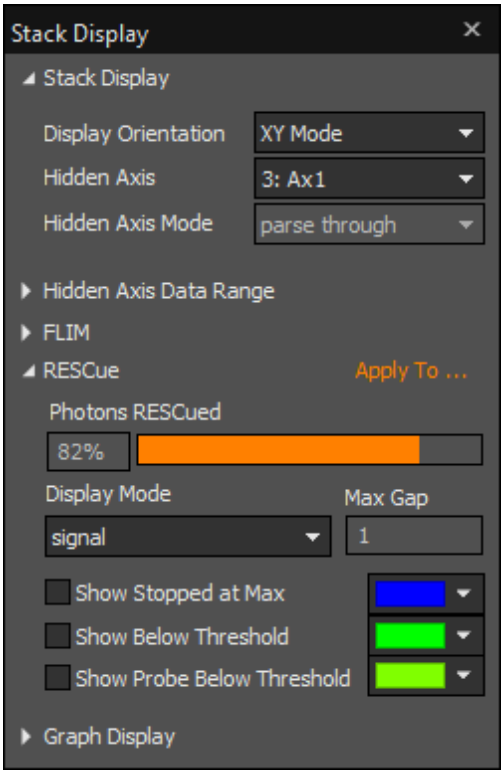



*Figure 2.68: 'Stack Display' – Adaptive illumination thresholds*

**'Show Stopped at Max':**

This tick bog toggles the display of the upper threshold mask for the currently selected image. The upper threshold mask indicates all pixels, where illumination was stopped early (before the full dwell time) since the user-defined upper pixel intensity threshold had been reached early.

This tick box has a function only if the currently selected image was acquired with Adaptive Illumination using an upper threshold. One example are STED images recorded with the 'STED Level' option activated in the easyCommander live dialog (known as RESCue STED). Other examples include confocal images with the 'Confocal Level' option activated in the easyCommander live dialog (known as Confocal RESCue).

**'Show Below Threshold':**

This tick box toggles the display of the lower threshold mask for the currently selected image. The lower threshold mask indicates all pixels, where the lower pixel intensity threshold, as defined by the user for this image channel, had not been reached.

The result of not reaching the threshold can be different for different Adaptive Illumination modes. Not reaching the threshold may simply disable illumination of the pixel in the following line step (e.g., confocal probing before STED, see below). For RESCue STED or Confocal RESCue (see above), the lower threshold mask can indicate pixels, where illumination was stopped early (before the full dwell time) since the user-defined lower pixel intensity threshold was not expected to be reached at the end of the dwell time.

This tick box has a function only if the currently selected image was acquired with Adaptive Illumination using a lower threshold.

**'Show Below Probe Threshold':**

This tick bog toggles the display of the probe mask for the currently selected image. The probe mask indicates all pixels which were not illuminated at all because a user-defined intensity threshold had not been reached in a preceding line step ('probe step').

This tick box has a function only if the currently selected image was acquired with Adaptive Illumination using a probe step (e.g., DyMIN or STED with a confocal probe step).

**'Photons Rescued':**

The 'Stack Display' dialog indicates the reduction of sample illumination achieved by Adaptive Illumination compared to conventional imaging. The value the illumination reduction in the currently selected image as a percentage of illumination that would have been used in conventional illumination.

## 2.10.5 Image right-click menu

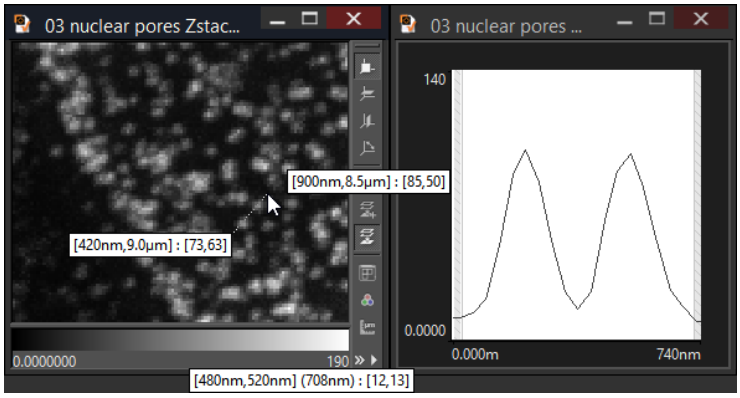
A right-click on images in measurement windows offers various functions. These include more detailed ROI functions (e.g., copying the whole ROI or just a ROI center position) and image manipulation.

Here, the user can also find visualization functions which are also available via toolbars (e.g., zoom, gallery mode/overlay, show/hide info pane, show/hide scale bar etc.). In addition, line plots and axis plots can be activated.

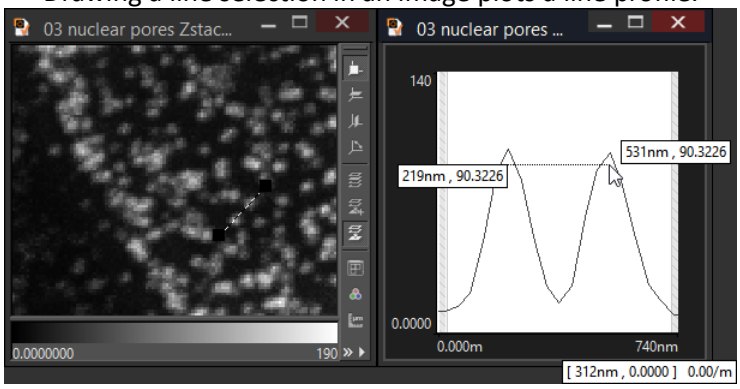
### *2.10.5.1 'Line Profile'*

After selecting the 'Line Profile' tool from the image right-click menu, the user can draw a line on any of the images in the measurement window (drawing via mouse holding the left mouse button). This produces a graph window next to the measurement window plotting the pixel intensity values along the line selection (see *FIGURE* 2.69). The plot style can be adjusted via the 'Stack Display' dialog (see *2.10.4.1*). The exact way of plotting the pixel intensities (pixel distance, pixel step, smart neighborhood) can also be selected via the image right-click menu ('Line Profile Modes').

While drawing the line selection in the image, the image coordinates of the line end points (physical values and pixel values) are displayed in two rectangular boxes next to the line. A rectangular box at the bottom of the measurement window displays the line length (in round brackets) and range covered along the horizontal and vertical axis (in rectangular brackets).




Drawing a line selection in an image plots a line profile.




Drawing a line in the plot to measure peaks.

*Figure 2.69: 'Line Profile' - Plot pixel intensity along a line selection*

The user can draw another line within the plot window, e.g., to measure exact the distance between peaks of the plot. While drawing the line selection in the plot, the coordinates of the line end points (e.g., peak position and amplitude) are displayed in two rectangular boxes next to the line. A rectangular box at the bottom of the measurement window displays the line length (left), the amplitude difference (center) between the line end points as well as the slope between both points (right).

> ***Note:*** *Standard rectangular ROI functions are only available if line plot mode has been disabled via the image right-click menu.*

### *2.10.5.2 Axis Profiles*

Similar to the 'Line Profile' across pixels in an image, the user plot the pixel intensities along a third (or fourth) data axis that is currently being scrolled through or visualized as a hidden axis. The relevant profile type (e.g., 'T-Profile' or 'Z-profile') can be selected via the image right-click menu. A rectangular ROI is required to generate the axis plot. For each point along the plotted axis, all pixels/points within the rectangular ROI are summed.

The plot style can be modified via the 'Stack Display' dialog (***2.10.4.1***) and the plot data can be inspected as described for 'Line Profiles' (***2.10.5.1***).

### *2.10.5.3 Histograms*

In addition to profiles, Imspector can generate histograms of image pixel values. Histograms can be directly selected via the image right-click menu. A rectangular ROI is required to generate the histogram of all pixels of the currently displayed image within the given ROI.

The plot style can be modified via the 'Stack Display' dialog (***2.10.4.1***) and the plot data can be inspected similar to 'Line Profiles' (***2.10.5.1***).

> ***Note:*** *If currently displaying an image with 'Hidden Axis' visualization a third (or fourth) data axis, the histogram will contain the pixel the values of the current visualization mode, not the values pixels at a single point of the third (or fourth) axis.*

### 2.10.6 Drag, drop, copy and paste

Graphs, image stacks and colormaps can be transferred between windows via drag and drop. This is done by pressing keyboard shortcuts while dragging with the left mouse button. Colormaps can also be transferred to stacks which will then be displayed using the dropped colormap (see section *2.10.1*).

For transferring data between windows via drag and drop, the following rules apply:

- **<Ctrl>:** Copy all slices of the current selection to the new window. If no area is selected, the whole stack is copied.
- **<Ctrl> + <Shift>:** Copy only the current slice of the current selection to the new window
- **<Ctrl> + <Alt>:** Instead of creating a new copy of the data, a new representation (view) of the same data is created in the target window. This allows the new view to refresh whenever the measurement is recorded again.

In the graph window the selection is determined by the vertical bars you can drag in from the border (they turn red when in use), **NOT** the rectangle selection. For 4D stacks the following additional rule applies:

- In add-up and maximum intensity projection mode all layers (along the hidden axis) are copied. In parse-through mode only the current layer is copied.

***Note:*** *Copying and inserting data into a vector-based graphics software should be done via export as a TIF file.*

### 2.10.7 The Change Stack Size Dialog

This dialog allows you to change the physical size, offset and pixel dimensions of the stack as well as its data type. It can be accessed using the button at the side of the image or via the shortcut <Ctrl> + T.

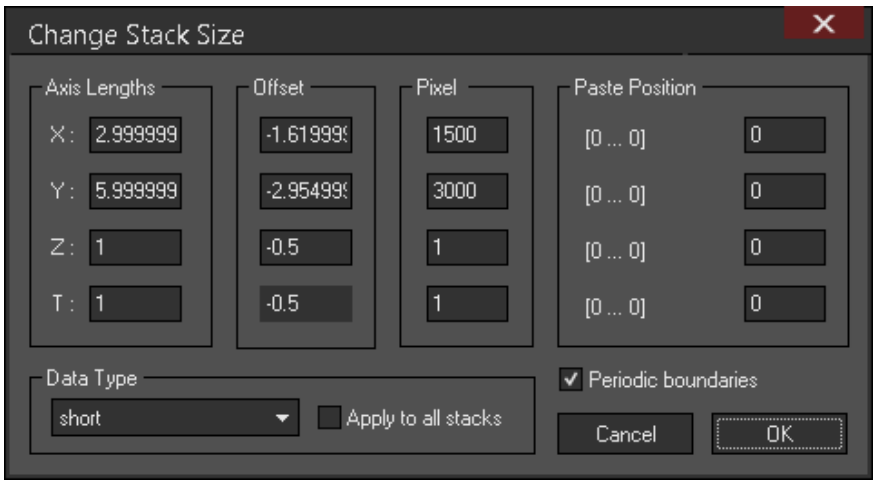



*Figure 2.70: Change Stack Size dialog*

## 2.10.8 Analysis: Image Arithmetics

Imspector offers image arithmetic functionality. To access this, please open the 'Arithmetic Operations' dialog ('Analysis' → 'Arithmetic Operations'). Here basic functionalities as 'Scale', 'Add', 'Subtract', 'Multiply', 'Divide', 'Offset', 'Invert' and 'Normalize' are offered (*FIGURE 2.71*).

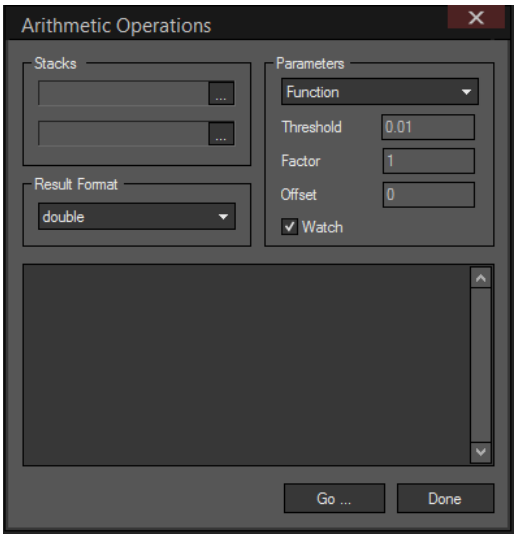



*Figure 2.71: 'Arithmetic operations' dialog*

Further several advanced functions can be applied to the images. To use this, select 'Function' as parameters in the dialog box on the right side. They you a function can be inserted in the lower panel of the 'Arithmetic Operations' live dialog. To see which functions are available click into the lower panel of the dialog and press <F2>

**Parameters:**

x; y; z = individual position

X; Y; Z = range or size of the image

x0; y0; z0 = offset of the position of a pixel

r = intensity value of the first (selected) stack

R =intensity value of the second (selected) stack

**Examples**

Image arithmetic:

sqrt(r) take the square root of the count values in the individual pixels

r^2 square of the count values in the individual pixels

r-R subtract the intensity value from the second selected image (stack) from those in the first selected image

**Thresholding an Image (one sided thresholding):**

First select the image to threshold at 'Stacks' using the pipette. Then set parameters to function. Then insert the formula: '**r>15?r:0**'. Finally Press 'Go …'

The formula contains following function: If a gray value is larger than 15 counts, leave it as it is. Otherwise set it to 0.

**Thresholding an Image (two-sided thresholding):**

First select the image to threshold at 'Stacks' using the pipette. Then set parameters to function. Then insert the formula: '**r>=15&r<20?r:0**'. Finally Press 'Go …'

The formula contains following function: If a gray value is larger than or equal to 15 counts and smaller than 20 counts, leave it like it is. Otherwise set it to 0.

**Creation of an Image with poissonian noise (no structure):**

Insert the formula: 'poidev(n)', where n is the average intensity value within the resulting image stack.

Creation of an Image with poissonian noise on an existing structure:

First select the image to threshold at 'Stacks' using the pipette. Then set parameters to function. Insert the formula: 'poidev(r)', where r is the intensity value within pixel of the selected image stack.

## 2.10.9 Analysis: Smoothing

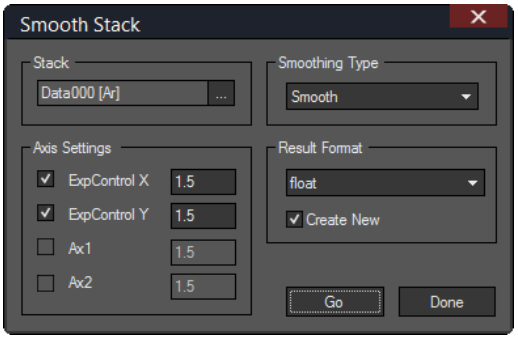



*Figure 2.72: 'Smooth Stack' dialog*

In Imspector images can be smoothed using the 'Smooth stack' dialog (*FIGURE 2.72*).

To smooth an image, the stack that should be smoothed is selected on the left side of the dialog.

Then the type of smoothing must be selected on the right side. Imspector offers different methods to smooth images:

- Typically smoothing is performed using Gaussian kernels with different sizes. To smooth an image using a Gaussian select 'Smooth' on the right side as 'Smoothing Type'.
- 'Median' allows to apply a median filter. This is commonly done to reduce 'salt & pepper noise'.
- Lowpass gauss
- Lowpass cutoff

## 2.10.10 Analysis: Interpolation

Imspector offers to interpolate images. The interpolation can be done in up to four directions. In the 'Interpolation' dialog (*FIGURE 2.73*) first the resolution of the resulting image is given. Here typically integers are used as scaling factors.
Imspector offers to use two algorithms that can be used to interpolate the data: 'Resample' and 'Lagrange Interpolation'. The results of the interpolation directly show the differences of the algorithms. The 'Lagrange Interpolation' leads to rather smooth images where the 'Resample' algorithm preserves the data.

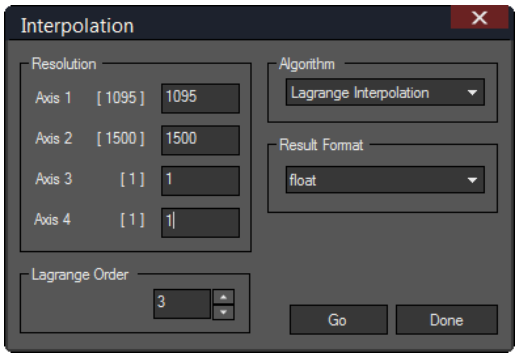



*Figure 2.73: 'Interpolation' dialog.*

## 2.10.11 Analysis: Deconvolution and Convolution

Deconvolution is often used in image processing to remove the influence of the properties of your imaging device (represented by the PSF) on the acquired image. The image is thereby assumed to be a convolution of the original object (dye molecules) distribution and the imaging device's PSF (aka blurring function), all of which is additionally distorted by ever present noise of mostly gaussian or poissonian statistics.

While the resolution of your image cannot be improved by deconvolution in a strict sense (i.e., information cannot be added if it is not already there), the deconvolved image is often less affected by imaging noise as a side effect. The obtained representation is often interpreted as the best estimate for the unknown object which was to be measured. The goal of deconvolution is to find this best estimate.

However, especially for very dark images with a low number of collected photons the usual deconvolution algorithms have to be applied with great care - the removal of noise also removes part of the information, and we can end up with an artifact-containing estimate. Also be aware that the deconvolved image is only an estimate of the object and can have systematic differences depending on the parameter of the deconvolution algorithm. Even small distances between peaks are normally preserved, however widths of objects in the order of the resolution can hardly be quantified (and indeed should not) and are somewhat arbitrary.

Imspector offers two options to deconvolve the imaging data: Linear deconvolution (***2.10.11.4***) and iterative Richardson-Lucy deconvolution (***0***).

The deconvolution method allows to incorporate certain a-priori knowledge, like the positivity of the object or boundaries on the variation of objects. These are implemented for example by regularization parameter in Imspector.

Following steps have to be performed for deconvolution:

- Interpolation of the images with a factor of 2 (in all directions) (***0***)
- Creation of idealized PSFs (***2.10.11.1***)
- Deconvolution with the linear or iterative deconvolution algorithm (***2.10.11.4***, ***0***)

> ***Note:*** *Please be careful with the interpretation of the results of deconvolution.*
> *Deconvolution should only be used for the display of images.*
> *If not applied in the right way, deconvolution may lead to artefacts in the images.*
> *The size and shape of structures in the deconvolved images may be different from those in the raw data. Sizes measurements of structures in deconvolved images may be inaccurate.*

### *2.10.11.1 Idealized PSFs*

For a deconvolution the knowledge about the PSF of the imaging system is important. A measurement of the PSF is often compromised by noise. Frequently only the width of the PSF can be determined with high significance. The shape of the PSF can frequently be approximated by a gaussian or a Lorentzian peak function in 1D, 2D or 3D. These idealized PSFs - as an estimate of the resolution of the microscope - can be computed in Imspector.

### *2.10.11.2 Measurement of PSFs*

The size of the PSF has to be estimated in measured image data.

To this end the size of fluorescent particles (i.e., point emitters) with a size smaller than the expected resolution of the microscope is determined using line profiles.

This can be done either in images of samples of fluorescent nanoparticles (which are smaller than the expected resolution) which are dispersed on a glass surface, which are measured under imaging conditions, which are identical to those the image that should be deconvolved was created. Alternatively, the fluorescent background within an acquired image – which frequently consists of very small fluorescent aggregates – may be used as basis of resolution measurements.

| Technique | Resolution | | |
|---|---|---|---|
| | X | Y | Z |
| Confocal | ~230 nm | ~230 nm | ~500 nm |
| 2D STED or 2D RESOLFT | Please measure | Please measure | ~500 nm |
| 3D STED or 3D RESOLFT | Please measure | Please measure | Please measure |

*Table 2.10-1: Resolutions*

### *2.10.11.3 Creating idealized PSFs*

Idealized PSFs can be created using imspector's arithmetic plugin (***2.10.8***).

Select an image stack that should be deconvolved and for which the PSF needs to be calculated. This is typically a measured image data stack.

Please make sure that the stack has been interpolated (using the interpolation → rescale function; ***0***) beforehand.

Open the Arithmetic plugin (***2.10.8***) and select 'Function' as parameters in the dialog box on the right side.

In the large edit box in the 'Arithmetic Operations' dialog you can input the formula to calculate your PSF (***TABLE 2.10-1***). It will be fit into a new stack exactly the size as your image.

When clicking on '*Go'* a new stack appears with a single centered peak (in 3D stacks one can see it only after scrolling to the central frame) which can be used in the following for deconvolving images.

| Formula | Example | PSF shape that is created in the center of the image |
|---|---|---|
| gaussian2D(x,y,w) | gaussian2D((x-x0-X/2),(y-y0-Y/2),30e-9) | 2D Gaussian with a width of 30 nm |
| gaussian3D(x,y,z,wx,wy,wz) | gaussian3D((x-x0-X/2),(y-y0-Y/2),(z-z0-Z/2),90e-9,90e-9,500e-9) | 3D Gaussian with a width of 90 nm x 90 nm x 500 nm |
| lorentzian2D(x,y,w) | lorentzian2D((x-x0-X/2),(y-y0-Y/2),30e-9) | 2D Lorentzian with a width of 30 nm |
| lorentzian3D(x,y,z,wx,wy,wz) | lorentzian3D((x-x0-X/2),(y-y0-Y/2),(z-z0-Z/2),90e-9,90e-9,500e-9) | 3D Lorentzian with a width of 90 nm x 90 nm x 500 nm, in the center of the image. |

*Table 2.10-2: Formulas for creating idealized PSFs.*

### 2.10.11.4 Linear Deconvolution

Linear deconvolution or Wiener Filtering is the optimal procedure when the image is compromised with gaussian noise.

Its algorithm is based in Fourier space where the convolution of PSF and object is represented by a simple multiplication. The reverse operation, the division is therefore simple to implement and will fail only where the Fourier transform of the PSF (the optical transfer function, OTF) is zero or has a small amplitude. These is unfortunately true for many high spatial frequencies in all practical cases; therefore a regularization parameter has to be added that dampens frequencies that were not transmitted very well and are dominated by noise and cannot be restored therefore.

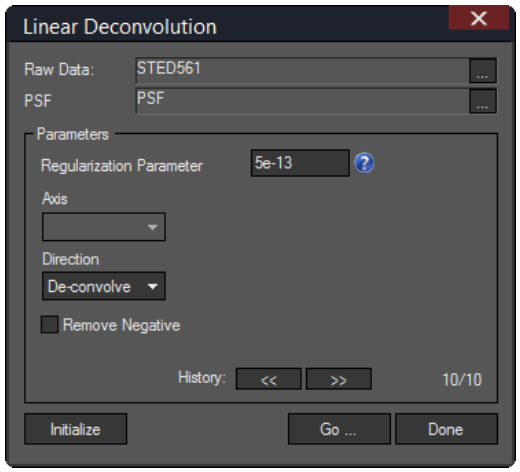



*Figure 2.74: Linear Deconvolution dialog.*

In Imspector the linear deconvolution dialog (**FIGURE 2.74**) needs a 'Raw Data' image stack and a 'PSF' image stack as an input. Please make sure that the format of the image data and the PSF data is the same – typically the 'float' format is used. It can be changed in the 'Change Stack Size' dialog.

The regularization parameter must be adjusted so that the outcome is regularized properly. The scale for adjusting is mostly logarithmic, we advice to try 1E-1, 1E-2, ... 1E-10 and values between. A lower regularization parameter will result in largely overshooting positive and negative signal with many artifacts. A larger than optimal regularization parameter will result in a smoothed version of the image. Because of the necessary regularization the resulting estimate is smoothed but sometimes does not get significantly smaller as expected when removing the PSF influence (noise prevents hard deconvolution in this case).

The direction of the operation can be selected to 'Deconvolution' or 'Convolution'.

During deconvolution, frequently negative intensity values appear in the image. These values can be removed by checking 'Remove Negative'.

The following workflow allows linear deconvolution:

- Select a raw data image
- Select a PSF image
- Set a staring regularization parameter ( e.g., 5e-13)
- Set the direction of the operation – here 'Deconvolution'
- First do not check 'Remove negative' (for the optimization of the 'Regularization Parameter')
- 'Initialize' Deconvolution
- Press 'Go...' to start the Deconvolution
- Wait for the deconvolved image to show up.
- Check the max. intensity value and the min. (here typically negative) intensity value of the colormap.
- Optimize the 'Regularization parameter'.
  As a rule of thumb, we advise to adjust the parameter so that the smallest negative value present in the result is not more than 10% in absolute value of the highest positive value (*FIGURE 2.75*).
- Further check the images carefully for potential deconvolution artifacts:
- In the background of the images, artifacts may appear as a punctuated noise pattern that is created from random bright and dark pixels (*FIGURE 2.77*).
- In the foreground of images, artifacts may appear as well. Here, very small intensity variations many be (over-) emphasized. This may lead to a situation, where compact structures appear to fall apart. As an example, fibers, that are logically connected and appear to be connected before deconvolution are looking like beads on a string after deconvolution.
- If you have achieved a desired (i.e., optimal) value, select 'Remove negative' to delete negative intensity values in the image.
- Select 'DONE' to close the linear deconvolution dialog.

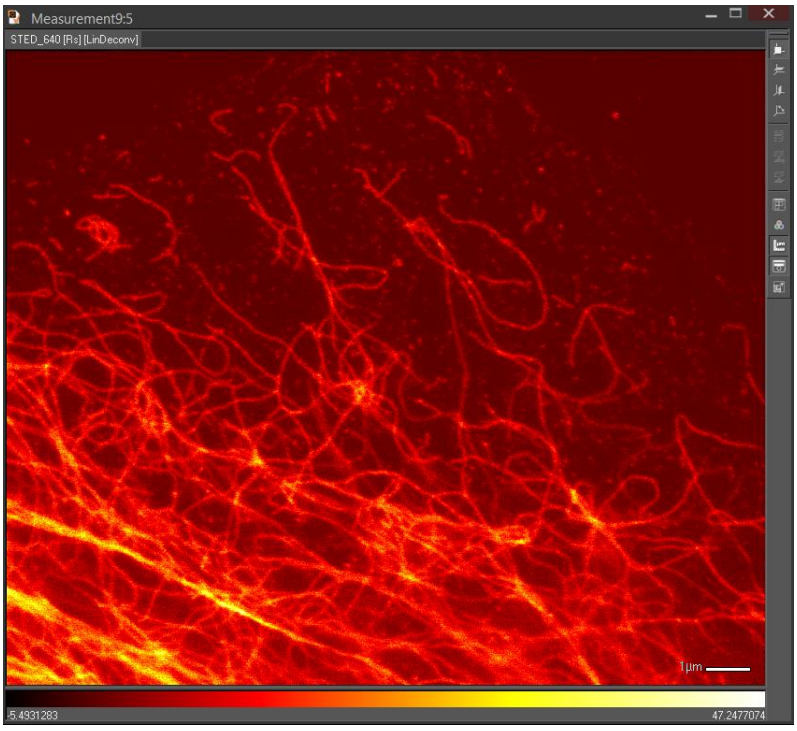



*Figure 2.75: Deconvolved Image with negative Intensity values (without subtracting negative values).*

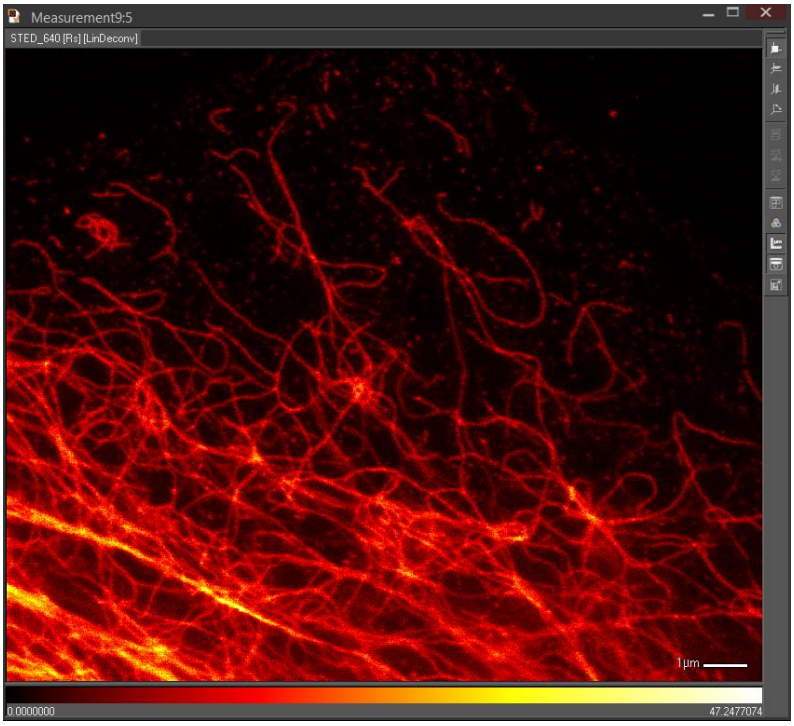



*Figure 2.76: Deconvolved Image without negative Intensity values (after subtracting negative values).*

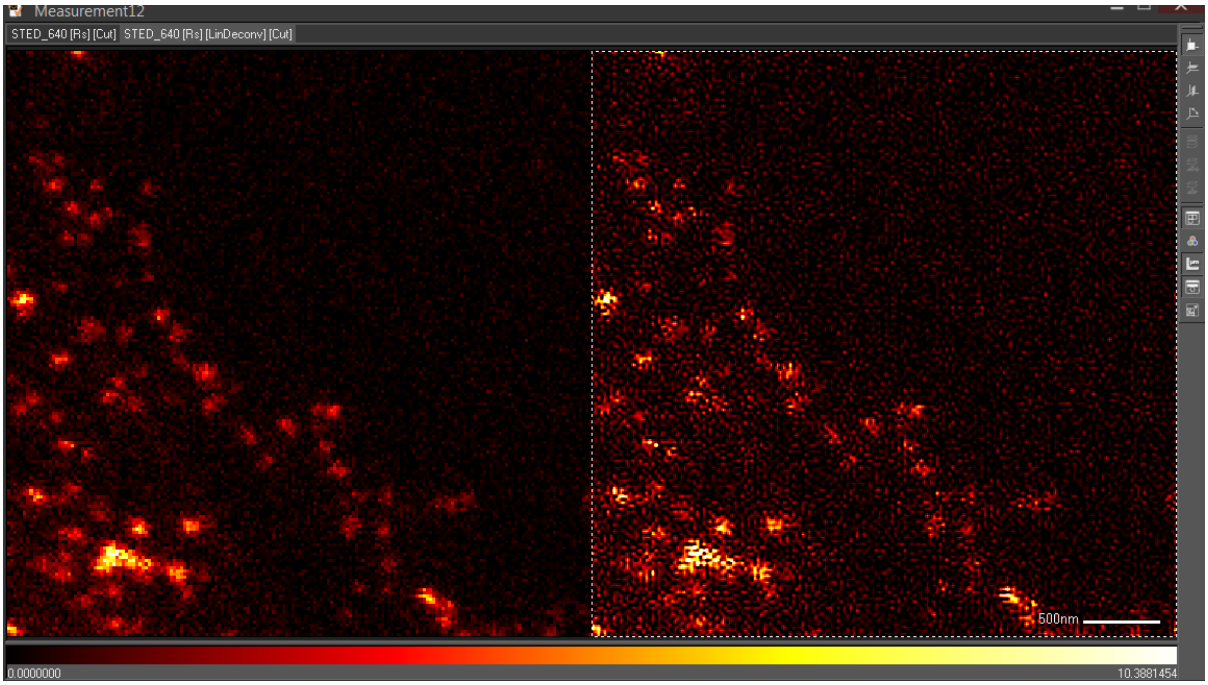



*Figure 2.77: Deconvolution artifacts in the background.*
*Left side: raw data; right side: deconvolved data.*

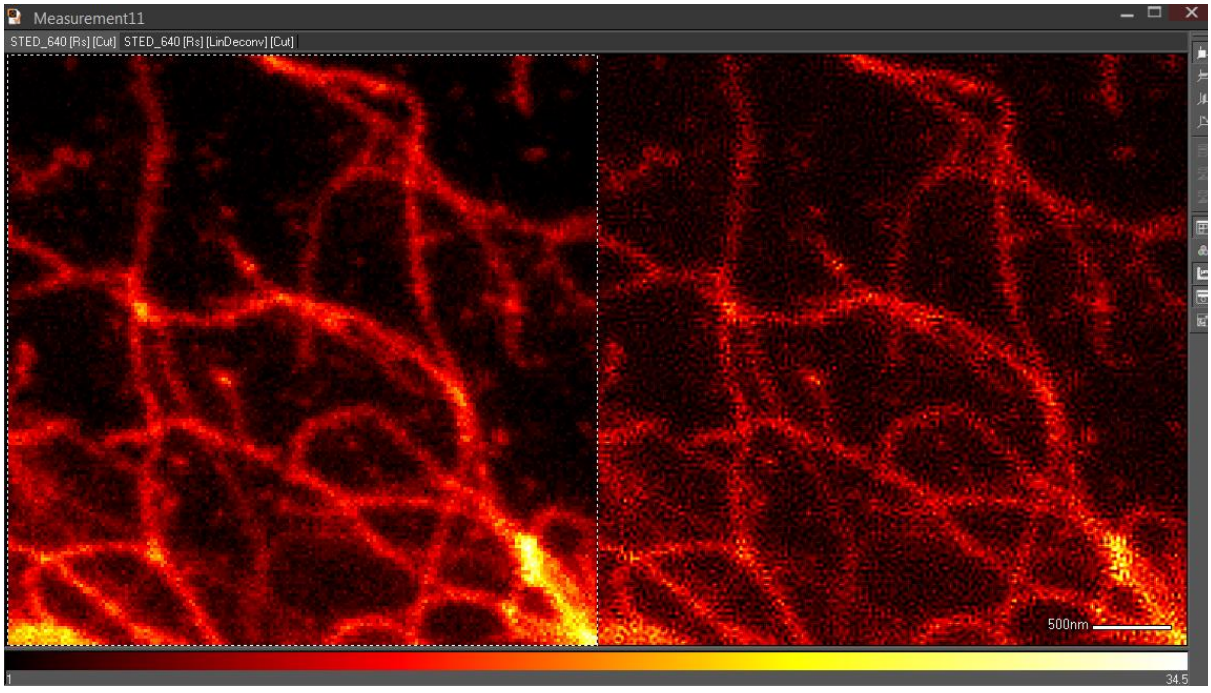



*Figure 2.78: Deconvolution artifacts in the foreground.*
*Left side: raw data; right side: deconvolved data.*

### 2.10.11.5 Iterative Richardson-Lucy Deconvolution

In addition to linear deconvolution, Imspector allows to use an iterative Richardson-Lucy deconvolution algorithm. It is used, when additionally, to Wiener Filtering additional restriction of purely positive image objects (e.g., dye concentrations) on the deconvolution process should be to be imposed.

The Richardson-Lucy deconvolution algorithm is an iterative algorithm. This means that next to a regularization parameter (as described above, **2.10.11.4**), the number of iterations is made as an additional parameter to dampen the influence of high spatial frequencies (which are dominated by noise).

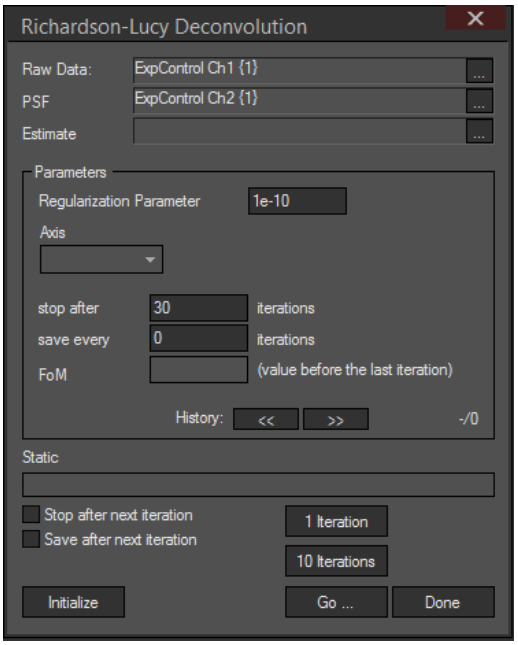



*Figure 2.79: Richardson-Lucy deconvolution dialog*

Although in principle the optimal regularization parameter can be estimated from statistical theory, this is almost never done in applications. If the optimal regularization parameter would be found, the algorithm could run forever, every number of iterations which is high enough would be sufficient. Another, more practical approach is to save the resulting image after a fixed number of iterations each and choose from the images. In the beginning they will show too much blur, in the end, even the noise in the image will be translated to a crumbling structure, clearly representing artifacts. For most applications, a regularization parameter of 1E-10 and up to 100 iterations with stopping every 10 iterations has been found to be sufficient.

The following workflow allows iterative Richardson-Lucy deconvolution:

- Select a raw data image
- Select a PSF image
- Set a starting regularization parameter ( e.g., 1e-10)
- Set a number of iterations that should be performed until the operation is stopped.
- Set a number of iterations after that a resulting image should be saved.
- 'Initialize' Deconvolution
- Press 'Go...' to start the Deconvolution
- Wait for the deconvolved image to show up.
- Optimize the 'Regularization parameter'.
  Check the images carefully for potential deconvolution artifacts:
- In the background of the images, artifacts may appear as a punctuated noise pattern that is created from random bright and dark pixels (*FIGURE 2.77*).
- In the foreground of images, artifacts may appear as well. Here, very small intensity variations many be (over-) emphasized. This may lead to a situation, where compact structures appear to fall apart. As an example, fibers, that are logically connected and appear to be connected before deconvolution are looking like beads on a string after deconvolution (*FIGURE 2.78*).
- Select 'DONE' to close the linear deconvolution dialog.

### 2.10.11.6 Convolution

Smoothing (Sect. *2.10.9*) is probably the easiest way to improve an image and is recommended especially for images with only a few counts where noise is the largest problem.

Alternatively, to smoothing using the smoothing dialog, an image can be convoluted with defined PSFs. This is the inverse way as a deconvolution is performed. It may be done using the same functions as the deconvolution.

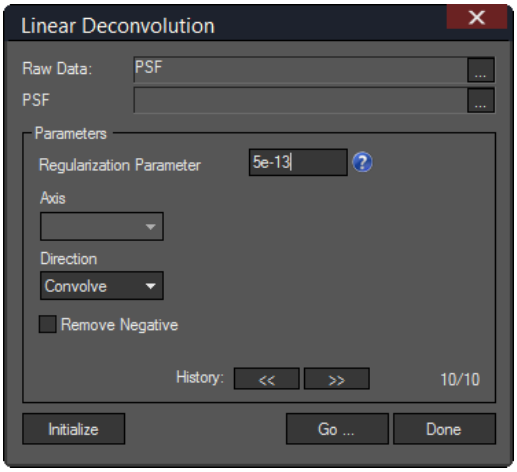



*Figure 2.80: Convolution of images*

The blurring effect of the PSF is here not removed but even more enhanced. However, the noise is greatly reduced. The smoothing kernel will be in most cases a gaussian function. That means we have to provide a stack with equal physical and logical dimensions as the image stack (up to 4D possible) containing a centered gaussian function of certain width. Convolution of these two stacks (the order of the stacks can be exchanged thereby) is then performed with the same algorithms as deconvolution.

## 2.11 Imspector File Format

The Imspector '.msr' file format is a native binary format and now api exists to read or write it in its entirety. As window positions, data dependencies, hardware configuration and settings measurement state and configuration settings are saved in each document reading the entirety of this information is hardly desirable for other programs.

The data and directly associated meta-data contained in any Imspector file is organized according to the specification of the OBF (*.obf) file format and Imspector also reads OBF files in an intuitive way. There are bindings for OBF to both c++, MATLAB and JAVA and we are working on Labview and Python bindings to allow reading of Imspector data into most applications used in scientific data analysis and to allow writing to an Imspector-readable format from Labview.

The C++ API for OBF will be rather stable as the file format evolves. The format itself is forward and backwards compatible between all versions released.

> ***Note:*** *The fact that OBF is compatible in both directions helps when reading a newer version .msr file with an old Imspector copy. While window positions and some analysis related data is omitted, all physical data and the associated meta-data remains readable. Just (temporally) rename the file to .obf and open it in imspector.*

## 2.12 Compatibility with 3rd Party Software

The abberior software Imspector is designed as an open platform that ensures compatibility with 3rd party software to ensure maximum usability of our system (*FIGURE 2.81*).

Imspector .msr files are compatible with common 3rd party software for data analysis (e.g., ImageJ/Fiji, Cell Profiler, Icy…), 3D data analysis software (Amira, Imaris, Image Surfer2, Paraview…) and data processing software (e.g., Matlab, Python).

Please ask us for compatibility information for other 3rd party software.

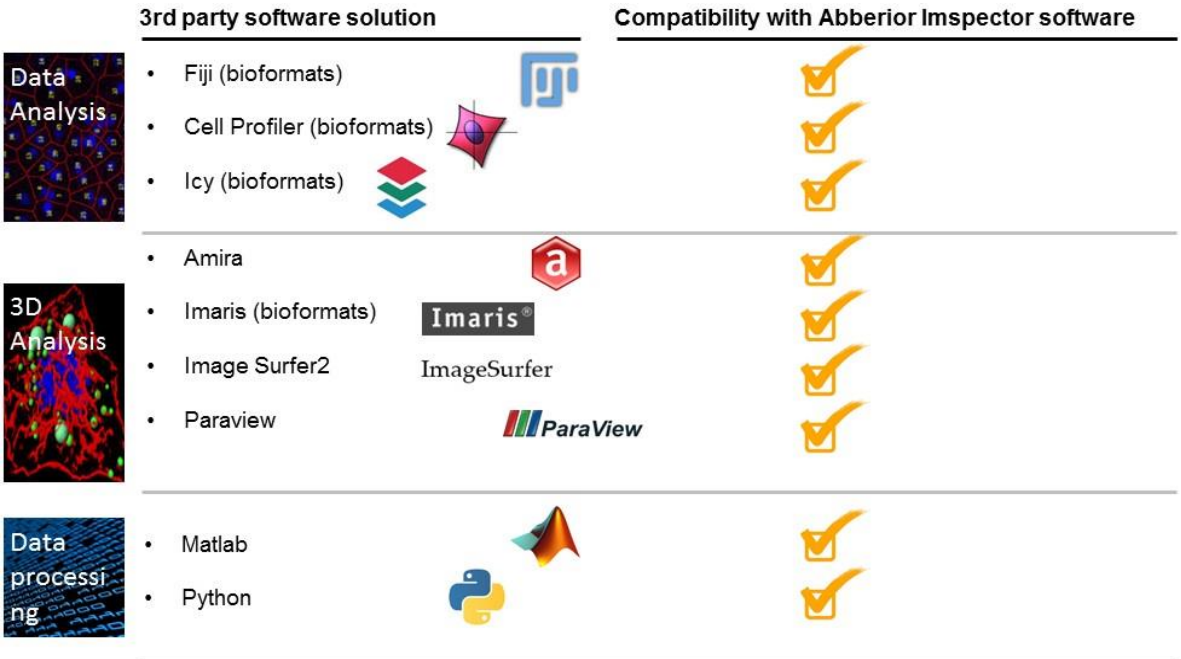



*Figure 2.81: Compatible 3rd party software.*

## 2.13 Export of Files

Data from Imspector measurements can be exported into several file formats:

- Colormap Tiff files/stacks (.tif/.tiff)
- RGB Tiff files/stacks (.tif)
- binary double files (.dbl)
- Avi files/movies (.avi)
- Visualization toolkit files (.vtk)
- MRC files (.st, .map, .ccp4, .mrc)
- ASCII data files (.dat, .asc)
- Becker&Hickl data files (.sdt)

To export data select the data Stack and select 'File → Export' or e use the shortcut <Ctrl> + E to open the Export Data dialog.

## 2.14 Import of Files

Imaging data from other vendors can be opened with Imspector:

- Leica .lif files (created by the Leica TCS SP5 & TCS SP8)
- Leica .lei files in conjunction with Tiff stacks (.tif) and a .txt file containing the meta data.

## 2.15 Python and Matlab Scripting (optional)

The abberior software Imspector is implemented in C++. Nevertheless, it has several interfaces to other programs: software libraries with Python and Matlab bindings are included – to allow customer plugins/drivers to control further external devices, e.g., further detectors, further laser sources etc.

This enables further the remote control via Python and Matlab scripts (*FIGURE 2.82*):
Virtually any parameter of the microscope that can be accessed via Imspector can also be accessed, set and changed via Python. This allows e.g., for long-term measurements without the user being present as well as remote control of the microscope from a remote location. Please approach us for details and/or support.

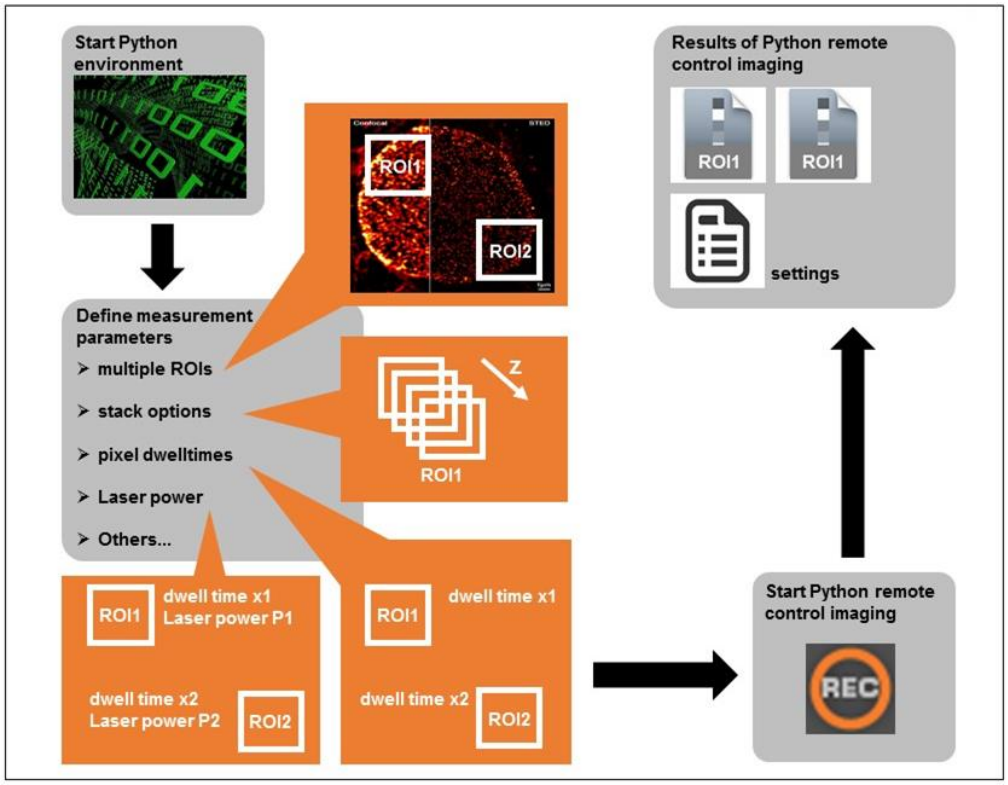



*Figure 2.82: Imspector Python & Matlab scripting.*

### 2.15.1 Imspector Python Interface

Imspector comes with a Python Interface named SpecPy which can be used either from the embedded Python console or from an external Python console running on the same computer (to enable sharing of measurement data) or even running on a different computer.

**Setup**

First the required modules have to be installed. Then you have to start the python environment.

**Start**

Working from an external Python console you need to start the Imspector Server (with requires administrator permissions the first time):

To load the Python Interface just say

```
from specpy import *
```

**Interface**

**Imspector**

`Imspector()`

first tries to return a local Imspector object (living in the same process) or else returns a proxy Imspector object connected to the Imspector application running on `localhost`.

`Imspector(host)`

where `host` is a host name returns a proxy Imspector object connected to the Imspector application running on the corresponding host.

If `imspector` is an Imspector object then

- `imspector.host()`
  returns the name of the host the Imspector application is running on or an empty string in case the Imspector object is local (living in the same process).
- `imspector.version()`
  returns the current Imspector version.
- `imspector.device_drivers()`
  returns the Imspector device drivers as a dictionary of name value pairs.
- `imspector.parameters(path)`
  where `path` is of the form *device/.../parameter_name* and returns the corresponding Imspector parameter value (the empty path returns a dictionary of name value pairs of all parameters).
- `imspector.set_parameters(path, value)`
  where `path` is of the form *device/.../parameter_name* and `value` is a value, sets the corresponding Imspector parameter value (the empty path sets a dictionary of name value pairs of all parameters).
- `imspector.measurement_names()`
  returns the list of names of all open measurements in Imspector.
- `imspector.active_measurement()`
  for the currently active measurement in Imspector, returns the corresponding Measurement

object or throws a RuntimeError if no measurement is active.

- `imspector.measurement(name)`
  where `name` is the name of an open measurement in Imspector, returns the corresponding Measurement object.
- `imspector.create_measurement()`
  creates an empty measurement in Imspector and returns the corresponding Measurement object.
- `imspector.open(path)`
  where `path` is the path to a measurement file, opens it in Imspector (if it is not already open) and returns the corresponding Measurement object.
- `imspector.activate(measurement)`
  where `measurement` is a Measurement object, activates the corresponding measurement in Imspector.
- `imspector.start(measurement)`
  where `measurement` is a Measurement object, starts the corresponding measurement in Imspector and returns immediately.
- `imspector.pause(measurement)`
  where `measurement` is a Measurement object, pauses the corresponding measurement in Imspector.
- `imspector.stop(measurement)`
  where `measurement` is a Measurement object, stops the corresponding measurement in Imspector.
- `imspector.run(measurement)`
  where measurement is a Measurement object, runs the corresponding measurement in Imspector (starts it and returns when it has finished).
- `imspector.close(measurement)`
  where `measurement` is a Measurement object, closes the corresponding measurement in Imspector.
- `imspector.active_stack()`
  for the currently active stack (from the currently active measurement) in Imspector, returns the corresponding Stack object or throws a RuntimeError if no stack is active.
- `imspector.connect_begin(callable, flag)`
  where `callable` is a callable Python object, connects it to the corresponding begin signal in Imspector (if `flag` is 0 the begin of the whole measurement and if `flag` is 1 the begin of one measurement step).
- `imspector.disconnect_begin(callable, flag)`
  where `callable` is a callable Python object, disconnects it from the corresponding begin signal in Imspector (if `flag` is 0 the begin of the whole measurement and if `flag` is 1 the begin of one measurement step).
- `imspector.connect_end(callable, flag)`
  where callable is a callable Python object, connects it to the corresponding end signal in Imspector (if `flag` is 0 the end of the whole measurement and if `flag` is 1 the end of one measurement step).
- `imspector.disconnect_end(callable, flag)`
  where `callable` is a callable Python object, disconnects it from the corresponding end signal in Imspector (if `flag` is 0 the end of the whole measurement and if `flag` is 1 the end of one measurement step).

**Measurement**

If `measurement` is a Measurement object then

- `measurement.name()`
  returns the name of the measurement.
- `measurement.number_of_configurations()`
  returns the number of configurations in the measurement.
- `measurement.configuration_names()`
  returns the list of names of all configurations in the measurement.
- `measurement.active_configuration()`
  for the currently active configuration in the measurement, returns the corresponding Configuration object.
- `measurement.configuration(position)`
  where `position` is in the range from zero to the number of configurations in the measurement minus one, returns the corresponding Configuration object.
- `measurement.configuration(name)`
  where `name` is one of the configuration names in the measurement, returns the corresponding Configuration object.
- `measurement.activate(configuration)`
  where `configuration` is a Configuration object, activates the corresponding configuration in the measurement (if the measurement contains only one configuration, this configuration is activated by default).
- `measurement.clone(configuration)`
  where `configuration` is a Configuration object, clones the corresponding configuration in the measurement and activates and returns the clone.
- `measurement.remove(configuration)`
  where `configuration` is a Configuration object, removes the corresponding configuration in the measurement.
- `measurement.parameters(path)`
  where `path` is of the form *device/.../parameter_name* returns the corresponding measurement parameter value for the currently active configuration (the empty path returns a dictionary of name value pairs of all parameters).
- `measurement.set_parameters(path, value)`
  where `path` is of the form *device/.../parameter_name* and `value` is a value, sets the corresponding measurement parameter value for the currently active configuration (the empty path sets a dictionary of name value pairs of all parameters).
- `measurement.number_of_stacks()`
  returns the number of stacks in the measurement.
- `measurement.stack_names()`
  returns the list of names of all stacks in the measurement.
- `measurement.stack(position)`
  where `position` is in the range from zero to the number of stacks in the measurement minus one, returns the corresponding Stack object.
- `measurement.stack(name)`
  where `name` is one of the stack names in the measurement, returns the corresponding Stack object.

- `measurement.create_stack(type, sizes)`
  where `type` is a NumPy ***array data type*** and `sizes` is a list of exactly four sizes of dimensions, creates a new stack and returns the corresponding Stack object.
- `measurement.update()`
  redraws all corresponding stacks in Imspector (useful when the stack content was changed from Python).
- `measurement.save_as(path[, compression])`
  where `path` is a file path and `compression` is `True` by default or `False` saves it into a file.

### Configuration

If `configuration` is a Configuration object then

- `configuration.name()`
  returns the name of the configuration.
- `configuration.parameters(path)`
  where `path` is of the form *device/.../parameter_name* returns the corresponding measurement parameter value for this configuration (the empty path returns a dictionary of name value pairs of all parameters).
- `configuration.set_parameters(path, value)`
  where `path` is of the form *device/.../parameter_name* and `value` is a value, sets the corresponding measurement parameter value for this configuration (the empty path sets a dictionary of name value pairs of all parameters).
- `configuration.number_of_stacks()`
  returns the number of stacks in this configuration.
- `configuration.stack_names()`
  returns the list of names of all stacks in this configuration.
- `configuration.stack(position)`
  where `position` is in the range from zero to the number of stacks in the configuration minus one, returns the corresponding Stack object.
- `configuration.stack(name)`
  where `name` is one of the stack names in this configuration, returns the corresponding Stack object.

### File

`File(path, mode)`

Where `path` is the path to an *.obf* or *.msr* file and `mode` is either `File.Read` or `File.Write` or `File.Append`, opens it and returns the corresponding File object.

If `file` is a File object then

- `file.description()`
  returns the description of the file.
- `file.set_description(string)`
  where `string` is a string sets the description of the file.
- `file.number_of_stacks()`
  returns the number of stacks in the file.

- `file.read(position)`
  where `position` is in the range from zero to the number of stacks in the file minus one, reads and returns the corresponding Stack object.
- `file.write(stack[, compression])`
  where `stack` is a Stack object and `compression` is `True` by default or `False` writes it to the file.
- `del file`
  closes it.

**Stack**

`Stack(array)`

where `array` is a NumPy ***array*** returns a new local Stack object with data values from the array.

`Stack(type, sizes)`

where `type` is a NumPy ***array data type*** and `sizes` is a list of sizes of all dimensions returns a new local Stack object.

If `stack` is a Stack object then

- `stack.name()`
  returns the name of the stack.
- `stack.set_name(string)`
  where `string` is a string, sets the name of the stack. If another stack in the same measurement already has the same name, suffixes of the form *[1], [2], ..* are added.
- `stack.description()`
  returns the description of the stack.
- `stack.set_description(string)`
  where `string` is a string, sets the description of the stack.
- `stack.type()`
  returns the type of the stack elements as NumPy ***array data type***.
- `stack.number_of_elements()`
  returns the number of elements of the stack.
- `stack.number_of_dimensions()`
  returns the number of dimensions of the stack (including singleton dimensions, i.e. a shortcut for `len(stack.sizes()))`.
- `stack.size(dimension)`
  where `dimension` is one of the dimensions, returns the corresponding size of the stack (the number of steps/positions in that dimension).
- `stack.sizes()`
  returns the list of sizes of all dimensions of the stack.
- `stack.label(dimension)`
  where `dimension` is one of the dimensions, returns the corresponding label of the stack.
- `stack.set_label(dimension, string)`
  where `dimension` is one of the dimensions and `string` is a string, sets the corresponding label of the stack.
- `stack.labels()`
  returns the list of labels of all dimensions of the stack.

- `stack.set_labels(strings)`
  where `strings` is a list of strings for all dimensions, sets the corresponding labels of the stack.
- `stack.length(dimension)`
  where `dimension` is one of the dimensions, returns the corresponding length of the stack.
- `stack.set_length(dimension, number)`
  where `dimension` is one of the dimensions and `number` is a number, sets the corresponding length of the stack.
- `stack.lengths()`
  returns the list of lengths of all dimensions of the stack.
- `stack.set_lengths(numbers)`
  where `numbers` is a list of numbers for all dimensions, sets the corresponding lengths of the stack.
- `stack.offset(dimension)`
  where `dimension` is one of the dimensions, returns the corresponding offset of the stack.
- `stack.set_offset(dimension, number)`
  where `dimension` is one of the dimensions and `number` is a number, sets the corresponding offset of the stack.
- `stack.offsets()`
  returns the list of offsets of all dimensions of the stack.
- `stack.set_offsets(numbers)`
  where `numbers` is a list of numbers for all dimensions, sets the corresponding offsets of the stack.
- `stack.data()`
  returns the data of the stack as a NumPy ***array***.
- `stack.set_data(array)`
  where `array` is a NumPy ***array***, sets the data values of the stack to those of the array,
- `stack.meta_data()`
  returns the meta data of the stack as a dictionary of name value pairs (amongst others the units of the pixels are given there and are valid for the pixel sizes and well as the stack lengths, i.e. the pixel sizes are the division of the stack lengths by the stack sizes).

## Examples

### Camera control

Connects to Imspector, displays the parameters, and doubles the exposure time of the sample camera if a camera is existing as device.

Requirements: Latest Python 2.X or 3.X, specpy (*https://pypi.python.org/pypi/specpy/1.0.2*)

which itself requires NumPy >= 1.8.1, Imspector with "Run Server" checked and an open measurement

See also: *https://imspector.mpibpc.mpg.de/documentation/specpy_examples.html*

```
# Python 2/3 compatibility
from __future__ import absolute_import, division, print_function

# import specpy
import specpy as sp

# connect to local Imspector
im = sp.Imspector()

# print Imspector host and version
print('Connected to Imspector {} on {}'.format(im.version(), im.host()))

# get active measurement and print nicely
msr = im.active_measurement()
from pprint import pprint
print('Parameters of measurement {}'.format(msr.name()))
pprint(msr.parameters())

# read and write a parameter
if 'SimCam' in msr.parameters():
    time = msr.parameters('SimCam/ExposureTime')
    msr.set_parameters('SimCam/ExposureTime', 2 * time)

# now have a look at Imspector, the parameter should have changed in the measurement properties of this measurement
```

**Python Data Analysis Example**

Connects to Imspector, opens a recorded measurement, computes statistical measures on this data and prints them in a file.

Requirements: Latest Python 2.X or 3.X, specpy (***https://pypi.python.org/pypi/specpy/1.0.2***)

which itself requires NumPy >= 1.8.1, Imspector with "Run Server" checked and the file "data_analysis_example.msr".

See also: ***https://im.mpibpc.mpg.de/documentation/specpy_examples.html***

```
# Python 2/3 compatibility
from __future__ import absolute_import, division, print_function

# import numpy
import numpy as np

# import specpy
import specpy as sp

# connect to local Imspector and open an example file
im = sp.Imspector()
measurement = im.open("data_analysis_example.msr")

# set threshold
threshold = 410

# open output file
file = open('data_analysis_example_output.txt', 'w')

# for each stack in the measurement
for name in measurement.stack_names():
        # get the stack (as our stub object)
        stack = measurement.stack(name)
        # get the data array from the stack
        data = stack.data()
        # compute mean and std
        mean = data.mean()
        standard_deviation = data.std()
        # print mean and std to console and to file
        print('stack {} has mean {} and std {}'.format(name, mean, standard_deviation))
        print('stack {} has mean {} and std {}'.format(name, mean, standard_deviation), file=file)
```

```
        # apply mask (all values smaller threshold)
        masked_data = np.ma.masked_less(data, threshold)
        # compute mean and std on masked image
        mean = masked_data.mean()
        standard_deviation = masked_data.std()
        # print mean and std on masked image to console and to file
        print('masked stack {} with threshold {} has mean {} and std {}'.format(name, threshold, mean,
standard_deviation))
        print('masked stack {} with threshold {} has mean {} and std {}'.format(name, threshold, mean,
standard_deviation), file=file)
        # change pixels (you can see this in Imspector) below threshold to another value
        np.putmask(data, data < threshold, 4095)

file.close()
```

**Python Measurement Example**

Example of how to traverse the measurements, configurations, stacks hierarchy of Imspector with Python.

Also runs a measurement.

Requirements: Latest Python 2.X or 3.X, specpy (https://pypi.python.org/pypi/specpy/1.0.2)

which itself requires NumPy >= 1.8.1, Imspector with "Run Server" checked and an open measurement.

March, 2015, Jan Keller (*jan.keller@mpibpc.mpg.de*)

See also: *https://imspector.mpibpc.mpg.de/documentation/specpy_examples.html*

```
# Python 2/3 compatibility
from __future__ import absolute_import, division, print_function

import pprint
pp = pprint.PrettyPrinter(indent=2)

# import NumPy
import numpy as np

# import specpy
import specpy as sp

# get imspector object and print version
im = sp.Imspector()
print('Imspector {} on {}'.format(im.version(), im.host()))

# print existing device drivers
print('devices')
pp.pprint(im.device_drivers())

# print current imspector parameters
print('parameters')
pp.pprint(im.parameters())

# list all open measurements
print('names of open measurements')
pp.pprint(im.measurement_names())

# get currently active measurement
msr = im.active_measurement()
```

```
print('name of active measurement: {}'.format(msr.name()))
print('configurations in active measurement')
pp.pprint(msr.configuration_names())
print('number of stacks: {}'.format(msr.number_of_stacks()))
pp.pprint(msr.stack_names())

# get another measurement (the first one, should be at least one open)
measurement_name = im.measurement_names()[0]
msr = im.measurement(measurement_name)
print('name of chosen measurement: {}'.format(msr.name()))
print('configurations in chosen measurement')
pp.pprint(msr.configuration_names())
print('active configuration: {}'.format(msr.active_configuration().name()))
print('number of stacks: {}'.format(msr.number_of_stacks()))
pp.pprint(msr.stack_names())

# set active configuration (the first one, should contain at least one)
configuration_name = msr.configuration_names()[0]
configuration = msr.configuration(configuration_name)
msr.activate(configuration)
print('active configuration: {}'.format(msr.active_configuration().name()))

# create new stack (will be added to active measurement)
s = msr.create_stack(np.int32, [100, 100, 1, 1])
print('number of stacks: {}'.format(msr.number_of_stacks()))
pp.pprint(msr.stack_names())

# runs the measurement
print('active configuration {}'.format(msr.active_configuration().name()))
im.run(msr)
print('number of stacks: {}'.format(msr.number_of_stacks()))
pp.pprint(msr.stack_names())

# set a configuration and runs it
configuration = msr.configuration(configuration_name)
msr.activate(configuration)
print('active configuration {}'.format(msr.active_configuration().name()))
im.run(msr)
```

```
print('number of stacks: {}'.format(msr.number_of_stacks()))
pp.pprint(msr.stack_names())
```

### 2.15.2 Imspector Matlab Interface

Imspector comes with a Matlab Interface named SpecMx which can be used by Matlab running on the same computer (to enable sharing of measurement data) or even running on a different computer.

The usage is very similar to ***Imspector Python Interface***. The syntax is identical with the obvious conversions (dictionaries to structs, None to []).

The data content of a Stack in Imspector (obtained by Stack.data()) can only be read, not written from Matlab.

## 2.16 Shortcut Keys

| Category | Shortcut | Function |
|---|---|---|
| | Right mouse click | Within an image: context menu appears for selection of e.g., line profile vs. ROI |
| | Left mouse button drag | Select a ROI area (or line profile position) within the image |
| | <Ctrl> + left mouse button drag | Copy selected image (or ROI of the image) into a new window; only measurement data is moved, not the hardware settings |
| | <Ctrl> + <Alt> + left mouse button drag | Copy the view of the data (independently of the selected ROI) to a new window. Both displays will use the same data (i.e. will be updated when new data is acquired) |
| | Select ROI + press 'R' + left mouse button drag to existing scan window | This selects the ranges of the ROI and copies it as scanning parameters. The following measurements uses the selected ROI |
| | Tab | Toggles on/off all live dialogs. This function can avoid blocking display space. |
| | <Alt> + left mouse button drag | Zoom into selected region |
| | Pg Up/Down | Go up/down one slice along the z-axis. Pressing <Shift> moves 10, pressing <Shift> + <Alt> moves 100 slices along the z-axis. |
| | <Ctrl> + Pg Up/Down | Go up/down one layer along the *hidden* axis. Pressing <Shift> moves 10, pressing <Shift> + <Alt> moves 100 slices along the hidden axis. |
| Image view; Size & Info | <Ctrl> + T | Shows information about the stack size (physical dimensions, pixel number, Offsets of the active document |
| File | <Ctrl> + N | Create new measurement |
| File | <Ctrl> + M | Edit comment |
| File | <Ctrl> + O | Open measurements |
| File | <Ctrl> + W | Close measurements |
| File | <Ctrl> + E | Export measurements |
| File | <Ctrl> + X | Crop stack/crop graph |
| File | <Ctrl> + V | Paste stack/paste graph |
| File | <Ctrl> + I | Import measurements |
| File | <Ctrl> + <Shift> + S | Save measurements |
| File | <Ctrl> + S | Save measurements as |
| Graph view | <Ctrl> + R | Rename data container/line profile |
| Graph view | F9 | Adapt the maximum (grey) value limit of the shown graph |
| Graph view | F10 | Adapt the minimum (grey) value limit of the shown graph |
| Graph view | <Ctrl> + T | Change graph size |
| Graph view | <Ctrl> + L | Toggle limits on/off |

| Category | Shortcut | Function |
|---|---|---|
| Graph view | <Ctrl> + A | Select all |
| Graph view | <Ctrl> + D | Change graph style |
| Graph view | <Ctrl> + <Shift> + E | Edit graph properties |
| Graph window | <Ctrl> + <Shift> + G | Open graph window |
| Region of interest | <Ctrl> + <Shift> + U | Set selection as ROI |
| Region of interest | <Ctrl> + <Shift> + R | Copy ROI |
| Image view | F5 | Fit image to frame |
| Image view | <Ctrl> + R | Re-name data container/re-name stack |
| Image view | <Ctrl> + <Shift> + C | Change pixel value |
| Image view; Colormap | F9 | Adapt the upper colormap limit to the current image or selection |
| Image view; Colormap | F10 | Adapt the lower colormap limit to the current image or selection |
| Image view; Colormap | <Alt> + <Shift> + E | Equalize colormap between different measurements in the stack view |
| Image view; Colormap | <Shift> + F9 | Adapt the upper colormap limit to the current image or selection; do it for all measurements |
| Image view; Colormap | <Shift> + F10 | Adapt the lower colormap limit to the current image or selection; do it for all measurements |
| Image view; Gallery | <Ctrl> + 1 ...<br><Ctrl> + 9 | Number of stacks (images) displayed in a row (in Gallery mode) |
| Image view; Gallery | <Ctrl> + G | Toggle Gallery mode/ overlay images: All images in the selected measurement window are overlayed vs. displayed next to each other |
| Image view; Projections | <Ctrl> + H | Hidden axis;<br>Projections |
| Image view; RGB | <Ctrl> + <Shift> + B | RGBize data: For 3 images in one window, colormaps red, blue and green are automatically selected. |
| Image view; Zoom | <Ctrl> + Num– | Zoom out |
| Image view; Zoom | <Ctrl> + Num+ | Zoom in |

*Table 2.16-1: List of shortcut keys*

# 3 Hardware components and quick guide

The following pages contain a list of the main components encountered during everyday imaging. Short instructions for each component are intended to provide a quick reference for users. For full hardware information, safety instructions and laser safety information, please refer to the printed hardware manual provided with your microscope (see printed hardware safety instructions). For start-up and shut-down procedures and typical workflows, please refer to the section Microscope operation.

## 3.1 Hardware overview

The microscope setup consists of several units. The exact layout may differ due to specific configurations and customizations.

- Electronics rack with PC power switch and laser safety keys
- Vibration isolation table carrying the optics box and microscope body (large air suspension table for INFINITY, active isolation for FACILITY setups)
- Table for PC screen, keyboard, and peripherals
- Olympus touch panel control (TPC) and focus control unit (focus/stage/modes) are located on table.
- Accessory units below the vibration isolation table: Controllers for microscope body and stage (here: Olympus), active vibration isolation controller (FACILITY only), Adaptive Optics controller (optional)
- Behind/below the vibration isolation table: Large accessory box containing one or more STED lasers (location may differ in older microscopes)

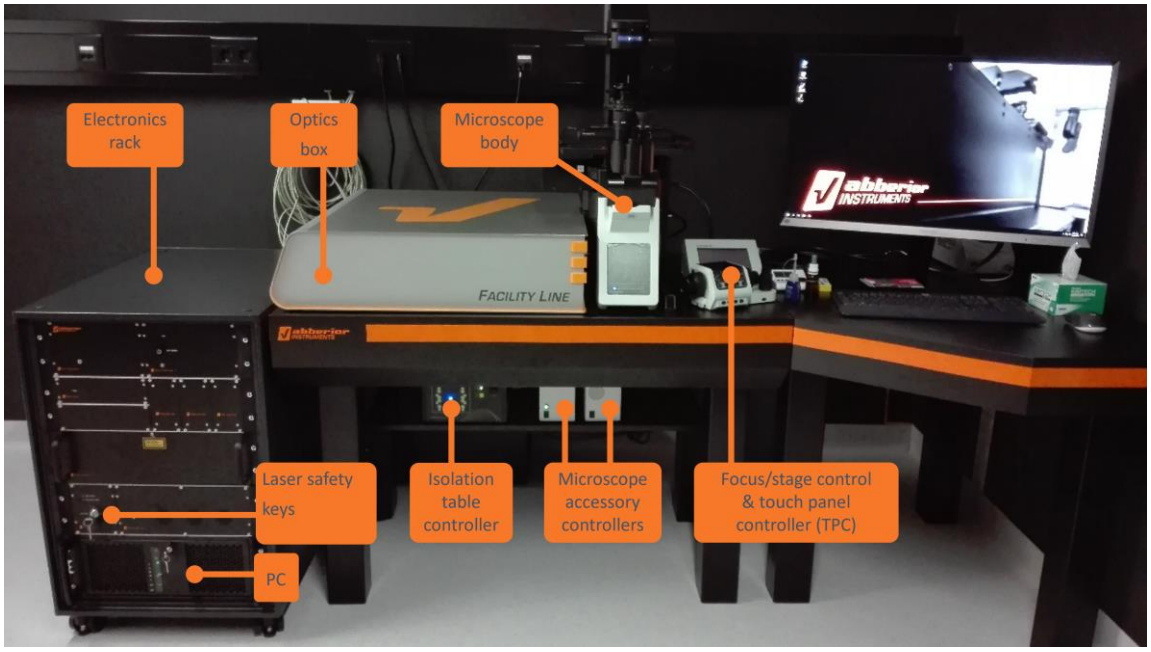



*Figure 3.1: Microscope layout*

The laser safety key can be used to securely disable any laser emission. It can be switched between the Off State (O) and the ON state (I) which is required for imaging. Please refer to the printed hardware manual for more information on laser safety.

The PC switch can be used to switch on the computer, while the reset switch can be used to reset the computer and restart it. A switched-on computer is indicated by an active green light directly below the reset key.

## 3.2 Olympus focus control unit (MCZ)

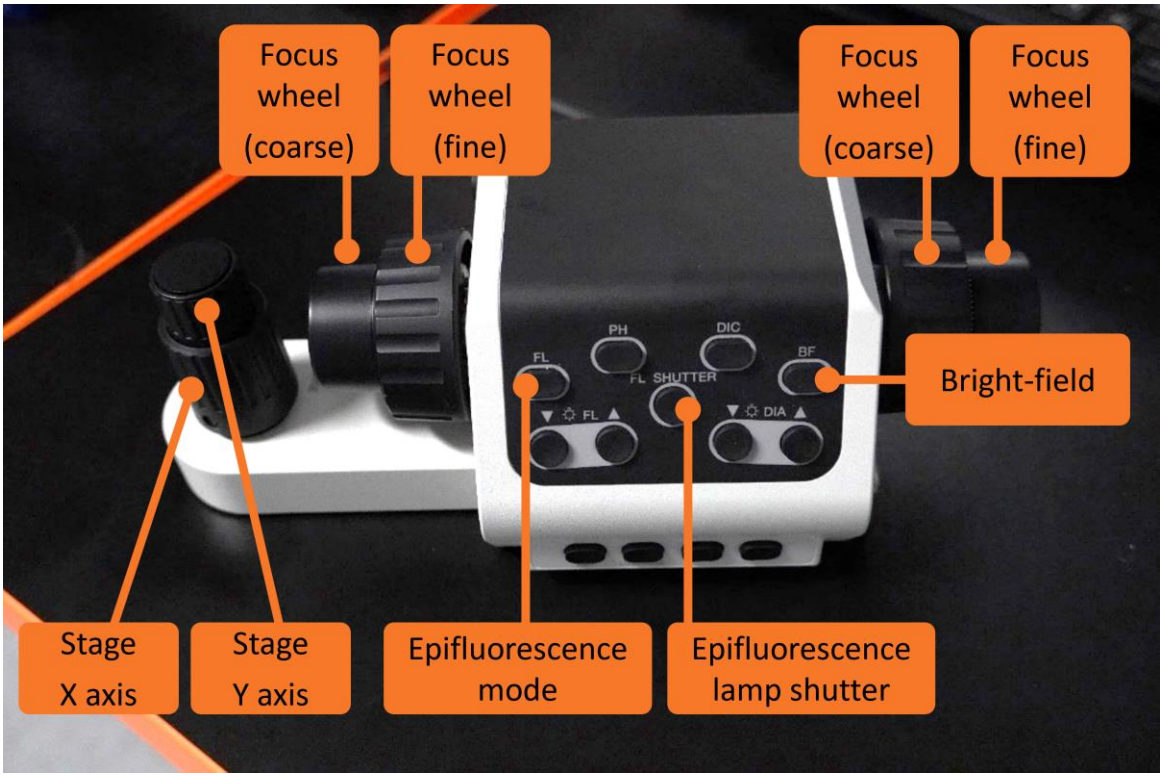



*Figure 3.2: Olympus focus control unit (MCZ)*

The Olympus focus control unit (MCZ) is used to control the microscope stage (X/Y position), the nosepiece (z-focus) and can be used to switch between the Modi Epifluorescence, Scanning (default), and Brightfield illumination. The fluorescence shutter (FL shutter) can be found in the center of the control and can be used to open and close the shutter for the fluorescence lamp. The pair of buttons labelled with the lightbulb DIA can be used to increase or decrease the lamp intensity in brightfield.

The XY control dials can be positioned for either right-handed preference or left-handed preference (shown in *FIGURE* 3.2) by lifting the upper unit and rotating bottom unit. The lower dial is controlling the x-axis movement, the upper the y-axis movement. On the backside of the control unit, a button can be activated to switch between coarse and fine xy mode.

The large dials attached to the central part of the unit can be used to adjust z-focus position of the objective nose piece. The larger wheel controls the coarse focus while the smaller wheel controls fine focus.

> ***Note:*** *Some setups may have different stage / nosepiece controllers (e.g., setups with Nikon microscope body) or manual stage / nosepiece control (custom and some upright setups without motorised stage or nosepiece).*

## 3.3 Olympus TPC (touch panel control)

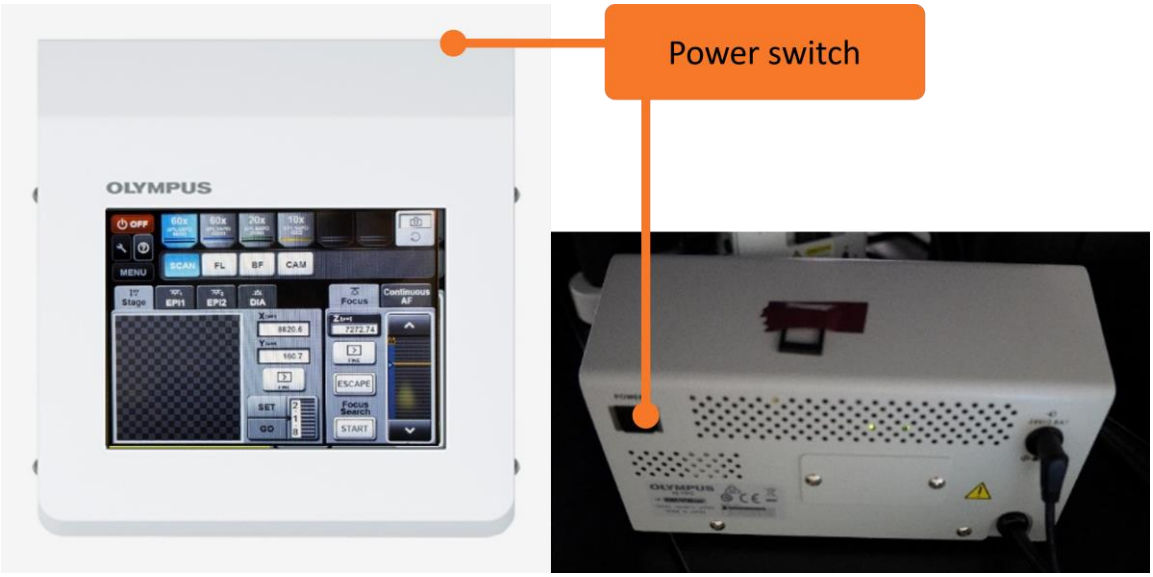



*Figure 3.3: Olympus TPC (touch panel control)*

The Olympus TPC (Touch panel control) can be switched on/off by pressing the flat button at the back of the unit. Booting the TPC takes 1-2 minutes; afterwards it can be initialized by starting Lightbox/Imspector.

> ***Note:*** *It is possible to manually press "Start Operation" on the TPC screen. This sends the microscope body into standalone mode with an independent configuration without connection to the Imspector software. To reconnect to the software, please press "OFF" in the upper left corner to get back to the main screen.*

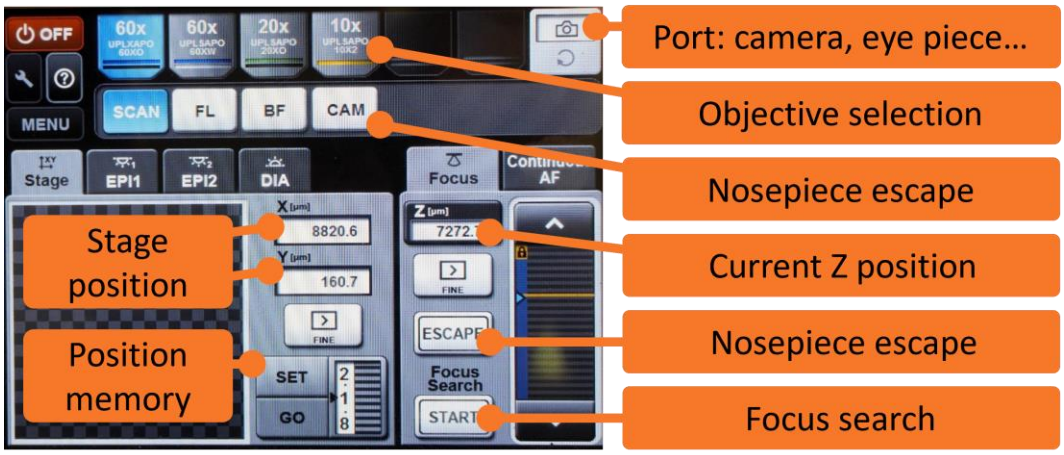



*Figure 3.4: Olympus TPC – General menu & Scan mode*

The touch panel control (TPC) is displaying the available objective lenses at the top and allows to change these by touch.

Several pre-configured imaging modes can be selected in the second row:

- **Scan**
- **Fl**uorescence
- **B**right**F**ield
- **CAM**era.

Scan mode is the default mode for confocal and STED imaging, ensuring that the correct cubes and viewing mode/port are selected. The Imspector (or Lightbox) software will automatically switch to Scan mode once a standard confocal or STED measurement is started (unless the scan mode has been deliberately changed in the measurement settings, see *MICROSCOPE BODY LIVE DIALOGS*).

The stage position can be changed by touch in the lower left part of the TPC. The “ESCAPE” button on the right part of the TPC moves the objective fully lens down and into its escape position. Upon pressing it again, the objective lens moves back to the previous Z-position.

> ***Note:*** *The TPC has its own position memory. The positions 1 to 8 can be selected, the position can be stored by “SET” and recalled by “GO”. This memory is retained as long as the TPC is switched on.*
>
> ***Note:*** *Please keep in mind that large escape movements in Z may cause increased z-drift.*
>
> ***Note:*** *The Z-position can be set by touch (Z[µm]) and entered manually if needed.*

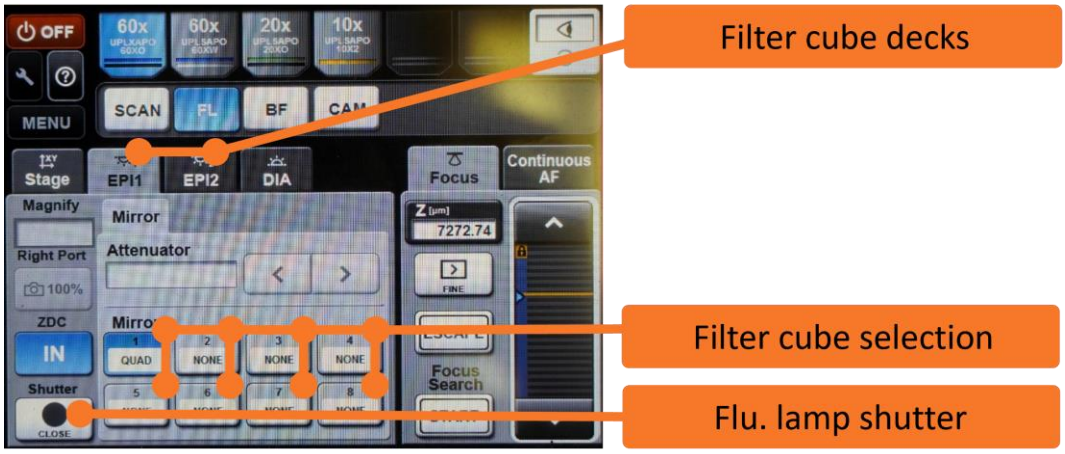



*Figure 3.5: Olympus TPC - Epifluorescence mode*

The epifluorescence mode can be activated by pressing “FL” on the TPC or the respective FL button on the Olympus Focus control unit (called MCZ). Automatically, the first filter deck (EPI1) is selected, and the QUAD band filter is selected. This filter transmits a broad range of fluorescent light to the detector or eye pieces. The fluorescence to be excited is selected by activating the appropriate LED on the separate LED control unit (. A button to control the fluorescent shutter can be found on the Olympus focus control unit (MCZ) or in the lower left corner of the TPC.

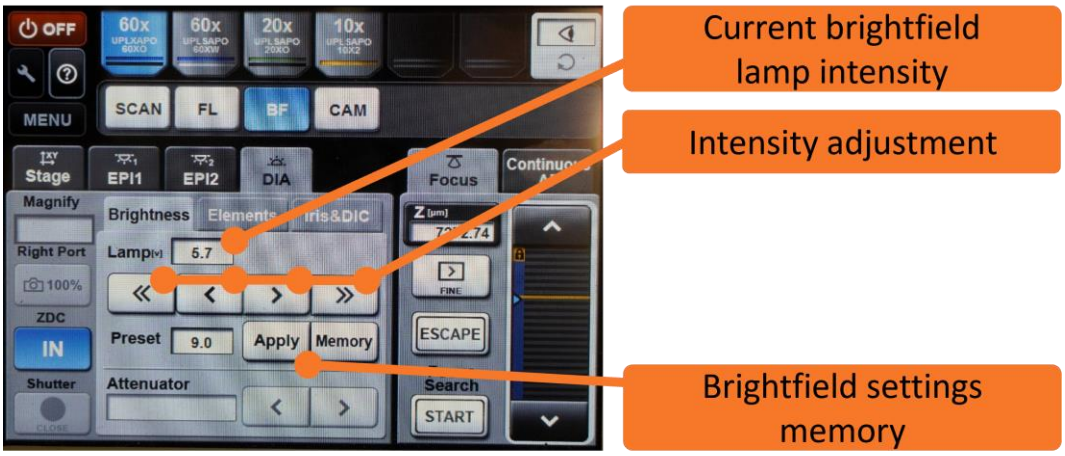



*Figure 3.6: Olympus TPC – Brightfield mode*

The brightfield (BF) mode activates the inbuild brightfield lamp. The lamp controls and memory functions can be found in the DIA ribbon. The brightness of the lamp can be controlled and applied via the arrowhead buttons.

***Note:*** *See condenser instructions for more information on brightfield mode.*

## 3.4 Olympus ZDC: Focus Search / Continuous Autofocus

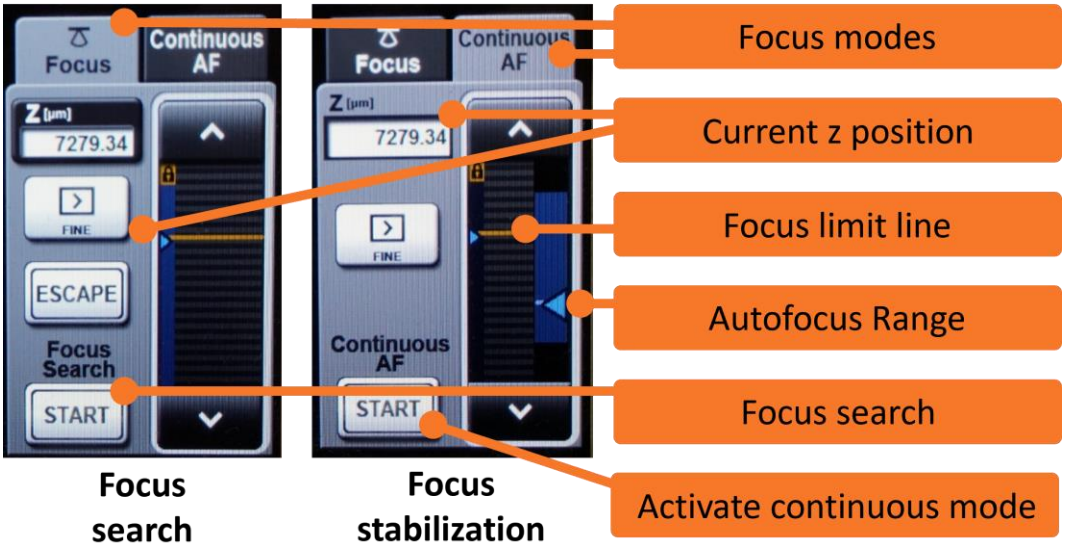



*Figure 3.7: Autofocus functions on the TPC*

The Autofocus Unit (ZDC) enables two functions: Finding the coverslip surface level (Focus Search) and keeping a defined sample plane in focus (Continuous Autofocus). Both functions rely on the reflection of a red-shifted autofocus laser at the interface between the coverslip and the sample medium.

This requires at least a small difference in refractive index between the coverslip glass and the sample medium. Both functions are compatible with simultaneous confocal or STED imaging. The autofocus functions can be controlled directly via the Olympus touch panel control (see below) or via the Imspector software (see *MICROSCOPE BODY LIVE DIALOGS*).

**Focus Search:** The focus finder moves the nosepiece position to bring the objective lens focus point close to the position of the interface between coverslip and sample medium. For many samples, this is a good starting point, as most cultured cells are grown directly on coverslips / glass bottom dishes and larger samples (e.g., tissue) are usually fixed onto coverslip. You will hear one beep if Focus Search was successful and three beeps if the autofocus cannot detect the coverslip position.

**Continuous Autofocus:** The continuous autofocus is designed to keep the objective lens focus point at a defined sample position. The continuous autofocus is a two-step process. Activation via the “Start” button first locates the coverslip position, then the current z position in the sample is locked. Here, the sample position is defined by an offset from interface of coverslip and sample medium. Using the focus wheel the focus on the Olympus focus control unit (MCZ), the z-position can be adjusted within the Autofocus range (not the full nosepiece range).

> ***Note:*** *Optimal autofocus performance can be obtained by using mounting media with a refractive index differing from the refractive index of the glass coverslip (RI ~1.518). Examples are abberior Mount Solid AF/Mowiol, other hardening embedding media, abberior Mount Liquid AF, glycerol based embedding media, or water-based media such as PBS or cell culture media. Mounting media as TDE, ProLong Glass, methyl salicylate or DPX may sometimes match the refractive index of the coverslip glass so well that the autofocus cannot detect the coverslip.*

**Trouble shooting:** If the focus finder fails to locate the coverslip, you have several options: Ensure there are no bubbles in the immersion liquid and/or sample medium. Remove potential dirt from the coverslip surface (e.g., cell culture residues) and try again. Check that the refractive index of the sample medium is not too close to the refractive index of the coverslip glass (RI ~1.518). Ensure that the autofocus unit has been calibrated for objective currently in use (if not, please contact support). If working within the Imspector software, ensure that the focus search range covers the current sample position (see hardware settings section Microscope body). Check whether the current focus limit (yellow line in *FIGURE* 3.7) prevents bringing the coverslip surface into focus.

> ***Note****: When using the autofocus within the Lightbox or Imspector software, please refer to additional information in the software manual (*MICROSCOPE BODY LIVE DIALOGS*), especially if your microscope also features a Z-piezo (see* EXPERIMENTAL CONTROL – SCANNING*).*

## 3.5 Epifluorescence illumination

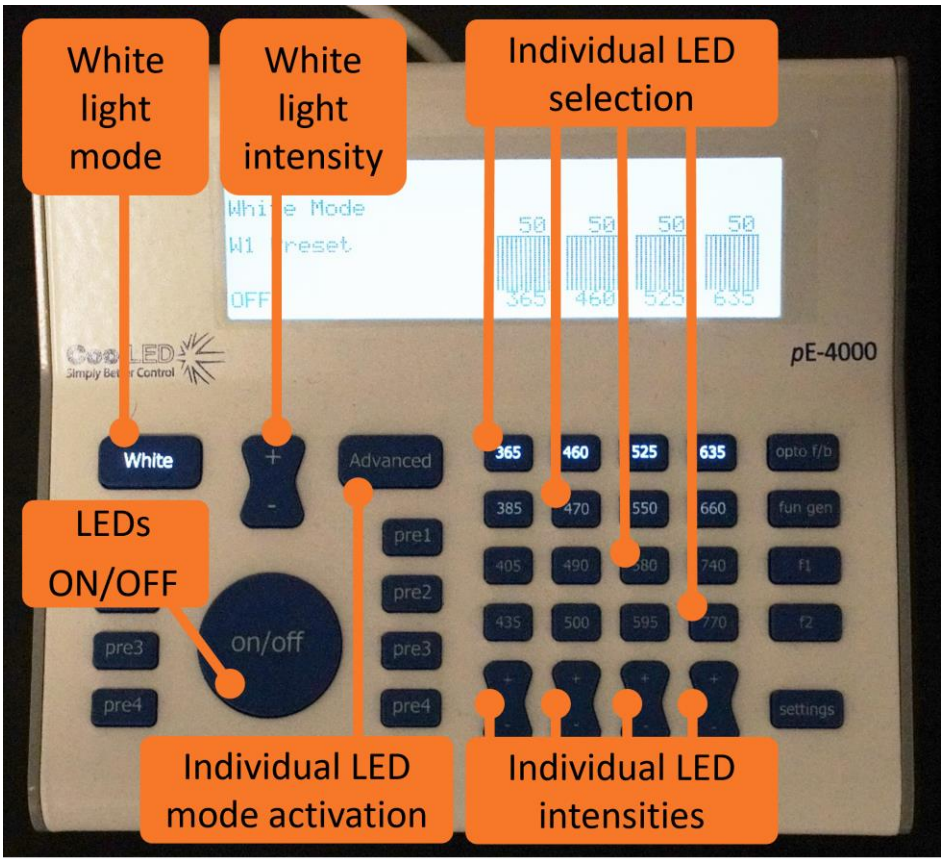



*Figure 3.8: Epifluorescence illumination (CoolLED pE-4000)*

A variety of fluorescent light sources are available including e.g., a 4-lamp Thorlabs LED or a CoolLED pE-4000 with 16 wavelength-specific LEDs. For the CoolLED depicted in ***FIGURE*** 3.8, the individual LEDs can be selected after pressing “Advanced” followed by the buttons on the right. Alternatively, the “white” option can be activated to illuminate with white light (mixture of multiple LEDs). The on/off button is used to activated/deactivate the emission of the currently selected LEDs. The +/- button above controls the intensity level of the LEDs. For further information please consult the complementary LED manual of the manufacturer (as found in the collection of printed manuals).

## 3.6 Illumination column

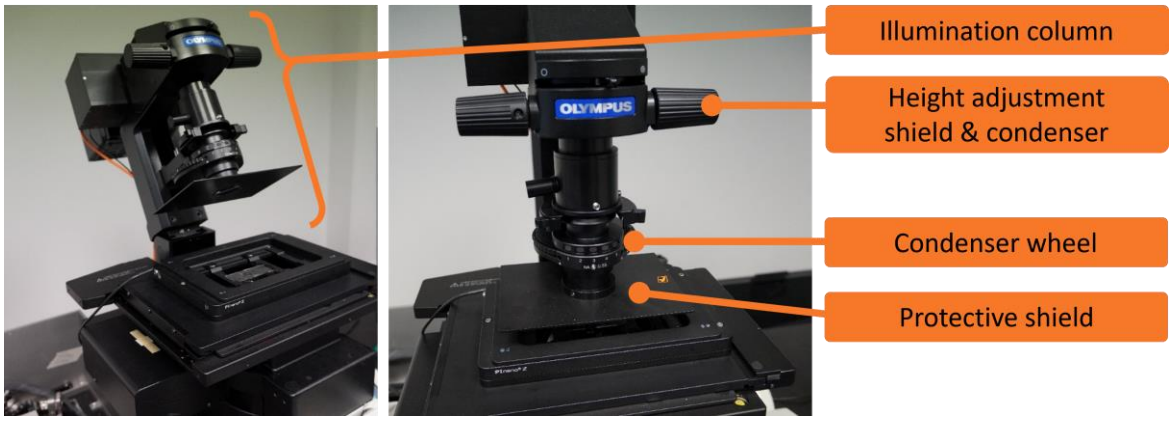



*Figure 3.9: Illumination column (Olympus IX83)*

For sample mounting, the illumination column of the microscope needs to be tilted back (see **FIGURE** 3.9). Then the slide or dish can be mounted in the respective sample holder and the boom is tilted back to the neutral position. The interlock for laser safety allows laser emission only in upright position of the illumination column.

The installed protective shield should be lowered to cover the sample and minimize the escape of scattered light.

> ***Note:*** *Microscopes with microscope bodies other than an inverted Olympus IX83 body may have different illumination columns or no column at all.*

## 3.7 Condenser

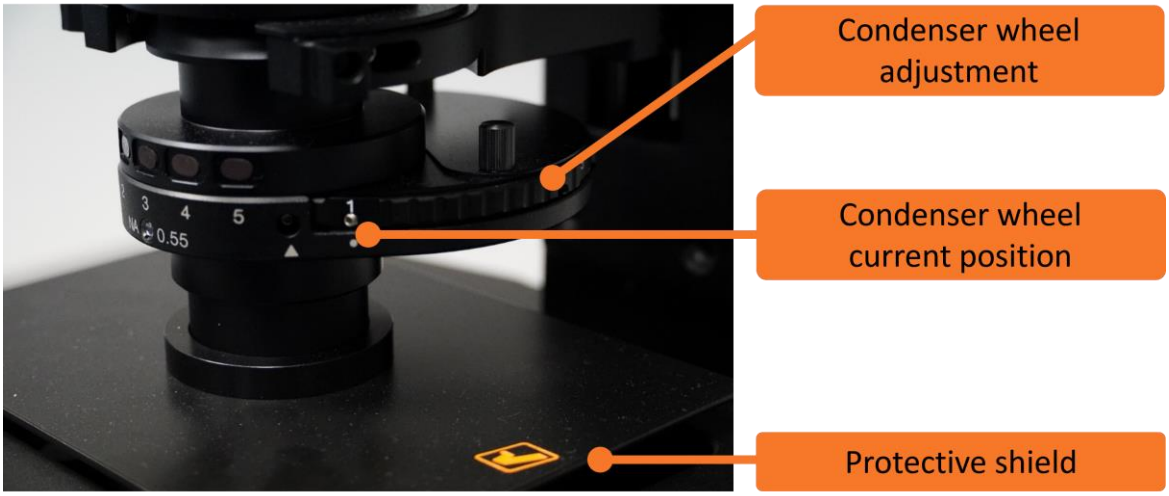



*Figure 3.10: Condenser wheel (Olympus IX83)*

The condenser wheel needs to be in position (#1) for confocal and STED imaging (see **FIGURE** 3.10). This will block the brightfield illumination for laser safety reasons.

For brightfield imaging, please turn the condenser wheel to position #2 to allow brightfield illumination to pass through the condenser.

If the Imspector or Lightbox software reports an issue with the “System Status” preventing laser emission, it is very likely that condenser wheel position, the illumination column angle, the observation mode (eyepiece, camera) or the laser safety key position need to be changed into a laser safe state.

> ***Note:*** *In microscopes with transmitted light detection, multiple condenser wheel positions may be compatible with confocal and STED imaging. This is due to a more protected beam path for the transmitted excitation and STED laser light.*

The vertical position of the condenser can be adjusted to optimise brightfield illumination. If installed on your system, the vertical position can also be adjusted for transmitted light detection (in confocal and STED mode) or DIC (camera or eyepiece). Please refer to the manuals of the microscope body manufacturer for details on optimising brightfield or DIC (if installed).

> ***Note:*** *Microscopes with microscope bodies other than an inverted Olympus IX83 body may have different condensers or no condenser at all. In upright setups, a condenser may be located below the objective and stage.*

## 3.8 Electronics rack

The electronics rack houses the PC, fast electronics, AOM controllers, galvo controllers, excitation lasers and excitation laser controllers, safety interlock electronics as well as multiple power supplies.

The user can adjust a number of switches:

- PC power switch to start-up or reset the setup PC.
- Laser safety keys: All laser safety keys need to be in the ON position to allow confocal and STED imaging.
- Galvo shutdown button: This button can be pressed to shut down the galvo scanner in cases of emergency. Depress the button to re-enable galvo scanning.
- A number of power switches may be present at the front of the electronics rack. For older microscopes, these may be required during manual switch-ON and shut-down procedures (see section *4.1*).

## 3.9 STED laser rack

STED lasers are usually accommodated in a smaller rack located behind the microscope. The STED laser rack may feature additional laser safety keys that need to be in the 'ON' position to allow STED imaging. A lamp can indicate the status of each STED laser.

## 3.10 Vibration isolation table

A vibration isolation table is important for STED microscopes, as they reach resolutions of tens of nanometers making even small vibrations noticeable in images.

### 3.10.1 Air suspension table

Please contact support if the table is not properly suspended on all four edges.

**Risk of injury:** Limbs may be crushed/pinched if reaching between table breadboard and table suspension/legs. See hardware manufacturer information for more information.

### 3.10.2 Active suspension table

Please do not place heavy objects on breadboard/table insert to allow proper function of active vibration isolation. Please do not wedge any objects into the gap between the table and the breadboard as this will prevent proper function of the active vibration isolation.

# 4 Microscope operation

## 4.1 Startup and shutdown

The Expert Line, Infinity and Facility microscopes are turn-key systems with a simple startup routine. Microscopes delivered before 2015, may require a more complex routine for startup.

### 4.1.1 Complete Startup

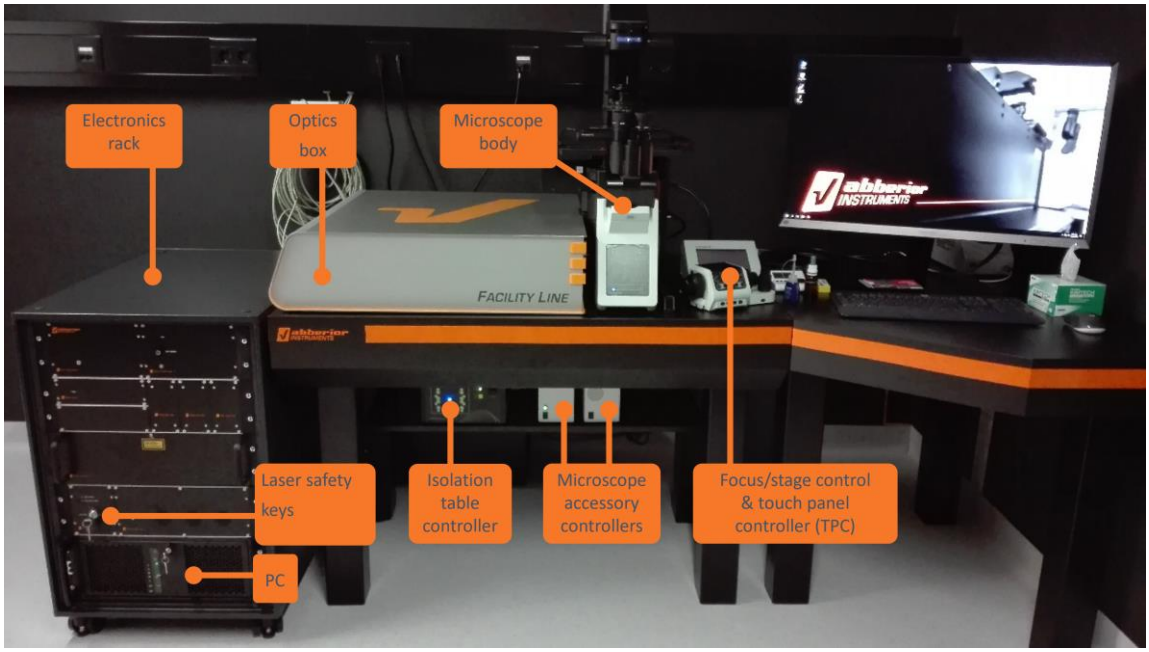



*Figure 4.1: Microscope layout*

To switch the microscope ON for imaging, please follow the steps below. All components are shown in Figure 3.1: Microscope layout. To switch the microscope OFF, please follow the same routine in reverse order to close/power down all components.

1. Switch ON PC (if currently OFF).
2. Switch ON active isolation table (for FACILITY microscopes only).
3. Switch ON all microscope units*
   Microscope accessory controllers, touch screen panel (TPC)
4. Optional: Switch ON fluorescence lamp or LED.
5. Ensure all keys and switches of the electronics rack are ON.
6. Start Imspector software (shortcut on desktop)
7. After several seconds of initialization, different dialogs may open, depending on whether the system is currently OFF (see *FIGURE 4.2*) or partially switched ON (see *FIGURE 4.3*). Now confirm hardware switch ON.

*Different units may need to be switched ON for microscopes based on microscope bodies other than an Olympus IX83.

***Note:** When starting the Imspector software, there may be additional options and variations.*

*For more details on Imspector startup options, see section Starting Imspector.*

- *Different configuration files for different users.*
- *Different software versions requiring an up-conversion or down-conversion of the current configuration file.*
- *Some microscopes initially load the Lightbox GUI. Users can then switch to Imspector via the keyboard shortcut "**Ctrl+Shift+ Alt+F11**".*

***Note:** The microscope allows confocal/STED imaging only if the microscope is in a laser-safe state. A yellow/red status symbol (top right screen corner in Imspector/Lightbox software) may indicate that the hardware needs to be adjusted:*

- *Ensure that the condenser wheel is set to position 1 (see CONDENSER).*
- *Ensure that the illumination column in an upright position (see FIGURE 3.9).*
- *Ensure all laser safety keys are in the ON position (see FIGURE 3.1).*
- *Ensure Scan mode is selected. (Usually selected via the touch-screen panel: FIGURE 3.3).*
- *Infinity microscopes: Ensure the lid of the optics box is closed properly.*

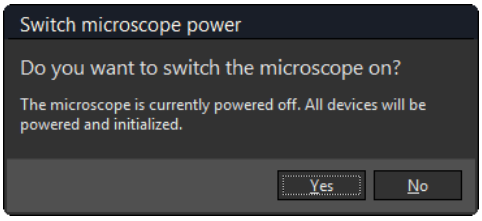



*Figure 4.2: Switch microscope power dialog (OFF to ON)*




*Figure 4.3: Switch microscope power dialog (partially OFF to ON)*

### 4.1.2 Startup when Imspector is already running

If the microscope control software Imspector is already running but the microscope hardware has been (partially) switched OFF, please follow the following steps for powering up:

1. Click the 'Power Switch' button (usually located near the top right of the Imspector window).

2. Different dialogs may open, depending on whether the system is currently OFF (see *FIGURE 4.2*) or partially switched ON (see *FIGURE 4.3*) Now confirm hardware switch ON.

### 4.1.3 Complete shutdown

The shutdown procedure is very similar to the complete startup procedure (section *4.1.1*) in reverse order:

1. Stop all measurements and save all data.
2. Close the Imspector software.
3. Confirm hardware switch OFF (see *FIGURE 4.4*). This process will take several seconds.
4. Shut down PC (if intended).
5. Switch OFF active isolation table (for FACILITY microscopes only).
6. Switch OFF fluorescence lamp / LED.
7. Depending on the policies of your facility, switch OFF all microscope units* (Microscope accessory controllers, touch screen panel (TPC)).
   abberior does not recommend switching off the microscope controllers on a daily basis.
8. Ensure all keys and switches of the electronics rack are OFF.

*Different units may need to be switched OFF for microscopes based on microscope bodies other than an Olympus IX83.

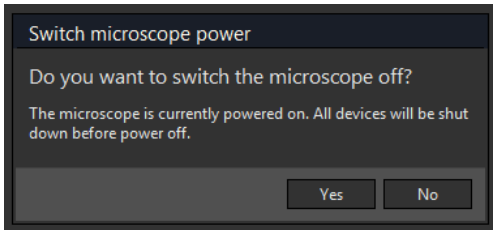



*Figure 4.4: Switch microscope power dialog (ON to OFF)*

### 4.1.4 Shutdown but keep Imspector running

Users may want to power down the hardware but keep the Imspector software running, e.g., to visualize or analyze previously recorded data. Please note that data may also be opened, visualized and analyzed on different PCs using the appropriate Imspector BASE version (see Imspector Full version and BASE version). To power down the hardware without closing the Imspector software, please follow the following steps:

1. Stop all measurements and save all data.
2. Click the 'Power Switch' button (usually located near the top right of the Imspector window
3. Different dialogs may open, depending on whether the system is currently OFF (see Figure 4.2) or partially switched ON (see ***FIGURE 4.3***)
   Now confirm hardware switch ON. Close the Imspector software.
4. Confirm hardware switch OFF (see ***FIGURE 4.4***). This process will take several seconds.
5. Switch OFF active isolation table (for FACILITY microscopes only).
6. Switch OFF fluorescence lamp / LED.
7. Depending on the policies of your facility, switch OFF all microscope units*
   (Microscope accessory controllers, touch screen panel (TPC)).
   abberior does not recommend switching off the microscope controllers on a daily basis.
8. Ensure all keys and switches of the electronics rack are OFF.

*Different units may need to be switched OFF for microscopes based on microscope bodies other than an Olympus IX83.

Figure 4.2: Switch microscope power dialog (OFF to ON)

### 4.1.5 Complete startup (manual procedure for old models only)

Microscopes delivered before 2015 may not feature a software-controlled power switch for automated startup and shutdown routines (if unsure, please check with support). Here, a manual startup procedure may be required in the order as described below. Not suitable for newer microscopes.

1. Switch ON PC (if currently OFF).
2. Switch ON all microscope units*
   Microscope accessory controllers, touch screen panel (TPC)
3. Optional: Switch ON fluorescence lamp or LED.
4. Switch ON the main power switch of the power supply (orange) and of the AOM controller (black), located at the front panel.
5. Switch ON the STED laser(s) by switching the main power switch and the key switch into the “1” position.
6. Switch ON the excitation lasers at the electronics rack.
7. Switch on the detectors at the power supply.
8. Start Imspector software (shortcut on desktop)

*Different units may need to be switched ON for microscopes based on microscope bodies other than an Olympus IX83.



***Important note:*** *If you have both a 775 nm and a 595 nm STED laser the switching order is relevant and depends on the actual laser models built into your system. During installation, you will be informed about the correct switching order for your system.*

### 4.1.6 Complete Shutdown (manual procedure for older models only)

Microscopes delivered before 2015 may not feature a software-controlled power switch for automated startup and shutdown routines (if unsure, please check with support). Here, a manual shutdown procedure may be required in the order as described below. Not suitable for newer microscopes.



***Important notes:***

*Powering OFF of a STED laser without a controlled shutdown process can cause catastrophic damage to the STED laser. Always make sure to follow the shut down process of the STED laser(s) exactly to prevent any kind of damage. Therefore, use only the 'Pow' button in the Imspector control software!*

*If you have both a 775 nm and a 595 nm STED laser the switching order is relevant and depends on the actual laser models built into your system. During installation, you will be informed about the correct switching order for your system.*

1. Switch ON PC (if currently OFF).
2. Switch ON all microscope units*
   Microscope accessory controllers, touch screen panel (TPC)
3. Stop all measurements and save your data.
4. Click the ‘Power Switch’ button (usually located near the top right of the Imspector window, see

5. Power Switch and Microscope Status Toolbar). This will shutdown the STED laser in a controlled manner. Once the STED laser shutdown is completed, an error message will indicate the OFF state of the STED laser.
6. Switch OFF the STED lasers at the electronics rack by turning the key switch to '0' and switching OFF the lasers main power switch.
7. Switch OFF the detectors at the power supply.
8. Turn OFF the power switch of the AOM controller and of the main power supply.
9. Close the Imspector software.
10. Shut down PC (if intended)
11. Depending on the policies of your facility, switch OFF all microscope units*
    (Microscope accessory controllers, touch screen panel (TPC)).
    abberior does not recommend switching off the microscope controllers on a daily basis.

*Different units may need to be switched OFF for microscopes based on microscope bodies other than an Olympus IX83.

# 5 Further reading

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## 5.2 RESOLFT microscopy

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# 6 Abbreviations

| | |
|---|---|
| AOM | Acusto-Optical Modulator |
| AOTF | Acusto-Optical Tunable Filter |
| APD | Avalanche Photo Diode |
| cw | continuous wave |
| DM | Deformable Mirror |
| DYMIN | Dynamic Minimum |
| rsEGFP | reversible switchable enhanced GFP |
| EGFP | enhanced GFP |
| EYFP | enhanced YFP |
| FLIM | Fluorescence Lifetime IMaging |
| FWHM | Full-width of Half Maximum |
| GFP | Green Fluorescent Protein |
| IP | intellectual property |
| LUT | Look-Up-Table |
| NA | Numerical Aperture |
| PFS | Perfect Focusing System |
| PMT | Photomultiplier (Tube) |
| PSF | Point Spread Function |
| RESCUE | Reduction of State transition Cycles |
| RESOLFT | REversible Saturable Optical Fluorescence Transitions |
| RF | Radio Frequency |
| ROI | Region of Interest |
| RSFP | Reversible Switching Fluorescent Protein |
| SLM | Spatial Light Modulator |
| STED | STimulated Emission Depletion |
| TCSPC | Time Correlated Single Photon Counting |
| YFP | Yellow Fluorescent Protein |
| ZDC | IX3 Z-Drift Compensator (IX3-ZDC) |

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